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A systematic review on the instruments used for measuring the level of multimorbidity

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TITLE PAGE

A systematic review on the instruments used for measuring the level of multimorbidity

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ABSTRACT

Objectives: There are multiple instruments for measuring multimorbidity. The main objective of this systematic review was to provide a list of instruments for measuring level of multimorbidity. The secondary objective was to provide details of the requirements, strengths and limitations of these instruments.

Methods: A systematic review was conducted according to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses guidelines (PROSPERO registration number: CRD42018105297). We searched MEDLINE, Embase and CINAHL electronic databases and manually searched the Journal of Comorbidity. Studies published between January 2010 and August 2018 inclusive, in English language. Studies also had to select adult patients from primary care or the general population, and had at least one specified outcome variable. Two authors screened the titles, abstracts and full texts independently. Disagreements were resolved with a third author. The modified Newcastle Ottawa Scale was used for quality assessment.

Results: A total of 67 studies (74 articles) were identified, and 46 studies were rated to have a low risk of bias. In total, 27 unique instruments were described. Disease count and weighted indices like the Charlson Comorbidity Index were commonly used. Other approaches included case-mix or pharmaceutical-based instruments. Disease count was the common instrument used for measuring all three essential core outcomes of multimorbidity research: mortality, mental health and quality of life. There is a rise in the development of novel weighted indices by using prognostic models. The data obtained for measuring multimorbidity were from various sources including medical records, patient self-reports, and large administrative databases.

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Conclusions: We provided a list of 27 different instruments detailing the requirements, strengths and limitations of these instruments. We noted that a clear description of these instruments was lacking in the literature.

ARTICLE SUMMARY

Strengths and Limitations of this study

- This review builds upon Huntley et al.'s 2009 review paper and provides an updated, comprehensive list of instruments that measure levels of multimorbidity for community-dwelling individuals
- Thorough literature search of three major electronic databases with the involvement of a health science librarian
- Reporting of search as per Preferred Reporting Items for Systematic Reviews and Meta-Analysis guidelines
- Exclusion of non-English language papers and grey literature

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MAIN TEXT

1. BACKGROUND

Multimorbidity is defined as the co-occurrence of two or more chronic medical conditions in an individual.[1] It is a growing public health challenge and accounts for most of the expenditures in the healthcare system.[2] The complex interactions of several co-existing diseases have profound implications on individuals[3, 4] and their healthcare providers.[5,6]

There are multiple instruments for measuring multimorbidity due to a lack of standardisation of the definition of multimorbidity and its measurements. Lefevre et al.[7] listed four common methods of measuring multimorbidity. They are: by simple counts of chronic diseases (i.e., disease count), by grouping chronic diseases into dyads or triads (i.e., dyad and triad patterns), by identifying groups of people with common disease and characteristics that occur more often than by chance (i.e. non-random associative patterns), and by using an index of variable complexity (i.e., weighted indices). The first three methods are usually used to measure the prevalence or patterns of multimorbidity. The fourth method is usually used to measure the different levels of multimorbidity to predict an outcome or to evaluate an intervention for a desired outcome. The set of core outcomes of multimorbidity (COSmm) was established after consulting a Delphi panel of experts.[8] Core outcome sets (COS) represent the minimum that should be measured and reported in all clinical trials of multimorbidity.[9]

One major criticism of the first three methods is that the severity of individual conditions is not usually specified.[10] Even then, reporting the severity of individual conditions does not accurately reflect the combined effects of multiple conditions because the combined effect may be additive, multiplicative, or even subtractive.[11]

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Huntley et al.[12] published a systematic review describing the instruments used to measure the morbidity burden in primary care and the general population. They found 17 different instruments from 194 articles. The most widely used instruments and those with the most significant evidence of validity were the Charlson Comorbidity Index (CCI), disease counts, and the Adjusted Clinical Groups (ACG) system.[12] However, this review was conducted in 2009 and the multimorbidity research has increased exponentially since then.

The first objective of the present review was to build upon Huntley et al.'s 2009 review paper in order to provide a current and comprehensive list of instruments that measure levels of multimorbidity for community-dwelling individuals. We used the term 'level of multimorbidity' to refer to the combined effects of multiple conditions on an individual. We specifically looked at instruments that measure the prediction of a specific outcome with different levels of multimorbidity in community-dwelling individuals. Our second objective was to provide details of the requirements, strengths and limitations of these instruments so that clinicians and researchers can select or develop instruments that match their needs.

2. METHODS

A protocol for this systematic review (CRD42018105297) was published online on PROSPERO.[13] We searched MEDLINE, EMBASE, CINAHL and also manually searched the Journal of Comorbidity for potential articles. The medical subject headings and keywords used for the search are shown in Appendix 1.

We selected studies that included: (1) adult patients from primary care or the general population; (2) studies with at least one specified outcome variable; and (3) full-text articles published from January 2010 to August 2018. Studies were excluded if they (1) selected

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patients from the hospital or nursing home only or patient data were drawn solely from the hospital or the nursing home; or (2) selected patients with an index condition; or (3) used level of multimorbidity as a covariate and not the main independent variable; or (4) were not written in English. We did not include a specific definition of multimorbidity instrument because, given a lack of consensus in the literature on the use of this term, we wanted to include a diverse range of papers on the above topic. We included instruments that measured multimorbidity for the purposes of establishing prevalence as well as instruments that measured multimorbidity for the purposes of predicting outcomes.

One reviewer (ESL) conducted a preliminary screen of titles and abstracts to exclude records that were irrelevant. Abstracts of the remaining records were screened independently by two reviewers (ESL & EH) according to the eligibility criteria. Disagreements were resolved through discussion until a consensus was reached. The full-text articles were then retrieved for the agreed list and independently assessed according to the eligibility criteria by the same reviewers. Disagreements were resolved through discussion with a third reviewer (TSH) until a consensus was reached. After agreement on the list of articles, the reference lists of included articles were hand-searched for additional eligible papers.

The selected articles were next appraised independently by three reviewers (ESL, EH and TSH) for risk of bias. Each article was assessed using the modified Newcastle-Ottawa Scale (NOS)[14,15] which examined three broad categories: a) Selection; b) Comparability; and c) Outcome (Appendices 2 & 3). We reported only instruments that were described in studies with a low risk of bias.

We contacted the authors, as needed, for additional information or clarification. Any disagreements on the risk of bias were resolved among the three reviewers through regular

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meetings. HLK and FYW were responsible for tracking and updating the final outcome of the risk-of-bias assessment. We anticipated that the studies would be heterogeneous and therefore the decision was made *a priori* not to conduct a meta-analysis.

3. RESULTS

The number of included articles was 74 articles, representing 67 unique studies. A summary of the PRISMA flow chart is depicted in Figure 1. Thirty studies selected participants from the general population and 37 studies selected participants from primary care. There were 28 cohort studies, 36 cross-sectional studies, and three that combined the two methods (i.e., cross-sectional and cohort).

We found 33 unique instruments of which 27 were from studies of low risk of bias. The instruments were categorised according to Sarfati[16,17] into: 1) simple counts of individual conditions; 2) organ or system-based approaches; 3) conditions that have been weighted and combined into indices; and 4) other approaches. A total of 112 outcomes were reported from all the studies. The most commonly reported outcomes were health care use (n=34), mortality (n=14), and health-related quality of life (n=14). Appendices 4 & 5 summarises the risk of bias assessment of each article. Table 1 provides a summary of the study design, population source, age group, multimorbidity measurements and outcome measures of all the 54 articles with low risk of bias.

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Table 1. Summary of included articles with good risk of bias judgement

Author (Year)	Study Design	Population source	Age	Multimorbidity Measurement	Outcomes measured
Agborsangaya (2013)[18]	CS	GP	≥ 18	DC	HRQoL
*Barile (2013)[19]	Cohort	GP	≥ 65	DC	ADL limitations, Physically unhealthy days, Mentally unhealthy days
*Barile (2012)[20]	CS	GP	≥ 65	DC	Physical HRQoL, Mental HRQoL
Barnett (2012)[21]	CS	PC	≥ 0	DC	Presence of mental health disorder
Biehl (2016)[22]	Cohort	PC	≥ 65	ERA, CCI	Presence of critical illness
+Brilleman (2014)[23]	Cohort	PC	≥ 18	QOF count, CCI, EDC count, ACG, RUB	Primary healthcare cost
+Brilleman & Salisbury (2013)[24]	Cohort	PC	≥ 18	QOF count, CCI, EDC count, ACG, RUB, Prescribed drugs count	Mortality and Number of primary care consultations (3-years period)
Carey (2013)[25]	Cohort	PC	≥ 60	Standard QOF, Extended QOF, CCI (Khan)	Mortality (1-year period)
Chapman (2015)[26]	Cohort	GP	≥ 18	CCI, CCI-PSR	Mortality (5, 10, 15, 20, 25-years period)
Chen (2011)[27]	CS	GP	≥ 18	DC	General Health, Mental Distress, Physical Distress, Activity limitations
Crane (2010)[28]	Cohort	PC	≥ 60	ERA	No. of hospital visits, ED visits, Hospital admissions, Days hospitalised (1-year period)
#Crooks (2016)[29]	Cohort	PC	20 to 100	Co-morbidity linked score, CCI, Elixhauser Index	Mortality (1-year period)
#Crooks (2015)[30]	Cohort	PC	≥ 20	CCI (Read), CCI (ICD-10), CCI (Read & ICD-10)	All-cause mortality (1-5 years)
DiNapoli (2017)[31]	CS	PC	≥ 50	Organ systems with chronic disease	Presence of depressive or anxiety disorder
Formiga (2013)[32]	Cohort	PC	85	CCI	Mortality (3-years period)
Formiga (2011a)[33]	Cohort	GP	90 to 99	CCI	Mortality (5-years period)
**Formiga (2011b)[34]	CS	PC	85	CCI	Successful aging
**Formiga (2016)[35]	Cohort	PC	85	CCI	Mortality (5-years period)
Fraccaro (2016)[36]	Cohort	PC	≥ 18	CCI (Khan)	Mortality (1, 5, 10-years period), Mortality (3, 6, 12-months period)
Galenkamp (2011)[37]	CS	GP	57 to 98	DC	SRH
Garin (2014)[38]	CS	GP	≥ 50	DC	QOL, Disability
Glynn (2011)[39]	CS	PC	> 50	DC	Primary Care Consultations, Hospital outpatient visits, Hospital admissions, Healthcare cost (all 1-year period)
Haas (2013)[40]	Cohort	PC	≥ 18	ACG, Minnesota Health Care Home Tiering, HCC, ERA, CCC, CCI, Hybrid Model	Hospitalisation, ED visits, Readmission within 30 days, Healthcare expenditure (all 1-year period)
Hwang (2015)[41]	Cohort	GP	≥ 0	ACE-27, ACE-27 count	Healthcare expenditure
Kristensen (2014)[42]	CS	PC	> 0	RUB	Fee-for-services expenditures
Lapi (2015)[43]	CS	PC	≥ 15	HSMI	Total mean healthcare cost per year
Lawson (2013)[44]	CS	GP	≥ 20	DC	Preference_ Weighted HRQoL

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Table 1. Summary of included articles with good risk of bias judgement (Continued)

Author (Year)	Study Design	Population source	Age	Multimorbidity Measurement	Outcomes measured
Lemke (2012)[45]	Cohort	GP	≥ 0	CCI, ACG	Inpatient hospitalisations
Marengoni (2011)[46]	CS	GP	≥ 75 (baseline) ≥ 77 (follow-up)	DC	Disability
McDaid (2013)[47]	CS	GP	≥ 50	DC	Disability, QoL, SRH
++Payne (2014)[48]	Cohort	PC	≥ 20	DC	Unplanned hospital admissions (1-year period)
++Payne (2013)[49]	Cohort	PC	≥ 20	DC	Unplanned hospital admission, Potentially preventable admission (all 1-year period)
Quail (2011)[50]	Cohort	GP	≥ 20	DC, CCI (Quan), Elixhauser (Quan), No. of different dispended drugs, CDS	Mortality (1-year period), One or more hospitalisations, Two or more hospitalisations
Ranstad (2014)[51]	CS	GP	≥ 0	RUB	Registered active listing in primary care and all healthcare
Reyes (2014)[52]	Cohort	PC (Men)	≥ 65	CCI	Hip fractures
Ryu (2015)[53]	CS	PC	≥ 18	DC	Deficits of perceived general health, Depressive symptoms
Salisbury (2011)[54]	Cohort	PC	≥ 18	QOF count, EDC count	Primary Care consultation rates, Continuity of care (all 3-year period)
Saver (2014)[55]	Cohort	GP	≥ 65	CCI (Romano) + Hypertension	Acute ACSH, Chronic ACSH
Shadmi (2011)[56]	CS	GP	≥ 18	ADG, CCI	No. of primary care physician visits, specialist visits, hospitalisation
Stanley & Sarfati (2017)[57]	Cohort	PC	≥ 18	M3 Index, CCI, Elixhauser (van Walraven)	Mortality, Overnight hospitalisation (all 1-year period)
Streit (2014)[58]	Cohort	PC	50 to 80	CCI, DC	Quality of cardiovascular preventive care, Quality of preventive care
Sullivan (2012)[59]	CS	GP	≥ 18	DC	Preference-based HRQoL
Takahashi (2016)[60]	Cohort	PC	≥ 18	Minnesota Tiering (ACG), Enhanced model	Hospitalisation / ED visits
Takahashi (2012)[61]	Cohort	PC	> 60	ERA	Mortality, Nursing home placement (all 2-year period)
Ubalde-Lopez (2016)[62]	CS	GP	F (Mean) 35.9, M (Mean): 37.9	MDMS	Sickness absence episodes taken in last 2 years
van den Bussche (2011)[63]	CS	PC	≥ 65	DC	Frequency of contacts with physicians, No. of different ambulatory physicians contacted (all 1-year period)
van Oostrom (2014)[64]	CS	PC	≥ 55	DC	No. of contacts with general practice, medications prescribed and referrals
##Wallace (2016a)[65]	Cohort	PC	≥ 70	Pra tool, Modified Pra tool	Emergency hospital admission (1-year period)
##Wallace (2016b)[66]	Cohort	PC	≥ 70	DC, Barnett conditions DC, CCI, Prescribed drugs count, RxRisk-V	Emergency admission, Functional decline (all 2-year period)
Wei (2018)[67]	CS	GP	≥ 51	MWI	Subjective physical functioning, Grip strength, Gait speed, Cognitive performance, ADL limitations, IADL limitations

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Table 1. Summary of included articles with good risk of bias judgement (Continued)

Author (Year)	Study Design	Population source	Age	Multimorbidity Measurement	Outcomes measured
Wei & Mukamal◊ (2018)[68]	Cohort	GP	≥ 36	MWI, DC, CCI	Mortality (10-years period), Future physical functioning
Wikman (2011)[69]	CS	GP	≥ 50	DC	QoL, Affective well-being
Wister (2015)[70]	CS	GP	≥ 65	MM additive scale, MM weighted by HUI3, MM weighted by ADL Scale, MM weighted by HUI3 betas	Life satisfaction, Perceived health status

Note. ACE-27 = Adult Comorbidity Evaluation-27; ACG = Adjusted Clinical Groups; ACSH = Ambulatory Care Sensitive Hospitalisation; ADG = Aggregated Diagnosis Groups; ADL = Activities of Daily Living; CCC = Chronic Condition Count; CCI = Charlson Comorbidity Index; CCI-PSR = Charlson Comorbidity Index-Psychosocial Risk; CDS = Chronic Disease Score; CS = Cross-Sectional; DC = Disease Count (Unweighted); ED = Emergency Department; EDC = Expanded Diagnosis Clusters;ERA = Elder Risk Assessment; GP = General Population; HCC = Hierarchical Condition Categories; HRQoL = Health-related Quality of Life; HSMI = Health Search Morbidity Index; HUI3 = Health Utility Index; IADL = Instrumental Activities of Daily Living; ICD-10 = International Classification of Diseases, Tenth Revision; MDMS = Multidimensional Multimorbidity Score; MM = Multimorbidity; MWI = Multimorbidity-Weighted Index; PC = Primary Care; Pra tool = Probability of repeated admission risk prediction tool; QOF = Quality and Outcomes Framework; QoL = Quality of Life; RUB = Resource Utilisation Band; RxRisk-V = A Veterans Association adapted pharmacy-based case-mix instrument; SRH = Self-Rated Health

◊ The youngest participant was from the NHS II cohort, aged 25 in 1989. Data collection for this study started in 2000 (NHS and HPFS cohorts) and 2001 (NHS II cohort) and hence, the age of the participants recruited in this study is 36 years old and above.

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3.1 Description of all the instruments used

Table 2 summarises all the 27 instruments that were identified from the studies with low risk of bias. Table 3 provides a summary of multimorbidity instruments and their associations with outcome measurements. Table 3a lists those with positive associations and Table 3b lists those with negative and no associations between measurements of multimorbidity and the outcomes.

3.1.1 Simple counts of individual conditions

Disease count was based on the total number of all the conditions an individual had, usually from a pre-specified list of chronic conditions. It was used in 42 out of the 67 studies (62.7%). Disease count was positively associated with activity limitations, continuity of care, disability, healthcare cost, healthcare utilisation, medications, mental disorders, and mortality; and negatively associated with general health, physical function, quality of life, and self-rated health (Table 3).

3.1.2 Organ or system-based approaches

There were three instruments in this category. They were Cumulative Illness Rating Scale (CIRS),[71,72] Cornell Medical Index (CMI),[73] and Organ Systems with chronic disease count (Organ-CDC),[31] and each instrument was described by one study. The two studies that used the instruments CIRS and Cornell Medical Index were not rated 'good' after the risk of bias assessment and were not described further in Table 3.

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Table 2. Description of instruments used for measurement of multimorbidity and the data sources and resources required

Category	Instrument	System/Condition based	Weightage; Scoring Method	Data Sources and resources required
A: Count of Individual Conditions				
A-1	DC	Condition (7-147)	Unweighted; Condition count	EMR, GP records, Health service database, Hospital discharge abstract, Insurance claims or Questionnaires - telephone, face-to-face, mailed
A-2	CCC	Condition (6)	Unweighted; Based on AHRQ's clinical classification software & no. of conditions for each category	EMR
B: Organ or System-based Approaches				
B-3	Organ systems with CDC	Organ system (17)	Unweighted Sum of organ systems	EMR
C: Weighted Indices				
C-4	ACE	Condition (27)	1-3 (based on severity of most severe condition); Highest score of single item	Insurance Claims' database
C-5	CCI	Condition (19)	1-6 (based on impact on 1-year mortality (RR) – original); Sum of weighted conditions	Administrative database, EMR, Medical chart review, Interviews or postal questionnaire.
C-6	CLS	Condition (98)	Based on impact for mortality (hazard ratio); Sum of beta coefficients of each category	Linked patients' records of all primary care events, hospital admissions and causes of death.
C-7	EI (original & modified)	Condition (21-31)	Based on impact on in-hospital mortality; Summing of beta coefficients	Insurance Claims' or Medical services database
C-8	ERA	Condition (6-9)	Weighted (based on impact on future hospitalisation); Sum of weighted regression coefficients	EMR and administrative database
C-9	HCC	Condition (70)	Based on Medicare capitation payments for health expenditure.; More severe manifestations of a condition dominating (and zeroing out the effect of) less serious ones. Other diseases are summed additively.	EMR & HCC software licensing
C-10	MMI	Condition (55)	Weighted based on 1-year mortality; Summing of beta coefficients	Linked patients' records
C-11	MDMS	Condition (7 chronic conditions, 2 health behaviours for 1st dimension & 5 symptoms for 2nd dimension)	Weighted but not based on any specific outcome; Sum of the value for the weighted absolute contributions of each of the dimensions.	Standardised medical evaluation (Interviewer-administered)
C-12	MM weighted by ADL Scale	Condition (19)	Weighted based on OARS functional status scale measuring ADL; Sum of weighted conditions	Face-to-face or telephone interviews
C-13	MM weighted by HUI	Condition (19)	Weighted based on correlation with health utility index; Sum of weighted conditions	Face-to-face or telephone interviews
C-14	MM weighted by HUI betas	Condition (19)	Weighted based on correlation with health utility index and adjusted for age and sex; Summing of beta coefficients	Face-to-face or telephone interviews

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Table 2. Description of instruments used for measurement of multimorbidity and the data sources and resources required (Continued)

Category	Instrument	System/Condition based	Weightage; Scoring Method	Data Sources and resources required
C-15	MWI	Condition (81)	Weighted based on impact on SF-36 physical functioning scale; Sum of weights	Interviewer-administered or mail questionnaire
C-16	Extended QOF (weighted)	Condition (14)	0-6, based on impact on 1-year mortality (RR); Sum of weighted conditions	EMR
C-17	Standard QOF (weighted)	Condition (9)	1-3, based on impact on 1-year mortality (RR); Sum of weighted conditions	EMR
D: Other Approaches (D1 = Case Mix, D2 = Pharmaceutical-based)				
D1-18	ACG	Condition (93 mutually exclusive ACGs. Some are modified to 68 ACGs)	Incorporated into ACGs based on impact on resource use (proprietary); Variable	EMR & ACG software licensing
D1-19	ADG	Condition (32 groups)	Based on duration, severity, diagnostic certainty, aetiology, and need for specialty care; Variable	EMR & ACG software licensing
D1-20	HM	Condition (NS)	Only MN tier 4 + MN tier 3 with ERA > 10; Variable	EMR, HCC software licensing & administrative data
D1-21	HSMI	Condition (73 chronic and acute conditions)	Based upon yearly health care costs directly derived from primary care setting; Sum of regression coefficients (range from -0.06 to 1.04)	EMR
D1-22	Minnesota Tiering	Condition (NS)	Grouping patients into 'complexity tiers' based on the number of major condition categories; Condition Count	EMR or administrative data & MN Tiering software licensing
D1-23	Resource Utilisation Band	Condition (6 mutually exclusive bands)	Based upon ACG algorithm on impact on resource use (proprietary); Variable	EMR & ACG software licensing
D2-24	CDS	Condition (17)	Weighted 1-5; Sum of weights based on pharmacological database	Prescription drug database
D2-25	Drug Count	NA; Variable. Some may be based on pharmacologic-therapeutic classification system	Weighted; Medication count	Self-reported questionnaire
D2-26	Modified Pra tool using RxRisk-V	NA; Pra tool + RxRisk-V	Weighted due to RxRisk-V; 4 categories	GP medical record + linked pharmacy claims database
D2-27	RxRisk-V	NA; WHO-ATC classification system	Weighted according to the diagnostic group of drugs to predict future health care costs; Sum of weights	GP medical record + linked pharmacy claims database

Note. ACE = Adult Comorbidity Evaluation; ACG = Adjusted Clinical Groups; ADG = Aggregated Diagnosis Groups; ADL = Activities of Daily Living; CCC = Chronic Condition Count; CCI = Charlson Comorbidity Index; CLS = Comorbidity Linked Score; CDC= Chronic Disease Count; CDS = Chronic Disease Score; DC = Disease Count; Elixhauser Index = EI; EMR = Electronic Medical Records; ERA = Elder Risk Assessment; GP = General Practitioner; HCC = Hierarchical Condition Categories; HM = Hybrid Model (MN Tier + ERA); HUI = Health Utility Index; HSMI = Health Search Morbidity Index; MDMS = Multidimensional Multimorbidity Score; MM = Multimorbidity; MMI = Multimorbidity Measure Index; MWI = Multimorbidity-Weighted Index; OARS = Older Americans Resources and Services; Pra tool = Probability of repeated admission risk prediction tool; QOF = Quality and Outcomes Framework; RR = Risk Ratio; RxRisk-V = A Veterans Association adapted pharmacy-based case-mix instrument; SF-36 = 36-item Short Form Survey

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Table 3a. Summary of multimorbidity instruments and their associations with outcomes measured: Positive association

Multimorbidity Measures	Positive Association between Outcomes and Multimorbidity
A = Count of Individual Conditions	
DC (Many different groupings ranging from 7 [37 61 74] to 147 [39] conditions and some are further categorised [40])	ADL limitations [19], Activity limitations [27], Continuity of care (3-years) [54], Deficits of perceived general health [53], Depressive symptoms [53], Disability [38,46,47], Emergency hospital admission (2-years) [66], Frequency of contacts with physicians (1-year) [63], Functional decline (2-years) [66], Healthcare costs [41], Hospital admissions (1-year) [48-50], Hospital outpatient visits (1-year) [60], Hospitalisation/Emergency Department Visits [60], Mental distress [27], Mortality (1-year) [50], (3-years) [24], (10-years) [68], No. of contacts with general practice (1-year) [64], No. of medications prescribed (1-year) [64], No. of mentally unhealthy days [19,20], No. of physically unhealthy days [19,20], No. of different ambulatory physicians contacted (1-year) [63], No. of primary care consultations (1-year) [24], (3-years) [24], No. of referrals (1-year) [64], Physical distress [27], Presence of mental health disorder [39], Primary care consultations (1-year period) [54], (3-years) [54], Primary healthcare cost [23], Potentially preventable unplanned admission (1-year period) [49]
CCC	Healthcare costs [40], Hospital admissions (1-year) [40], No. of emergency department visits (1-year) [40], Readmission within 30 days (1-year) [40]
B = Organ or System-Based Approaches	
Organ systems with CDC	Presence of depressive or anxiety disorder [31]
C = Weighted Indices	
ACE-27	Healthcare expenditure [41]
CCI	Ambulatory care sensitive hospitalisations (acute & chronic) [55], Emergency department visits (1-year) [40], Emergency hospital admission (2-years) [66], Functional decline (2-years) [66], Healthcare expenditure [40], Hip fractures [52], Hospitalisation (1-year) [37,40,45,50,57], Mortality (1-year) [25,36,50,57], (5-years) [26,36], (10-years) [26,36], (15, 20, 25-years) [26], No. of primary care consultations (3-years) [24], No. of primary care physician visits (1-year) [56], No. of specialist visits (1-year) [56], Potentially preventable unplanned admission (1-year) [48], Presence of critical illness [22], Primary healthcare cost [23], Mortality (1-year) [29,30], (3-years) [24,32], (5-years) [30,33,35], (10-years) [68], Readmission within 30 days (1-year) [40]
CLS	Mortality (1-year) [29]
ERA	Healthcare expenditure [40], Mortality (2-years) [61], No. of days hospitalised (1-year) [28], No. of emergency department visits (1-year) [28,40], No. of hospital admissions (1-year) [28,40], No. of hospital visits (1-year) [28], Nursing home placement (2-years) [61], Presence of critical illness [22], Readmission within 30 days (1-year) [40]
EI (Original & Modified)	Hospitalisation (1-year) [50,57], Mortality (1-year) [29,50,57]
HCC	Hospitalisation (1-year) [40], ED visits (1-year) [40], Readmission within 30 days (1-year) [40], Healthcare expenditure (1-year) [40]
MM3I	Hospitalisation (1-year) [57], Mortality (1-year) [57]
MDMS	Sickness absence episodes taken in 2 years (male) [62]
MWI	ADL limitations [67], IADL limitations [67], Mortality (10-years) [68]
QOF (Standard & Extended)	Mortality (1-year) [25]
D = Other Approaches (D-1 = Case-Mix, D2 = Pharmaceutical-based)	
D-1(1): ACG	Hospitalisation (1-year) [45], Mortality (3-years) [24], No. of primary care consultations (3-years) [24], Primary healthcare cost [23], Readmission within 30 days (1-year) [40]

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Table 3a. Summary of multimorbidity instruments and their associations with outcomes measured: Positive association (continued)

D-1(2): ADG	Hospitalisation (1-year) [56], No. of primary care physician visits (1-year) [56], No. of specialist visits (1-year) [56]
D-1(3): HM	Emergency department visits (1-year) [40], Healthcare expenditure [40], Hospitalisation (1-year) [40], Readmission within 30 days (1-year) [40]
D-1(4): HSMI	Healthcare cost (primary care) [43]
D-1(5): Minnesota Tiering	Emergency department visits (1-year) [40,60], Healthcare expenditure [40], Hospitalisation (1-year) [40,60], Readmission within 30 days (1-year) [40]
D-1(6): Resource Utilisation Band	Fee-for-service expenditures [42], Primary healthcare cost [23], Mortality (3-years) [24], Number of primary care consultations (3-years) [24], Registered active listing in primary care [51], Registered active listing in all healthcare [51]
D-2(7): CDS	Hospitalisation (1-year) [50], Mortality (1-year) [50]
D-2(8): Prescribed Drug Count	Emergency hospital admission (2-years) [66], Functional decline (2-years) [66], Hospitalisation (1-year) [50], Mortality (1-year) [50], (3-years) [24], No. of primary care consultations (3-years) [24]
D-2(9): Pra tool Modified	Emergency hospital admission (1-year) [65]
D-2(10): RxRisk-V	Emergency hospital admission (2-years) [66], Functional decline (2-years) [66]

Table 3b. Summary of multimorbidity instruments and their associations with outcomes measured: Negative or no association

Multimorbidity Measures	Association between Outcomes and Multimorbidity	
	Negative Association	**No Association**
A = Count of Individual Conditions		
DC	Affective well-being [69], Future physical functioning [68], General health [27], Life satisfaction [70], Perceived health status [70], Self-rated Health [37,47]	Quality of cardiovascular preventive care [58], Quality of preventive care [58]
C = Weighted Indices		
CCI	Future physical functioning [68], Successful aging [34]	Quality of cardiovascular preventive care [58], Quality of preventive care [58]
MDMS	-	Sickness absence episodes taken in 2 years (female) [62]
MM weighted by ADL Scale	Life satisfaction [70], Perceived health status [70]	-
MM weighted by HUI	Life satisfaction [70], Perceived health status [70]	-
MM weighted by HUI betas	Life satisfaction [70], Perceived health status [70]	-
MWI	Cognitive performance [67], Future physical functioning [68], Grip strength [67], Subjective physical functioning [67]	Gait speed [67]

Note. ACE-27 = Adult Comorbidity Evaluation-27; ACG = Adjusted Clinical Groups; ADG = Aggregated Diagnosis Groups; ADL = Activities of Daily Living; CCC = Chronic Condition Count; CCI = Charlson Comorbidity Index; CLS = Comorbidity Linked Score; CDC= Chronic Disease Count; CDS = Chronic Disease Score; DC = Disease Count; Elixhauser Index = EI; ERA = Elder Risk Assessment; HCC = Hierarchical Condition Categories; HM = Hybrid Model (MN Tier + ERA); HUI = Health Utility Index; HSMI = Health Search Morbidity Index; MDMS = Multidimensional Multimorbidity Score; MM3I = Multimorbidity Measure (M3) Index; MWI = Multimorbidity-Weighted Index; Pra tool = Probability of repeated admission risk prediction tool; QOF = Quality and Outcomes Framework; RxRisk-V = A Veterans Association adapted pharmacy-based case-mix instrumen

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3.1.3 Weighted Indices

There were 17 unique weighted instruments found in the included studies. The original Charlson Comorbidity Index (CCI) with its different modifications was the most frequently used instrument and was used in 22 studies. The CCI was based on disease count, but the 17 conditions were weighted originally based on their impact on one-year mortality.[75] The final score was derived by the summation of all the weighted conditions. There were many variations and modifications of the score including the addition of psychosocial factors. The instrument was found to be associated with multiple outcomes other than one-year mortality.

Most of the other weighted index instruments were novel, like the Multimorbidity-Weighted Index (MWI), in which the investigators built multivariable prognostic models from a set of potential predictor conditions and weighted the conditions based on an outcome of clinical interest. The most common outcomes chosen were mortality and physical function. Other outcomes included health expenditure,[40] health utility index,[70] and severity of the most severe condition.[41] The Multidimensional Multimorbidity Score (MDMS)[62] was unique as it was weighted based on health behaviours and patient symptoms and not based on any specific outcome.

3.1.4 Other Approaches to measuring multimorbidity

Other approaches included case-mix and pharmaceutical-based instruments. The study using clinical judgement was rated to be of high risk of bias.[76]

For case-mix approach, the Adjusted Clinical Group (ACG) and Resource Utilisation Band (RUB) were the most commonly used instruments.[77] Most of the case-mix instruments required proprietary software licenses from the United States of America and obtained data from electronic medical records or administrative data.

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The second group of instruments in this category was related to pharmaceutical data. The most frequent type was the unweighted drug count. The other three (Chronic disease score, A Veterans Association adapted pharmacy-based case-mix instrument like RxRisk-V, and modified Probability of repeated admission risk prediction tool using RxRisk-V) were all weighted indices. Except for the drug count that was based on a self-report questionnaire, the rest required a prescription drug database to obtain the data.

4. DISCUSSION

4.1 Summary of Findings

Twenty-seven unique instruments for measuring the level of multimorbidity with low risk of bias were identified and categorised according to the classification by Sarfati.[16] The most commonly used instrument was 'Disease Count'. It was also the only instrument that was associated with the three essential outcomes from the core outcomes set of multimorbidity research (COSmm)[8] i.e., quality of life, mental health, and mortality.

4.2 Comparison with previous research

Although the most common instrument identified in this systematic review was similar to that of Huntley et al.,[12] several instruments including Duke Severity of Illness Checklist (DUSOI) and Functional Comorbidity Index (FCI) identified in their paper were not found in this systematic review. The possible reasons for not finding these instruments in this review could be due to the lack of interest in the instrument by the research community in recent years (last publication using DUSOI was in 2004[78]), or the exclusion of studies specifying an index condition and studies not from the primary care or general population in the search strategy.

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4.3 Advantages and disadvantages of selected instruments

4.3.1 Disease count

Adding the number of conditions from a list of chronic conditions provides an ordinal score.[79] The advantage is simplicity and the ease of data ascertainment with minimal resources required. However, using 'disease count' incorrectly assumes a linear relationship on an ordinal scale, and ignores the fact that an additional unit of increase in 'disease count' will likely have a diminishing impact. The other disadvantage noted was the lack of transparency in the operational definition of multimorbidity, especially regarding the list of conditions considered for multimorbidity, and the cut-points used.

4.3.2 Weighted Indices

The common weighted indices identified in this systematic review were Charlson Comorbidity Index (CCI), Elders Risk Assessment (ERA), Elixhauser Index (EI) and Multimorbidity-weighted Index (MWI). These weighted indices were often utilised in prognostic models to build complex multivariable regression models in which the weights were calculated from hazard ratios, odds ratios, or regression coefficients.[80]

The advantage of these weighted indices is that the weights allow the adaptation of an index to a specific outcome. An investigator could recalibrate the correct weight by creating a prognostic model to produce a contextualised instrument for a different setting. Prognostic models can provide clinically relevant risk stratification and help to allocate resources.[81] The disadvantage of such indices is that calculated weights are greatly influenced by the population, outcomes used, and the instrument's original conception and purpose, hampering the ability to compare across studies.

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4.3.3 Case-Mix

The Adjusted Clinical Group (ACG) system has a good track record in the United States and several other countries, especially for measuring the outcomes of healthcare utilisation. However, the instrument is proprietary, and the exact algorithm of the instrument is not open to the public and may not be suitable in certain settings. Another disadvantage is the financial costs involved in obtaining the license.

4.3.4 Pharmaceutical-based instruments

Medication-based indices include versions of the Chronic Disease Score,[82] which later became known as the RxRisk,[83] and its adaptation for use in the veteran population, the RxRisk-V.[84] Like the disease count, its main advantage is the ease of use with minimal resources required. However, many studies were not transparent regarding which type of drugs were included.

4.4 Data sources

Data sources used by these instruments included medical record information, patient self-report, clinical judgment, and large administrative databases. Each data source has its inherent advantages and disadvantages. For patient self-report, patients with cognitive impairment may under-report symptoms and may be seen less frequently by their physicians, resulting in an under-recognition or under-treatment of conditions.[79] It has also been shown that health administrative data based on billing system underestimated the prevalence of many chronic conditions.[85]

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The available data in a particular setting may strongly influence the ultimate instrument chosen for multimorbidity research. As there is currently no consensus on the gold standard for sources of data, it is difficult to assess which data source was superior from this review.

4.5 Clinical Implications

Although there was no restriction on the type of study design in our search strategy, we only found observational studies in our search. The lack of interventional studies is reflective of the lack of interventional studies on multimorbidity in primary care and the general population generally.

There were 17 multimorbidity outcomes identified by a Delphi panel of international experts of multimorbidity intervention studies.[8] However, only ten out of the seventeen outcomes were reported in the 67 studies identified in this systematic review and the most common outcome that was investigated was health care use.

Ideally, a single instrument measuring the level of multimorbidity should be able to predict a variety of relevant outcomes if the same instrument could be utilised by using different weights in calculating the scores. Such multiple-scoring instruments may be the way forward for validation of prognostic models for different outcomes and different populations with established multimorbidity instruments. The choice of conditions included in such an instrument should include those with a high prevalence in the specific population and those with a significant impact on the affected population. The definition of high prevalence and significant impact would depend on the type of healthcare setting or geography. However, for pragmatic reasons, the final selection of the conditions to be included in such a multiple-scoring instrument may still have to take into account the availability of relevant and reliable data. A certain degree of reductionism will also have to be accepted because a single instrument will

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not be able to encompass all the nuances of the different interactions of chronic conditions on an individual living in his/her unique milieu.

4.6 Strengths and limitations of the study

The main strengths of this systematic review were the involvement of a health science librarian in our search strategy, a published protocol, adherence to the protocol without changes during the systematic review process,[86] and the critical appraisal of all the primary articles with a risk-of-bias assessment tool.

The systematic review had several limitations. We excluded grey literature and included only articles that were published in the English language. We also did not contact authors directly for a suggestion of studies, nor identified a list of instruments from the preliminary search and then performed an additional search using the same databases.[87] However, given the consistency of our results as shown in Table 3 where the associated outcomes were all in the same direction, it is unlikely that missed studies would significantly alter the main findings of our review.

5. CONCLUSIONS

In this systematic review, we found 27 instruments for measuring the level of multimorbidity of community-dwelling individuals that predict or explore the association of multimorbidity with at least one specified outcome. Disease count and weighted indices like the Charlson Comorbidity Index (CCI), the ERA, and Elixhauser Index (EI) were commonly used for measuring the level of multimorbidity. Other approaches to measuring the level of multimorbidity included case-mix or pharmaceutical-based instruments.

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We found continuing interest in measuring the level of multimorbidity with disease count and drug count. There has also been a rise in the development of novel weighted indices using prognostic models or validation of existing well-established instruments like the CCI over the last few years. There is currently an absence of a gold standard for where to obtain chronic disease information. The most suitable instrument will depend on the specified outcome of interest, the study population, and the type of data and resources available.

Finally, there is still much work to improve on the body of knowledge of multimorbidity when most investigators in the last decade measured multimorbidity without including some of the important outcome measures of multimorbidity. We also suggest that a clear description of the instruments is required in the publication of multimorbidity studies to counter the frequent lack of information currently seen so as to contribute to robust multimorbidity research in future.

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All authors made substantial contribution to the design of the work. ESL, together with a health science librarian, created the search strategy and conducted the search. ESL, EH, TSH, HLK and FYW screened papers, extracted and analysed data. ESL drafted the paper, and all authors reviewed drafts and approved the final version.

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Patient and public involvement: This research was done without patient involvement. Patients were not invited to comment on the study design and were not consulted to develop patient relevant outcomes or interpret the results. Patients were not invited to contribute to the writing or editing of this document for readability or accuracy.

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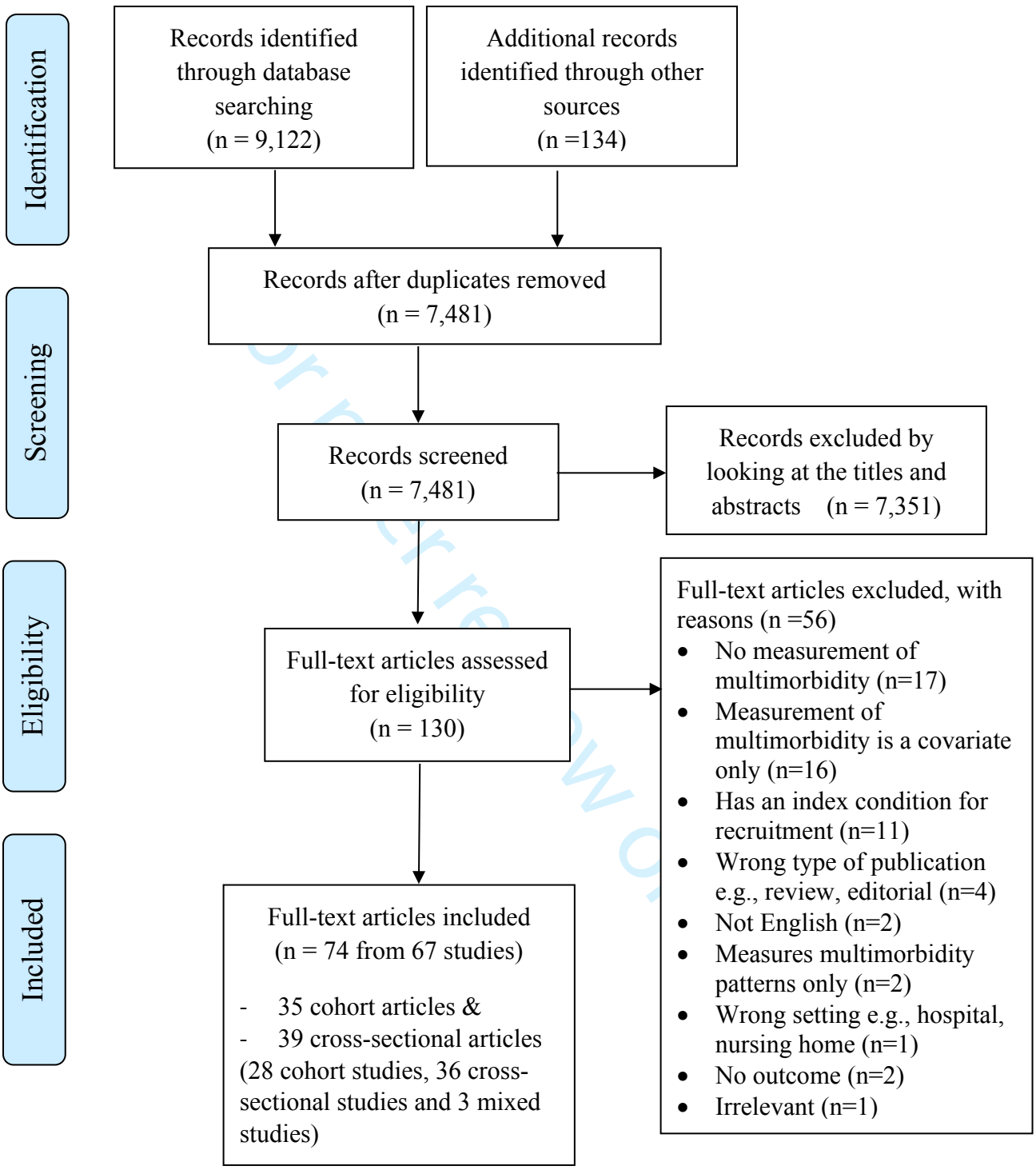
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Figure 1. PRISMA Flow Diagram

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Figure 1. PRISMA Flow Diagram



Appendix 1. Search Strategy

Table 1. Medline Search 1946 to August 13, 2018

#	Searches	Results
1	exp comorbidity/ or multiple chronic conditions/	95224
2	(multimorbid* or multi-morbid* or comorbid* or co-morbid*).ab,ti,kw.	147739
3	((multiple or coexist* or co-exist* or concurrent* or simultaneous*) adj2 (disease* or illness* or diagnos* or condition* or morbid* or disorder*)).ab,ti,kw.	45667
4	1 or 2 or 3	247802
5	primary health care/ or "continuity of patient care"/ or exp general practice/ or ambulatory care/ or physicians, family/ or physicians, primary care/ or community health services/ or general practitioners/	229791
6	((ambulatory or community or general or family or primary) adj2 (care or health* or practi* or physician* or medicine or center* or centre* or facilit* or clinic*)).ab,ti,kw.	350963
7	5 or 6	461463
8	epidemiologic measurements/ or risk assessment/ or "Outcome Assessment (Health Care)"/ or patient reported outcome measures/ or health status indicators/ or "severity of illness index"/ or sickness impact profile/ or diagnosis-related groups/ or case mix/	516146
9	((health status or risk or outcome* or sickness impact) adj2 (appraisal* or assessment* or evaluation* or measur* or index* or indic* or profile*)).ab,ti,kw.	344511
10	((severity or burden) adj2 (illness* or diseas* or disorder* or condition* or diagnos*)).ab,ti,kw.	72588
11	(charlson comorbidity index or charlson index or charlson score or elixhauser* or cumulative illness rating scale* or CIRS or adjusted clinical group* or ACG* or disease count or Duke severity* or DUSOI or casemix or case-mix).ab,ti,kw.	18192
12	8 or 9 or 10 or 11	808827
13	4 and 7 and 12	4172
14	limit 13 to (english language and yr="2010 -Current")	2173

Table 2. Embase Classic+Embase 1947 to August 13, 2018

#	Searches	Results
1	comorbidity/ or multiple chronic conditions/	196646
2	(multimorbid* or multi-morbid* or comorbid* or co-morbid*).ab,ti,kw.	244599
3	((multiple or coexist* or co-exist* or concurrent* or simultaneous*) adj2 (disease* or illness* or diagnos* or condition* or morbid* or disorder*)).ab,ti,kw.	66050
4	1 or 2 or 3	378500
5	primary health care/ or patient care/ or primary medical care/ or general practice/ or general practitioner/ or ambulatory care/ or family medicine/ or community care/ or community health services/	578870
6	((ambulatory or community or general or family or primary) adj2 (care or health* or practi* or physician* or medicine or center* or centre* or facilit* or clinic*)).ab,ti,kw.	453501
7	5 or 6	811904
8	risk assessment/ or outcome assessment/ or patient reported outcome/ or health status indicator/ or disease activity score/ or global disease burden/ or organ dysfunction score/ or "severity of illness index"/ or sickness impact profile/ or general health status assessment/ or disease severity/ or diagnosis related group/ or charlson comorbidity index/ or comorbidity assessment/ or elixhauser comorbidity index/ or case mix/	1257685

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9	((health status or risk or outcome* or sickness impact) adj2 (appraisal* or assessment* or evaluation* or measur* or index* or indic* or profile*)).ab,ti,kw.	438484
10	((severity or burden) adj2 (illness* or diseas* or disorder* or condition* or diagnos*)).ab,ti,kw.	106531
11	(charlson comorbidity index or charlson index or charlson score or elixhauser* or cumulative illness rating scale* or CIRS or adjusted clinical group* or ACG* or disease count or Duke severity* or DUSOI or casemix or case-mix).ab,ti,kw.	32221
12	8 or 9 or 10 or 11	1613925
13	4 and 7 and 12	9317
14	limit 13 to (english language and yr="2010 -Current")	6113

Table 3. CINAHL – since inception to August 14, 2018

#	Searches	Results
1	(MH “comorbidity”)	30423
2	multimorbid* or multi-morbid* or comorbid* or co-morbid*	49461
3	(multiple or coexist* or co-exist* or concurrent* or simultaneous*) N2 (disease* or illness* or diagnos* or condition* or morbid* or disorder*)	9374
4	1 or 2 or 3	57381
5	(MH “primary health care”) or (MH “family practice”) or (MH “ambulatory care") or (MH “ambulatory care facilities") or (MM “community health services”) or (MM “community health centers”) or (MH “physicians, family”) or (MH “continuity of patient care")	83299
6	(ambulatory or community or general or family or primary) N2 (care or health* or practi* or physician* or medicine or center* or centre* or facilit* or clinic*)	205406
7	5 or 6	211994
8	(MH “risk assessment”) or (MH "Outcome Assessment") or (MH “patient-reported outcomes) or (MH “health status indicators”) or (MH "severity of illness indices") or (MH “sickness impact profile”) or (MH “diagnosis-related groups”) or (MH “case mix”)	69377
9	(health status or risk or outcome* or sickness impact) N2 (appraisal* or assessment* or evaluation* or measur* or index* or indic* or profile*)	165689
10	(severity or burden) N2 (illness* or diseas* or disorder* or condition* or diagnos*)	51231
11	(charlson comorbidity index or charlson index or charlson score or elixhauser* or cumulative illness rating scale* or CIRS or adjusted clinical group* or ACG* or disease count or Duke severity* or DUSOI or casemix or case-mix)	4820
12	8 or 9 or 10 or 11	211895
13	4 and 7 and 12	1454
14	limit 13 to (english language and yr="2010 -Current")	800

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Appendix 2. Coding Description for the Modified Newcastle-Ottawa Scale for Cohort Studies

Category	Description
Selection	
1) Representativeness of the sample	This item assesses the representativeness of sample in the community, not from some general population. a) Truly representative* (e.g., everyone from the database) b) Somewhat representative* (with at least 2 criteria but selection method was convincing due to random sampling) c) Selected group (e.g., only certain socio-economic groups or areas) d) No description of sampling strategy
2) Ascertainment of multimorbidity	This item assesses the method by which multimorbidity was confirmed. a) Secure record* (e.g., GP questionnaire) b) Structured interview* (e.g., interviewer-administered questionnaire) c) Written self-report (e.g., mailed survey, if items are unable to be confirmed by objective measure) d) No description / Other
3) Demonstration that outcome of interest was not present at start of study	A statement of no history of disease earns a star. In the case of mortality studies, outcome of interest is still the presence of a disease, rather than death. a) Yes* b) No
Comparability	
1) Study controls for age and sex	Covariates must be adjusted for in the analysis. Statements of no differences between groups or that differences were not statistically significant are not sufficient for establishing comparability. Note: If the relative risk for the exposure of interest is adjusted for the covariates listed, then the groups will be considered to be comparable on each variable used. a) Yes* b) No
2) Study controls for other factors	a) Yes* b) No
Outcome	
1) Statistical test	Statistical test(s) must be clearly described and appropriate before assessing the other items under the Outcome category. a) Clearly described and appropriate b) Not described, incomplete or inappropriate
2) Assessment of outcome	This item assesses the method by which the outcome of interest was confirmed. For some outcomes, reference to the medical record is sufficient to satisfy the requirement for confirmation. This may not be adequate for other outcomes where reference to specific tests or measures would be required. a) Independent or blind assessment* (e.g., interviewer-administered questionnaire) b) Record linkage* (e.g., identified through ICD codes on database records) c) Self-report (i.e., no reference to original medical records) d) No description / Other

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3)	Was follow-up long enough for outcomes to occur?	An acceptable length of time was at least 1 year of follow-up. a) Yes* b) No
4)	Adequacy of follow-up of cohorts	This item assesses the follow-up of the sample to ensure that losses are not related to the outcome. a) Complete follow-up - all subject accounted for* b) Subjects lost to follow-up unlikely to introduce bias* (Number lost ≤20% or description of those lost suggested no different from those followed.) c) Follow-up rate less than 80% and no description of those lost d) No statement

Thresholds for converting the modified Newcastle-Ottawa scales to Good, Fair, and Poor Quality:

- Good Quality - 2-3 stars in Selection category (Representativeness of Sample item must be fulfilled) AND 1-2 stars in Comparability category AND 2-3 stars in Outcome category.
- Fair Quality - 1-2 stars in Selection category AND 1-2 stars in Comparability category AND 2-3 stars in Outcome category.
- Poor Quality - 0 star in Selection category OR 0 stars in Comparability category OR 0-2 stars in Outcome category.

Wells GA, Shea B, O'Connell D, Peterson J, Welch V, Losos M, Tugwell P. The Newcastle-Ottawa Scale (NOS) for assessing the quality of nonrandomized studies in meta-analysis. Available from: http://www.ohri.ca/programs/clinical_epidemiology/oxford.asp

Modifications from original NOS include:

- 'Statistical test' was added to the Outcome category
- 'Representativeness of sample' item under Selection category and 'Statistical test' item under Outcome category must both be fulfilled for study to be considered Good Quality.
- 'Selection of the non-exposed cohort' item was removed from Selection category as most studies did not describe a non-exposed cohort, and this review sought to compare the different levels of multimorbidity within the group.
- 'Representativeness of the exposed cohort' item under Selection category was renamed 'Representativeness of the sample' since the 'non-exposed cohort' was removed above.
- 'Ascertainment of exposure' under the Selection category was renamed as 'Ascertainment of multimorbidity' to specify that multimorbidity is the exposure in this review.
- 'Study controls for age and sex' and 'Study controls for other factors' were revised to be items under Comparability category
- The thresholds for converting the scales to quality were amended accordingly due to the above modifications.

Consensus:

- The study team decided that a period of one-year was a reasonable period of follow-up under 'Was follow-up long enough for outcomes to occur?' item under Outcome category.

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Appendix 3. Coding Description for the Modified Newcastle-Ottawa Scale for Cross-sectional Studies

Category	Description
Selection	
1) Representativeness of the sample	This item assesses the representativeness of sample in the specified population, not from some general population. e) Truly representative* (e.g., everyone from the database, random sampling) f) Somewhat representative* (with at least 2 criteria but selection method was convincing due to random sampling) g) Selected group (e.g., only certain socio-economic groups or areas) h) No description of sampling strategy
2) Ascertainment of multimorbidity	This item assesses the method by which multimorbidity was confirmed. e) Secure record* (e.g., Clinical records, GP questionnaire) f) Structured interview* (e.g., interviewer-administered questionnaire) g) Written self-report (e.g., mailed survey, if items are unable to be confirmed by objective measure) h) No description / Other
3) Sample Size	If there is no description, a reported sample size of 800 and above is satisfactory. c) Justified and satisfactory* d) Not justified
4) Non-respondents	Acceptable response rates for surveys through various methods† • In-person: 57% • Mail: 50% • Average: 33% • Email: 30% • Internet: 29% • Telephone: 18% • In-app: 13% a) Comparability between respondents and non-respondents characteristics is established, and the response rate is satisfactory.* b) The response rate is unsatisfactory, or the comparability between respondents and non-respondents is unsatisfactory. c) No description of the response rate or the characteristics of the responders and the non-responders
Comparability	
3) Study controls for age and sex	Confounders must be adjusted for in the analysis. Statements of no differences between groups or that differences were not statistically significant are not sufficient for establishing comparability. Note: If the relative risk for the exposure of interest is adjusted for the confounders listed, then the groups will be considered to be comparable on each variable used in the adjustment. c) Yes* d) No
4) Study controls for other factors	c) Yes* d) No

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Outcome	
5) Statistical test	Statistical test(s) must be clearly described and appropriate before assessing the other items under the Outcome category. c) Clearly described and appropriate d) Not described, incomplete or inappropriate
6) Assessment of outcome	This item assesses the method by which the outcome of interest was confirmed. For some outcomes, reference to the medical record is sufficient to satisfy the requirement for confirmation. This may not be adequate for other outcomes where reference to specific tests or measures would be required. e) Independent or blind assessment***** f) Record linkage***** (e.g., identified through ICD codes on database records) g) Self-report (i.e., no reference to original medical records) h) No description / Other

Thresholds for converting the modified Newcastle-Ottawa scales to Good, Fair, and Poor Quality:

- Good Quality – 3-4 stars in Selection category (Representativeness of Sample item must be fulfilled) AND 1-2 stars in Comparability category AND 1 star in Outcome category.
- Fair Quality - 2 stars in Selection category AND 1-2 stars in Comparability category AND 1 star in Outcome category.
- Poor Quality - 0 star in Selection category OR 0 stars in Comparability category OR 0 stars in Outcome category.

†Lindermann N. What's the average survey response rate? [2018 benchmark]. SurveyAnyplace. Available from: https://surveyanyplace.com/average-survey-response-rate/.

Alshabanat A, Zafari Z, Albanyan O, Dairi M, FitzGerald JM. Asthma and COPD Overlap Syndrome (ACOS): a systematic review and meta-analysis. PLoS ONE. 2015;10(9):e0136065. Doi:10.1371/journal.pone.0136065.

Modifications from original NOS include:

- 'Ascertainment of exposure' item under Selection category was renamed as 'Ascertainment of multimorbidity' to specify that multimorbidity is the exposure in this review.
- The ratings for the 'Ascertainment of exposure' item under Selection category were revised to 'secure record, structured interview, written self-report, no description/other' to align with the Modified NOS for cohort studies in this review.
- 'Study controls for age and sex' and 'Study controls for other factors' items were revised to be items under Comparability category.
- 'Was follow-up long enough for outcomes to occur' item under Outcome category was renamed as 'Statistical test'. This item must be fulfilled for the study to be considered as Good Quality.
- The thresholds for converting the scales to quality were amended accordingly due to the above modifications.

Consensus:

- The study team decided on a list of acceptable response rates for various survey methods for the 'Non-respondents' item under Selection category.
- The study team decided that a sample size of 800 was reasonable under 'Sample size' item under Selection category.

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Appendix 4. Summary of risk of bias (RoB) appraisal of included cohort articles using the modified Newcastle-Ottawa Quality Assessment Scale (35 articles)

	Barile et al. (2013)[40]	Biehl et al. (2016)[43]	Boeckxstaens et al. (2015b)[19]	Brilleman et al. (2014)[44]	Brilleman and Salisbury (2013)[45]	Carey et al. (2013)[46]	Chapman et al. (2015)[47]	Crane et al. (2010)[49]	Crooks et al. (2016)[50]	Crooks et al. (2015)[51]	Formiga et al. (2013)[52]	Formiga et al. (2011a)[53]	Formiga et al. (2016)[55]	Fraccaro et al. (2016)[56]	Haas et al. (2013)[23]	Hwang et al. (2015)[25]	Jennings et al. (2015)[88]	Jia and Lubetkin (2016)[89]	Jia et al. (2018)[90]	Lemke et al. (2012)[63]	Md Yusof et al. (2010)[20]	Payne et al. (2014)[66]	Payne et al. (2013)[67]	Quail et al. (2011)[68]	Reyes et al. (2014)[70]	Salisbury et al. (2011)[72]	Saver et al. (2014)[73]	Stanley and Sarfati (2017)[75]	Streit et al. (2014)[76]	Takahashi et al. (2016)[78]	Takahashi et al. (2012)[79]	Tyack et al. (2016)[91]	Wallace et al. (2016a)[82]	Wallace et al. (2016b)[83]	Wei and Mukamal (2018)[85]
Selection Criteria																																			
Representativeness of the sample	*	*	-	*	*	*	*	*	*	*	*	*	*	*	*	*	*	-	*	*	*	-	*	*	*	*	*	*	*	*	*	*	-	*	*	*
Ascertainment of Multimorbidity	-	*	*	*	*	*	-	*	*	*	*	*	*	*	*	*	*	*	-	*	*	-	*	*	*	*	*	*	*	*	*	*	-	*	*	-
Demonstration that outcome of interest was not present at the start of the study	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*
Comparability Criteria																																			
Study controls for age and sex	*	-	*	*	*	*	*	-	*	*	-	*	-	*	*	*	*	*	-	*	-	*	*	*	-	*	*	*	*	*	*	*	*	*	*	-
Study controls for others	*	*	-	*	*	*	*	*	-	*	*	*	*	*	*	*	*	-	-	*	*	*	*	*	*	*	*	-	*	*	*	*	*	*	*	*
Outcome Criteria																																			
Statistical test†	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	X	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Assessment of outcome	-	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*
Follow-up was long enough for outcomes to occur	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*
Adequacy of follow-up of cohorts	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	-	*	*	*	*
Overall RoB Judgement[a]	G	G	F	G	G	G	G	G	G	G	G	G	G	G	G	G	F	P	P	G	F	G	G	G	G	G	G	G	G	G	G	F	G	G	G

Note. G = Good; F = Fair; P = Poor.

†Statistical test must be clearly described and appropriate before assessing the other items under the Outcome category. [a] Good rating is given when there are 2 to 3 stars in Selection category (Representativeness of Sample must be fulfilled) AND 1 to 2 stars in Comparability category AND 2 to 3 stars in Outcome category; Fair rating is given when there are 1 to 2 stars in Selection category AND 1 to 2 stars in Comparability category AND 2 to 3 stars in Outcome category; Poor rating is given when there are 0 star in Selection category OR 0 star in Comparability category OR 0 to 2 stars in Outcome category.

"*" Study satisfies the criteria; "-" Study did not satisfy the criteria; "✓" Statistical test is clearly described and appropriate; "X" Statistical test is not described, incomplete or inappropriate.

NB: Articles highlighted in light grey are combined studies (see Table 3-2 with similar highlights) and articles highlighted in dark grey are cohort studies. Those not highlighted are single articles which are also single studies.

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Appendix 5. Summary of risk of bias (RoB) appraisal of included cross-sectional articles using the modified Newcastle-Ottawa Quality Assessment Scale (39 articles)

	Agborsangaya et al. (2013)[39]	Barile et al. (2012)[41]	Barnett et al. (2012)[42]	Boeckxstaens et al. (2015a)[18]	Chen et al. (2011)[48]	DiNapoli et al. (2017)[21]	Formiga et al. (2011b)[54]	Galenkamp et al. (2011)[57]	Garin et al. (2014)[58]	Glynn et al. (2011)[59]	Gunn et al. (2012)[92]	Hanmer et al. (2010)[93]	Hu et al. (2017)[94]	Isaacs et al. (2014)[95]	Kojima et al. (2011)[87]	Kristensen et al. (2014)[60]	Lapi et al. (2015)[61]	Lawson et al. (2013)[62]	Li et al. (2016)[96]	Marengoni et al. (2011)[64]	McDaid et al. (2013)[65]	Muggah et al. (2012)[97]	Mujica-Mota et al. (2015)[11]	Naessens et al. (2011)[98]	Østergaard and Foldager (2011)[27]	Peters et al. (2018)[99]	Ranstad et al. (2014)[69]	Renne and Gobbens (2018)[100]	Ryu et al. (2015)[71]	Shadmi et al. (2011)[74]	Sibley et al. (2014)[101]	Sullivan et al. (2012)[77]	Ubalde-Lopez et al. (2016)[26]	van den Bussche et al. (2014)[80]	van Oostrom et al. (2014)[81]	Vos et al. (2013)[102]	Wei et al. (2018)[84]	Wikman et al. (2011)[86]	Wister et al. (2015)[24]
Selection																																						
Representativeness of the sample	*	*	*	-	*	*	*	*	*	*	*	*	-	*	-	*	*	*	*	-	*	*	*	*	-	-	*	*	*	*	*	*	*	*	-	*	*	*
Ascertainment of Multimorbidity	*	-	*	*	*	*	*	*	*	*	-	-	*	*	*	-	*	*	*	-	*	*	*	-	*	*	-	*	-	*	*	*	*	*	*	*	*	*
Sample size	-	*	*	-	*	*	-	*	*	*	-	*	-	*	*	-	*	*	*	-	-	*	*	*	*	*	-	*	*	*	*	-	*	*	*	*	*	-
Non-respondents	-	*	*	-	*	*	*	-	-	*	-	-	*	*	*	-	*	*	*	-	-	*	*	-	*	-	-	*	*	*	-	-	*	*	*	*	-	-
Comparability																																						
Study controls for age and sex	*	*	*	*	*	*	-	*	*	*	*	*	*	-	*	*	*	*	*	-	*	*	-	*	-	-	*	*	*	*	*	-	*	*	-	*	*	*
Study controls for others	*	*	*	*	*	*	*	-	*	*	*	*	*	-	*	*	*	*	*	-	*	*	-	*	-	-	*	*	*	*	*	-	*	*	-	*	*	-
Outcome																																						
Statistical test†	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	X	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Assessment of outcome	*	✓	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	-	*	*	*	*	*	*
Overall RoB Judgement[a]	G	G	G	P	G	G	G	G	G	G	G	F	F	F	P	P	G	G	G	P	G	G	P	F	P	P	F	G	P	G	G	P	G	G	G	G	P	G	G	G

Note. G = Good; F = Fair; P = Poor.

†Statistical test must be clearly described and appropriate before assessing the other items under the Outcome category. [a] Good rating is given when there are 3 to 4 stars in Selection category (Representativeness of Sample must be fulfilled) AND 1 to 2 stars in Comparability category AND 1 star in Outcome category; Fair rating is given when there are 2 stars in Selection category AND 1 to 2 stars in Comparability category AND 1 star in Outcome category; Poor rating is given when there are 0 to 1 star in Selection category OR 0 star in Comparability category OR 0 star in Outcome category.

"*" Study satisfies the criteria; "-" Study did not satisfy the criteria; "✓" Statistical test is clearly described and appropriate; "X" Statistical test is not described, incomplete or inappropriate.

NB: Articles highlighted in light grey are combined studies (see Table 3-1 with similar highlights). Those not highlighted are single articles which are also single studies.

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	3
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	5
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	6
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	6
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	6-7
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	6-7
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	Appendix 1
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	7
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	7
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	8
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	7
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	NA
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	NA

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	NA
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	NA
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	Figure 1
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	9-11
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	Appendix 4,5
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	12-18
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	NA
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	NA
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	NA
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	18-20
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	22
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	21
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	24

From: Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(7): e1000097. doi:10.1371/journal.pmed1000097

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A systematic review on the instruments used for measuring the association of the level of multimorbidity and clinically important outcomes

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TITLE PAGE

A systematic review on the instruments used for measuring the association of the level of multimorbidity and clinically important outcomes

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ABSTRACT

Objectives: There are multiple instruments for measuring multimorbidity. The main objective of this systematic review was to provide a list of instruments that measure the association of a specific outcome with different levels of multimorbidity as the main independent variable in community-dwelling individuals. The secondary objective was to provide details of the requirements, strengths and limitations of these instruments, and the chosen outcomes.

Methods: A systematic review was conducted according to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses guidelines (PROSPERO registration number: CRD42018105297). We searched MEDLINE, Embase and CINAHL electronic databases and manually searched the Journal of Comorbidity published in English language between 1st January 2010 and 23rd October 2020 inclusive. Studies also had to select adult patients from primary care or the general population, and had at least one specified outcome variable. Two authors screened the titles, abstracts and full texts independently. Disagreements were resolved with a third author. The modified Newcastle Ottawa Scale was used for quality assessment.

Results: A total of 96 studies were identified, and 69 studies were rated to have a low risk of bias. In total, 33 unique instruments were described. Disease count and weighted indices like the Charlson Comorbidity Index were commonly used. Other approaches included case-mix or pharmaceutical-based instruments. Disease count was the common instrument used for measuring all three essential core outcomes of multimorbidity research: mortality, mental health and quality of life. There was a rise in the development of novel weighted indices by using prognostic models. The data obtained for measuring multimorbidity were from various sources including medical records, patient self-reports, and large administrative databases.

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Conclusions: We listed the details of 33 instruments for measuring the level of multimorbidity from studies with low risk of bias as a resource for investigators interested in the measurement of multimorbidity.

ARTICLE SUMMARY

Strengths and Limitations of this study

- This review builds upon Huntley et al.'s 2012 review paper and provides an updated, comprehensive list of instruments that measure levels of multimorbidity in community-dwelling individuals
- Thorough literature search of three major electronic databases with the involvement of a health science librarian
- Reporting of search as per Preferred Reporting Items for Systematic Reviews and Meta-Analysis guidelines
- Exclusion of non-English language papers and grey literature

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MAIN TEXT

BACKGROUND

Multimorbidity is defined as the co-occurrence of two or more chronic medical conditions in an individual.[1] It is a growing public health challenge and accounts for most of the expenditures in the healthcare system.[2] The complex interactions of several co-existing diseases have profound implications on individuals[3 4] and their healthcare providers.[5 6]

There are multiple instruments for measuring multimorbidity and many of them do not usually specify the severity of individual conditions.[7] No gold standard multimorbidity measurement instrument exists and there is also no agreed categorisation of the available instruments. Sarfati[8 9] classified the various measurement instruments into four broad approaches. They are: (1) by simple counts of individual conditions (i.e., disease count), (2) by organ or system-based approaches, (3) by weighting conditions and combining them into indices, and (4) by other miscellaneous approaches. Most of these measurements are used to measure the prevalence or patterns of multimorbidity. However, they can also be used to predict an outcome or to evaluate an intervention for a desired outcome. A set of core outcomes of multimorbidity (COSmm) was proposed after consulting a panel of international experts in multimorbidity intervention studies using a Delphi process.[10] Core outcome sets (COS) represent the minimum that should be measured and reported in all clinical trials of multimorbidity.[11]

Huntley et al.[12] published a systematic review in 2012 describing the instruments used to measure the morbidity burden in primary care and the general population. They found 17 different instruments from 194 articles. The most widely used instruments and those with the most significant evidence of validity were the Charlson Comorbidity Index, disease counts,

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and the Adjusted Clinical Groups (ACG) system.[12] However, this review was conducted in 2009 and the multimorbidity research has increased exponentially since then.

The present review was to build upon the review paper by Huntley et al.[12] in order to provide a current and comprehensive list of instruments that measure levels of multimorbidity for community-dwelling individuals. We used the term 'level of multimorbidity' to refer to the combined effects of multiple conditions on an individual. The main objective of this review was to list instruments for measuring the levels of multimorbidity where risk of bias of the study design was assessed to be low. We specifically look for studies that measure the association of a clinically important outcome with different levels of multimorbidity as the main independent variable in community-dwelling individuals. Our second objective was to provide details of the requirements, strengths and limitations of these instruments, and the chosen outcomes in the studies so that clinicians and researchers can select or develop instruments that match their needs.

METHODS

A protocol for this systematic review (CRD42018105297) was published online on PROSPERO.[13] We searched MEDLINE, EMBASE, CINAHL and also manually searched the Journal of Comorbidity for potential studies. The medical subject headings and keywords used for the search are shown in Appendix 1.

We selected studies that included: (1) adult patients from primary care or the general population as the majority of patients with multimorbidity are managed by primary care physicians[14]; (2) studies with at least one specified outcome variable; and (3) full-text articles published from 1st January 2010 to 23rd October 2020. Studies were excluded if they (1) selected patients from

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the hospital or nursing home only or patient data were drawn solely from the hospital or the nursing home; or (2) selected patients with an index condition; or (3) used level of multimorbidity as a covariate and not the main independent variable; or (4) were not written in English. We did not include a specific definition of multimorbidity instrument because, given a lack of consensus in the literature on the use of this term, we wanted to include a diverse range of papers on the above topic.

One reviewer (ESL) conducted a preliminary screen of titles and abstracts to exclude records that were irrelevant. Abstracts of the remaining records were screened independently by two reviewers (ESL & EH) according to the eligibility criteria. Disagreements were resolved through discussion until a consensus was reached. The full-text articles were then retrieved for the agreed list and independently assessed according to the eligibility criteria by the same reviewers. Disagreements were resolved through discussion with a third reviewer (TSH) until a consensus was reached. After agreement on the list of articles, the reference lists of included articles were hand-searched for additional eligible papers.

As the aim of this review was to provide a list of instruments for measuring the levels of multimorbidity where risk of bias of the study design was assessed, the selected articles were next appraised independently by three reviewers (ESL, EH and TSH) using the modified Newcastle-Ottawa Scale (NOS)[15 16]. Each article was assessed under the three broad categories: a) Selection; b) Comparability; and c) Outcome (Appendices 2 & 3). We only reported instruments that were described in studies with a low risk of bias.

We contacted the authors, as needed, for additional information or clarification up to three times spaced one week apart. We contacted 25 authors and 19 of them replied. Any disagreements on the risk of bias were resolved among the three reviewers through regular

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meetings. HLK and FYW were responsible for tracking and updating the final outcome of the risk-of-bias assessment. We anticipated that the studies would be heterogeneous and therefore the decision was made *a priori* not to conduct a meta-analysis.

Patient and public involvement: This research was done without patient involvement. Patients were not invited to comment on the study design and were not consulted to develop patient relevant outcomes or interpret the results. Patients were not invited to contribute to the writing or editing of this document for readability or accuracy.

RESULTS

The number of included studies was 96, of which 69 were assessed to have low risk of bias. A summary of the PRISMA flow chart is depicted in Figure 1. Thirty-four studies selected participants from the general population and 44 studies selected participants from primary care. Most of the studies in this systematic review were from Europe and North America with very few Asian studies. There were 44 cohort studies of which 36 were assessed to have low risk of bias, and 52 cross-sectional studies of which 33 were assessed to have low risk of bias. We found 33 unique instruments from the 69 studies with low risk of bias. The instruments were categorised according to Sarfati[8 9] into: 1) simple counts of individual conditions; 2) organ or system-based approaches; 3) conditions that have been weighted and combined into indices; and 4) other approaches. A total of 118 outcomes were reported from all the studies. No studies were excluded for an outcome that was not deemed to be clinically important. Appendices 4 & 5 summarises the risk of bias assessment of each study. Table 1 provides a summary of the study design, population source, age group, multimorbidity measurements and outcome measures of all the 69 studies with low risk of bias.

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Table 1. Summary of included studies with low risk of bias judgement

Author (Year)	Study Design	Population source	Age	Multimorbidity Measurement	Outcomes measured
Agborsangaya (2013)[17]	CS	GP	≥ 18	DC	HRQoL
Bahler (2015)[18]	CS	GP	≥ 65	DC - ATC classification system	Total number of consultations
Barile (2013)[19]	Cohort	GP	≥ 65	DC	ADL limitations, Physically unhealthy days, Mentally unhealthy days
Barile (2012)[20]	CS	GP	≥ 65	DC	Physical HRQoL, Mental HRQoL
Barnett (2012)[21]	CS	PC	≥ 0	DC	Presence of mental health disorder
Biehl (2016)[22]	Cohort	PC	≥ 65	ERA, CCI	Presence of critical illness
Brilleman (2014)[23]	Cohort	PC	≥ 18	QOF count, CCI, EDC count, ACG, RUB	Primary healthcare cost The EDC performed best followed by the QOF and ACG.
Brilleman & Salisbury (2013)[24]	Cohort	PC	≥ 18	QOF count, CCI, EDC count, ACG, RUB, Prescribed drugs count	Mortality: The CCI was the best performing measure followed by the number of prescribed drugs. Number of primary care consultations (3-years period): The number of prescribed drugs had the greatest predictive validity followed by the ACG based measures (ACG, EDC count, and RUB).
Carey (2013)[25]	Cohort	PC	≥ 60	Standard QOF, Extended QOF, CCI (Khan)	Mortality (1-year period) The standard QOF score outperformed the CCI(Khan). The extended QOF score produced only a modest improvement in overall model performance.
Chapman (2015)[26]	Cohort	GP	≥ 18	CCI, CCI-PSR	Mortality (5, 10, 15, 20, 25-years period) The CCI-PSR showed substantially better discrimination than the CCI.
Charlson (2014) [27]	Cohort	GP	≥ 0	Charlson Comorbidity Index	Health care cost, Utilisation of services
Chen (2011)[28]	CS	GP	≥ 18	DC	General Health, Mental Distress, Physical Distress, Activity limitations
Chu (2018) [29]	CS	PC	≥ 40	DC, CIRS	Health care utilization
Crane (2010)[30]	Cohort	PC	≥ 60	ERA	No. of hospital visits, ED visits, Hospital admissions, Days hospitalised (1-year period)
Crooks (2016)[31]	Cohort	PC	20 to 100	Co-morbidity linked score, CCI, Elixhauser Index	Mortality (1-year period) The linked score had significantly improved discrimination and fit compared to the CCI and the Elixhauser Index
Crooks (2015)[32]	Cohort	PC	≥ 20	CCI (Read), CCI (ICD-10), CCI (Read & ICD-10)	All-cause mortality (1-5 years) There was no large difference in the discrimination of the model for whichever codes that were used to derive the CCI.
DiNapoli (2017)[33]	CS	PC	≥ 50	Organ systems with chronic disease	Presence of depressive or anxiety disorder
Formiga (2013)[34]	Cohort	PC	85	CCI	Mortality (3-years period)
Formiga (2011a)[35]	Cohort	GP	90 to 99	CCI	Mortality (5-years period)
Formiga (2011b)[36]	CS	PC	85	CCI	Successful aging
Formiga (2016)[37]	Cohort	PC	85	CCI	Mortality (5-years period)
Fraccaro (2016)[38]	Cohort	PC	≥ 18	CCI (Khan)	Mortality (1, 5, 10-years period), Mortality (3, 6, 12-months period)

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Table 1. Summary of included studies with low risk of bias judgement (Continued)

Author (Year)	Study Design	Population source	Age	Multimorbidity Measurement	Outcomes measured
Galenkamp (2011)[39]	CS	GP	57 to 98	DC	SRH
Garin (2014)[40]	CS	GP	≥ 50	DC	QOL, Disability
Glynn (2011)[41]	CS	PC	> 50	DC	Primary care consultations, Hospital outpatient visits, Hospital admissions, Healthcare cost (all 1-year period)
Haas (2013)[42]	Cohort	PC	≥ 18	ACG, Minnesota Health Care Home Tiering, HCC, ERA, CCC, CCI, Hybrid Model	Hospitalisation, ED visits, Readmission within 30 days, Healthcare expenditure (all 1-year period) The ACG model outperformed the other 5 models in all outcomes.
Hwang (2015)[43]	Cohort	GP	≥ 0	ACE-27, ACE-27 count	Healthcare expenditure The model, using year 1 data to determine if an individual would be classified into the persistent high-user group for the following 3 years, indicates a very high level of accuracy in predicting membership in a high-user group.
Jindai (2016) [44]	CS	GP	≥ 65	DC	Functional limitations (ADL, IADL, Leisure and social activities, Lower-extremity mobility, General physical activities)
Kristensen (2014)[45]	CS	PC	> 0	RUB	Fee-for-services expenditures
Lapi (2015)[46]	CS	PC	≥ 15	HSMI	Total mean healthcare cost per year The HSMI explained 50.17% of the variation in costs
Lawson (2013)[47]	CS	GP	≥ 20	DC	Preference-weighted HRQoL
Lemke (2012)[48]	Cohort	GP	≥ 0	CCI, ACG	Inpatient hospitalisations ACG-based predictive model was superior to CCI model.
Loprinzi (2016) [49]	CS	GP	60-85	DC	Cognitive function
Macinko (2019)[50]	CS	GP	≥ 18	DC (categorical 2 and 3 or more) (Self-reported)	Primary care experience (self-reported)
Marengoni (2011)[51]	CS	GP	≥ 75 (baseline) ≥ 77 (follow-up)	DC	Disability
McDaid (2013) [52]	CS	GP	≥ 50	DC	Disability, QoL, SRH
Milla-Perseguer (2019)[53]	CS	PC	≥ 18	CRG	Health-related quality of life (HRQL) - EQ-5D-3L
Monterde (2020) [54]	Cohort	GP	≥ 18	Adjusted Morbidity Group (GMA), CCI, DC, CRG	Use of healthcare resources
Palladino (2019)[55]	CS	GP	≥ 50	DC	Primary care use, Reduced functional capacity, Self-perceived health, Hospital admissions, Quality of life
Pati (2019) [56]	CS	PC	≥ 18	Severity burden score (21 conditions)	Health-related quality of life (SF-12)
Payne (2014)[57]	Cohort	PC	≥ 20	DC	Unplanned hospital admissions (1-year period)

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Table 1. Summary of included studies with low risk of bias judgement (Continued)

Author (Year)	Study Design	Population source	Age	Multimorbidity Measurement	Outcomes measured
Payne (2013) [58]	Cohort	PC	≥ 20	DC	Unplanned hospital admission, Potentially preventable admission (all 1-year period)
Payne (2020)[59]	Cohort	PC	≥ 20	CCI, DC (37 read codes), Cambridge Multimorbidity Score	Mortality, Unplanned inpatient hospital admission, primary care consultations
Quail (2011)[60]	Cohort	GP	≥ 20	DC, CCI (Quan), Elixhauser (Quan), No. of different dispended drugs, CDS	Mortality (1-year period): Elixhauser (Quan) performed best followed by CCI. One or more hospitalisations; Two or more hospitalisations: DC was the best performing measure
Ranstad (2014)[61]	CS	GP	≥ 0	RUB	Registered active listing in primary care and all healthcare
Reinke (2019)[62]	CS	PC	30-94	DC	Symptom burden (MSAS-SF), quality of life (Veterans RAND 12)
Reyes (2014)[63]	Cohort	PC (Men)	≥ 65	CCI	Hip fractures
Ryu (2015)[64]	CS	PC	≥ 18	DC	Deficits of perceived general health, Depressive symptoms
Salisbury (2011) [65]	Cohort	PC	≥ 18	QOF count, EDC count	Primary Care consultation rates, Continuity of care (all 3-year period)
Saver (2014)[66]	Cohort	GP	≥ 65	CCI (Romano) + Hypertension	Acute ACSH, Chronic ACSH
Shadmi (2011)[67]	CS	GP	≥ 18	ADG, CCI	No. of primary care physician visits, specialist visits, hospitalisation ADG explained the largest percent of variance or in health care resource use
Stanley & Sarfati (2017)[68]	Cohort	PC	≥ 18	M3 Index, CCI, Elixhauser (van Walraven)	Mortality, Overnight hospitalisation (all 1-year period) M3 Index outperformed both CCI and Elixhauser (van Walraven)
St John (2014) [69]	Cohort	GP	≥ 65	DC (0-36 conditions)	Mortality in 5 years
St John (2019)	Cohort	GP	≥ 65	DC	Functional Impairment in 5 years
Streit (2014)[70]	Cohort	PC	50 to 80	CCI, DC	Quality of cardiovascular preventive care, Quality of preventive care
Sullivan (2012)[71]	CS	GP	≥ 18	DC	Preference-based HRQoL
Takahashi (2016)[72]	Cohort	PC	≥ 18	Minnesota Tiering (ACG), Enhanced model	Hospitalisation / ED visits The enhanced model is better
Takahashi (2012)[73]	Cohort	PC	> 60	ERA	Mortality, Nursing home placement (all 2-year period)
Ubalde-Lopez (2016)[74]	CS	GP	F (mean): 35.9, M (Mean): 37.9	MDMS	Sickness absence episodes taken in last 2 years
van den Bussche (2011)[75]	CS	PC	≥ 65	DC	Frequency of contacts with physicians, Number of different ambulatory physicians contacted (all 1-year period)
van Oostrom (2014)[76]	CS	PC	≥ 55	DC	No. of contacts with general practice, Medications prescribed, Referrals

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Table 1. Summary of included studies with low risk of bias judgement (Continued)

Wallace (2016a)[77]	Cohort	PC	≥ 70	Pra tool, Modified Pra tool	Emergency hospital admission (1-year period) Both models demonstrated poor model discrimination
Wallace (2016b)[78]	Cohort	PC	≥ 70	DC, Barnett conditions DC, CCI, Prescribed drugs count, RxRisk-V	Emergency admission, Functional decline (all 2-year period) All measures demonstrated poor discrimination
Wei (2018)[79]	CS	GP	≥ 51	MWI	Subjective physical functioning, Grip strength, Gait speed, Cognitive performance, ADL limitations, IADL limitations
Wei (2019) [80]	Cohort	GP	≥ 51	MWI	Physical Functioning - SF-36, Mortality
Wei (2020) [81]	Cohort	GP	≥ 51	MWI	Cognitive functioning
Wei & Mukamal[◊] (2018)[82]	Cohort	GP	≥ 36	MWI, DC, CCI	Mortality (10-years period), Future physical functioning MWI performed best in predicting mortality as compared to DC and CCI
Wikman (2011)[83]	CS	GP	≥ 50	DC	QoL, Affective well-being
Wister (2015)[84]	CS	GP	≥ 65	MM additive scale, MM weighted by HUI3, MM weighted by ADL Scale, MM weighted by HUI3 betas	Life satisfaction, Perceived health status

Note. ACE-27 = Adult Comorbidity Evaluation-27; ACG = Adjusted Clinical Groups; ACSH = Ambulatory Care Sensitive Hospitalisation; ADG = Aggregated Diagnosis Groups; ADL = Activities of Daily Living; CCC = Chronic Condition Count; CCI = Charlson Comorbidity Index; CCI-PSR = Charlson Comorbidity Index-Psychosocial Risk; CDS = Chronic Disease Score; CS = Cross-Sectional; DC = Disease Count (Unweighted); ED = Emergency Department; EDC = Expanded Diagnosis Clusters;ERA = Elder Risk Assessment; GP = General Population; HCC = Hierarchical Condition Categories; HRQoL = Health-related Quality of Life; HSMI = Health Search Morbidity Index; HUI3 = Health Utility Index; IADL = Instrumental Activities of Daily Living; ICD-10 = International Classification of Diseases, Tenth Revision; MDMS = Multidimensional Multimorbidity Score; MM = Multimorbidity; MWI = Multimorbidity-Weighted Index; PC = Primary Care; Pra tool = Probability of repeated admission risk prediction tool; QOF = Quality and Outcomes Framework; QoL = Quality of Life; RUB = Resource Utilisation Band; RxRisk-V = A Veterans Association adapted pharmacy-based case-mix instrument; SRH = Self-Rated Health

[◊] The youngest participant was from the NHS II cohort, aged 25 in 1989. Data collection for this study started in 2000 (NHS and HPFS cohorts) and 2001 (NHS II cohort) and hence, the age of the participants recruited in this study is 36 years old and above.

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Table 2 summarises all the 33 instruments that were identified from the studies with low risk of bias. Table 3 provides a summary of multimorbidity instruments and their positive and negative associations with all the different outcome measurements from the 69 studies with low risk of bias.

Simple counts of individual conditions

Disease count was based on the total number of all the conditions an individual had, usually from a pre-specified list of chronic conditions. It was used in 37 out of the 69 studies (53.6%). Disease count was positively associated with activity limitations, continuity of care, disability, healthcare cost, healthcare utilisation, medications, mental disorders, and mortality; and negatively associated with general health, physical function, quality of life, and self-rated health (Table 3).

Organ or system-based approaches

There were two instruments in this category. They were Cumulative Illness Rating Scale (CIRS)[85 86] and Organ Systems with chronic disease count (Organ-CDC),[33] and each instrument was described by one study.

Weighted Indices

There were 21 unique weighted instruments found in the included studies. The original Charlson Comorbidity Index (CCI) with its different modifications was the most frequently used instrument and was used in 25 studies. The CCI was based on disease count, but the 17 conditions were weighted originally based on their impact on one-year mortality.[27] The final score was derived by the summation of all the weighted conditions. There were many

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Table 2. Description of instruments used for measurement of multimorbidity and the data sources and resources required

Category	Instrument	System/Condition based	Weightage; Scoring Method	Data Sources and resources required
A: Count of Individual Conditions				
A-1	DC	Condition (7-147)	Unweighted; Condition count	ATC list of conditions, Elixhauser list of conditions, EMR, GP records, Health service database, Hospital discharge abstract, Insurance claims or Questionnaires - telephone, face-to-face, mailed surveys. Participant involvement required
A-2	CCC[87]	Condition (6)	Unweighted; Based on AHRQ's clinical classification software & no. of conditions for each category	EMR
B: Organ or System-based Approaches				
B-3	Organ systems with CDC	Organ system (17)	Unweighted sum of organ systems	EMR
B-4	CIRS[88 89]	Body systems (13)	1-5 (based on severity of the condition); different weightage for diseases	EMR
C: Weighted Indices				
C-5	ACE[90]	Condition (27)	1-3 (based on severity of most severe condition); Highest score of single item	Insurance Claims' database
C-6	Cambridge MM Score[59]	Condition (20)	Weighted based on three different outcomes – primary care consultation, unplanned admission & mortality	EMR linked to mortality, hospital admission and socioeconomic deprivation
C-7	CCI[27]	Condition (19)	1-6 (based on impact on 1-year mortality (RR) – original); Sum of weighted conditions	Administrative database, EMR, Medical chart review, orInterviews or postal questionnaire where participant involvement is required
C-8	CCI (Deyo)[91]	Condition (23)	Different weightage	Claims data repository
C-9	CLS[31]	Condition (98)	Based on impact for mortality (hazard ratio); Sum of beta coefficients of each category	Linked patients' records of all primary care events, hospital admissions and causes of death.
C-10	EI (original & modified)[92]	Condition (21-31)	Based on impact on in-hospital mortality; Summing of beta coefficients	Insurance Claims' or Medical services database
C-11	ERA[30]	Condition (6-9)	Weighted (based on impact on future hospitalisation); Sum of weighted regression coefficients	EMR and administrative database
C-12	HCC[93]	Condition (70)	Based on Medicare capitation payments for health expenditure.; More severe manifestations of a condition dominating (and zeroing out the effect of) less serious ones. Other diseases are summed additively.	EMR & HCC software licensing and fees
C-13	M3 Index[68]	Condition (55)	Weighted based on 1-year mortality; Summing of beta coefficients	Linked patients' records
C-14	MDMS[74]	Condition (7 chronic conditions, 2 health behaviours for 1st dimension & 5 symptoms for 2nd dimension)	Weighted but not based on any specific outcome; Sum of the value for the weighted absolute contributions of each of the dimensions.	Standardised medical evaluation (Interviewer-administered); Participant involvement is required
C-15	MM weighted by ADL Scale[94]	Condition (19)	Weighted based on OARS functional status scale measuring ADL; Sum of weighted conditions	Face-to-face or telephone interviews where participant involvement is required

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Table 2. Description of instruments used for measurement of multimorbidity and the data sources and resources required (Continued)

Category	Instrument	System/Condition based	Weightage; Scoring Method	Data Sources and resources required
C-16	MM weighted by HUI[95]	Condition (19)	Weighted based on correlation with health utility index; Sum of weighted conditions	Face-to-face or telephone interviews where participant involvement is required
C-17	MM weighted by HUI betas[95]	Condition (19)	Weighted based on correlation with health utility index and adjusted for age and sex; Summing of beta coefficients	Face-to-face or telephone interviews where participant involvement is required
C-18	MWI[96]	Condition (81)	Weighted based on impact on SF-36 physical functioning scale; Sum of weights	Interviewer-administered or mail questionnaire where participant involvement is required
C-19	Extended QOF (weighted)[25]	Condition (14)	0-6, based on impact on 1-year mortality (RR); Sum of weighted conditions	EMR
C-20	Severity Burden Score[97]	Condition (21)	Sum of weights of diseases by the level of interference for each condition	Interviewer-administered structured questionnaire by nurses where participant involvement is required
C-21	Standard QOF (weighted)[25]	Condition (9)	1-3, based on impact on 1-year mortality (RR); Sum of weighted conditions	EMR
D: Other Approaches (D1 = Case Mix, D2 = Pharmaceutical-based)				
D1-22	ACG[98 99]	Condition (93 mutually exclusive ACGs. Some are modified to 68 ACGs)	Incorporated into ACGs based on impact on resource use (proprietary); Variable	EMR & ACG software licensing and fees
D1-23	ADG[98 99]	Condition (32 groups)	Based on duration, severity, diagnostic certainty, aetiology, and need for specialty care; Variable	EMR & ACG software licensing and fees
D1-24	CRG[100]	NA; Diagnostic categories derived from organ systems or clinical category (37)	Pre-formulated based on the 3M clinical risk groups and consists of 9 core health ranks	EMR - inpatient and outpatient & 3M Clinical Risk Grouping software v1.6 and service fees
D1-25	Adjusted Morbidity Groups (GMA)[101]	NA; mutually exclusive categories (31)	Based on multimorbidity and levels of patient complexity	Registry data
D1-26	HM[42]	Condition (NS)	Only MN tier 4 + MN tier 3 with ERA > 10; Variable	EMR, HCC software licensing, fees & administrative data
D1-27	HSMI[46]	Condition (73 chronic and acute conditions)	Based upon yearly health care costs directly derived from primary care setting; Sum of regression coefficients (range from -0.06 to 1.04)	EMR
D1-28	Minnesota Tiering[98 99]	Condition (NS)	Grouping patients into 'complexity tiers' based on the number of major condition categories; Condition Count	EMR or administrative data & MN Tiering software licensing and fees
D1-29	Resource Utilisation Band[98 99]	Condition (6 mutually exclusive bands)	Based upon ACG algorithm on impact on resource use (proprietary); Variable	EMR & ACG software licensing and fees
D2-30	CDS[102]	Condition (17)	Weighted 1-5; Sum of weights based on pharmacological database	Prescription drug database
D2-31	Drug Count	NA; Variable. Some may be based on pharmacologic-therapeutic classification system	Weighted; Medication count	Self-reported questionnaire where participant involvement is required
D2-32	Modified Pra tool using RxRisk-V[77 103 104]	NA; Pra tool + RxRisk-V	Weighted due to RxRisk-V; 4 categories	GP medical record + linked pharmacy claims database

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Table 2. Description of instruments used for measurement of multimorbidity and the data sources and resources required (Continued)

Category	Instrument	System/Condition based	Weightage; Scoring Method	Data Sources and resources required
D2-33	RxRisk-V[104]	NA; WHO-ATC classification system	Weighted according to the diagnostic group of drugs to predict future health care costs; Sum of weights	GP medical record + linked pharmacy claims database

Note. ACE = Adult Comorbidity Evaluation; ACG = Adjusted Clinical Groups; ADG = Aggregated Diagnosis Groups; ADL = Activities of Daily Living; ATC = Anatomical Therapeutic Chemical; CCC = Chronic Condition Count; CCI = Charlson Comorbidity Index; CLS = Comorbidity Linked Score; CDC= Chronic Disease Count; CDS = Chronic Disease Score; CRG = Clinical Risk Groups; DC = Disease Count; Elixhauser Index = EI; EMR = Electronic Medical Records; ERA = Elder Risk Assessment; GP = General Practitioner; HCC = Hierarchical Condition Categories; HM = Hybrid Model (MN Tier + ERA); HUI = Health Utility Index; HSMI = Health Search Morbidity Index; MDMS = Multidimensional Multimorbidity Score; MM = Multimorbidity; M3 Index = Multimorbidity Measure Index; MWI = Multimorbidity-Weighted Index; OARS = Older Americans Resources and Services; Pra tool = Probability of repeated admission risk prediction tool; QOF = Quality and Outcomes Framework; RR = Risk Ratio; RxRisk-V = A Veterans Association adapted pharmacy-based case-mix instrument; SF-36 = 36-item Short Form Survey

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Table 3. Summary of multimorbidity instruments and their associations with outcomes measured from the 69 included studies with low risk of bias

Multimorbidity Measures	Association between Outcomes and Multimorbidity		
	Positive Association	Negative Association	No Association
A = Count of Individual Conditions			
DC (Many different groupings ranging from 7[39 73 105] to 147[41] conditions and some are further categorised[42]	ADL limitations[19], Activity limitations[28], Continuity of care (3-years)[65], Deficits of perceived general health[64], Depressive symptoms[64], Disability[40 51 52], Emergency hospital admission (2-years)[78], Frequency of contacts with physicians (1-year)[75], Functional decline (2-years)[78], Functional Impairment[106], Functional Limitations[44], Healthcare costs[43], Hospital admissions (1-year)[57 58 60], Hospital outpatient visits (1-year)[72], Hospitalisation/Emergency Department Visits[72], Mental distress[28], Mortality (1-year)[60], (3-years)[24], (5 years)[69], (10-years)[82], No. of contacts with general practice (1-year)[76], No. of medications prescribed (1-year)[76], No. of mentally unhealthy days[19 20], No. of physically unhealthy days[19 20], No. of different ambulatory physicians contacted (1-year)[75], No. of primary care consultations (1-year)[24], (3-years)[24], No. of referrals (1-year)[76], Physical distress[28], Presence of mental health disorder[41], Primary care consultations (1-year period)[65], (3-years)[65], Primary healthcare cost[23], Potentially preventable unplanned admission (1-year period)[58], Symptom burden[62]	Affective well-being[83], Cognitive Function[49], Future physical functioning[82], General health[28], Life satisfaction[84], Perceived health status[84], Primary care experience – Self-reported[50], Quality of life[55 62], Self-rated Health[39 52], Self-perceived health[55], Functional capacity[55]	Quality of cardiovascular preventive care[70], Quality of preventive care [70]
DC-ATC	Total number of consultation[18], total health care costs[18]		
CCC	Healthcare costs[42], Hospital admissions (1-year)[42], No. of emergency department visits (1-year)[42], Readmission within 30 days (1-year)[42]		
B = Organ or System-Based Approaches			
Organ systems with CDC	Presence of depressive or anxiety disorder[33]		
C = Weighted Indices			
ACE-27	Healthcare expenditure[43]		
Cambridge MM Score	Mortality[59], Primary care consultation[59], Unplanned admission[59]		

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Table 3. Summary of multimorbidity instruments and their associations with outcomes measured from the 69 included studies with low risk of bias (Continued)

Multimorbidity Measures	Association between Outcomes and Multimorbidity		
	Positive Association	Negative Association	No Association
CCI	Ambulatory care sensitive hospitalisations (acute & chronic)[66], Emergency department visits (1-year)[42], Emergency hospital admission (2-years)[77], Functional decline (2-years)[77], Healthcare expenditure[42], Hip fractures[63], Hospitalisation (1-year)[39 42 48 60 68], Mortality (1-year)[25 38 60 68], (5-years)[26 38], (10-years)[26 38], (15, 20, 25-years)[26], No. of primary care consultations (3-years)[24], No. of primary care physician visits (1-year) [67], No. of specialist visits (1-year)[67], Potentially preventable unplanned admission (1-year)[57], Presence of critical illness[22], Primary healthcare cost[23], Mortality (1-year)[31 32], (3-years)[24 34], (5-years)[32 35 37], (10-years)[82], Readmission within 30 days (1-year)[42]	Future physical functioning[82], Successful aging[36]	Quality of cardiovascular preventive care[70], Quality of preventive care[70]
CIRS	Health care utilisation[29]		
CLS	Mortality (1-year)[31]		
ERA	Healthcare expenditure [42], Mortality (2-years)[73], No. of days hospitalised (1-year)[30], No. of emergency department visits (1-year)[30 42], No. of hospital admissions (1-year)[30 42], No. of hospital visits (1-year)[30 42], Nursing home placement (2-years)[73], Presence of critical illness[22], Readmission within 30 days (1-year)[42]		
EI (Original & Modified)	Hospitalisation (1-year)[60 68], Mortality (1-year)[31 60 68]		
HCC	Hospitalisation (1-year)[42], ED visits (1-year)[42], Readmission within 30 days (1-year)[42], Healthcare expenditure (1-year)[42]		
MM3I	Hospitalisation (1-year)[68], Mortality (1-year)[68]		
MDMS	Sickness absence episodes taken in 2 years (male)[74]		Sickness absence episodes taken in 2 years (female)[74]
MM weighted by ADL Scale		Life satisfaction[84], Perceived health status[84]	
MM weighted by HUI		Life satisfaction[84], Perceived health status[84]	
MM weighted by HUI betas		Life satisfaction[84], Perceived health status[84]	

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Table 3. Summary of multimorbidity instruments and their associations with outcomes measured from the 69 included studies with low risk of bias (Continued)

Multimorbidity Measures	Association between Outcomes and Multimorbidity		
	Positive Association	Negative Association	No Association
MWI	ADL limitations[79], IADL limitations[79], Mortality (10-years)[82]	Cognitive performance[79], Future physical functioning[82], Grip strength[79], Subjective physical functioning[79]	Gait speed[82]
Severity Burden Score		Mental component score (SF-12)[56]	
QOF (Standard & Extended)	Mortality (1-year) [25]		
D = Other Approaches (D-1 = Case-Mix, D2 = Pharmaceutical-based)			
D-1(1): ACG	Hospitalisation (1-year)[48], Mortality (3-years)[24], No. of primary care consultations (3-years)[24], Primary healthcare cost[23], Readmission within 30 days (1-year)[42]		
D-1(2): ADG	Hospitalisation (1-year)[67], No. of primary care physician visits (1-year)[67], No. of specialist visits (1-year)[67]		
D1 (3): CRG		HRQoL using EQ-5D-3L[53]	
D-1(3): Adjusted Morbidity Groups (GMA)	Use of healthcare resources[54]		
D-1(3): HM	Emergency department visits (1-year)[42], Healthcare expenditure[42], Hospitalisation (1-year)[42], Readmission within 30 days (1-year)[42]		
D-1(4): HSMI	Healthcare cost (primary care)[46]		
D-1(5): Minnesota Tiering	Emergency department visits (1-year)[42 72], Healthcare expenditure[42], Hospitalisation (1-year)[42 72], Readmission within 30 days (1-year)[42]		
D-1(6): Resource Utilisation Band	Fee-for-service expenditures[45], Primary healthcare cost[23], Mortality (3-years)[24], Number of primary care consultations (3-years)[24], Registered active listing in primary care[61], Registered active listing in all healthcare[61]		
D-2(7): CDS	Hospitalisation (1-year)[60], Mortality (1-year) [60]		
D-2(8): Prescribed Drug Count	Emergency hospital admission (2-years)[78], Functional decline (2-years)[78], Hospitalisation (1-year)[60], Mortality (1-year)[60], (3-years)[24], No. of primary care consultations (3-years)[24]		
D-2(9): Pra tool Modified	Emergency hospital admission (1-year)[77]		

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Table 3. Summary of multimorbidity instruments and their associations with outcomes measured from the 69 included studies with low risk of bias (Continued)

Multimorbidity Measures	Association between Outcomes and Multimorbidity		
	Positive Association	Negative Association	No Association
D-2(10): RxRisk-V	Emergency hospital admission (2-years)[78], Functional decline (2-years)[78]		

Note. ACE-27 = Adult Comorbidity Evaluation-27; ACG = Adjusted Clinical Groups; ADG = Aggregated Diagnosis Groups; ADL = Activities of Daily Living; ATC = Anatomical Therapeutic Chemical; CCC = Chronic Condition Count; CCI = Charlson Comorbidity Index; CDC= Chronic Disease Count; CDS = Chronic Disease Score; CLS = Comorbidity Linked Score; CRG = Clinical Risk Groups; DC = Disease Count; Elixhauser Index = EI; ERA = Elder Risk Assessment; HCC = Hierarchical Condition Categories; HM = Hybrid Model (MN Tier + ERA); HRQoL = Health-Related Quality of Life; HSMI = Health Search Morbidity Index; HUI = Health Utility Index; MDMS = Multidimensional Multimorbidity Score; MM3I = Multimorbidity Measure (M3) Index; MWI = Multimorbidity-Weighted Index; Pra tool = Probability of repeated admission risk prediction tool; QOF = Quality and Outcomes Framework; RxRisk-V = A Veterans Association adapted pharmacy-based case-mix instrument; SF-12 = Short Form-12

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variations and modifications of the score including the addition of psychosocial factors. The instrument was found to be associated with multiple outcomes other than one-year mortality.

Most of the other weighted index instruments were novel, like the Multimorbidity-Weighted Index (MWI), in which the investigators built multivariable prognostic models from a set of potential predictor conditions and weighted the conditions based on an outcome of clinical interest. The most common outcomes chosen were mortality and physical function. Other outcomes included health expenditure,[42] health utility index,[84] and severity of the most severe condition.[43] The Multidimensional Multimorbidity Score (MDMS)[74] was unique as it was weighted based on health behaviours and patient symptoms and not based on any specific outcome.

Other Approaches to measuring multimorbidity

Other approaches included case-mix and pharmaceutical-based instruments. For case-mix approach, the Adjusted Clinical Group (ACG) and Resource Utilisation Band (RUB) were the most commonly used instruments.[107] Most of the case-mix instruments required proprietary software licenses from the United States of America and obtained data from electronic medical records or administrative data. The Clinical Risk Groups instrument was similar but took into account the severity of individual conditions.[100]

The second group of instruments in this category was related to pharmaceutical data. The most frequent type was the unweighted drug count. The other three (Chronic disease score, A Veterans Association adapted pharmacy-based case-mix instrument like RxRisk-V, and modified Probability of repeated admission risk prediction tool using RxRisk-V) were all weighted indices. Except for the drug count that was based on a self-report questionnaire, the rest required a prescription drug database to obtain the data.

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Outcomes

We classified the 118 outcomes into 17 categories as reported in the core outcomes set of multimorbidity research (COSmm).[10] The most commonly reported outcomes were health care use (n=41), mortality (n=15), health-related quality of life (n=10), and physical function (n=10). The different studies unanimously showed that higher levels of multimorbidity was associated with higher health care use and mortality, lower health-related quality of life, and poorer physical function. Seven outcomes in the COSmm were not found in all the 69 studies with low risk of bias. These were treatment burden, self-management behaviour, self-efficacy, adherence, communications, shared decision making and prioritisation. There were 17 outcomes found in the 69 studies that were not described in the COSmm. These included cognitive function, risk of suicide, frailty and falls. The outcomes not found to have any association with the instruments for measuring the level of multimorbidity were preventive care[70], sickness absence episodes (female)[74], and gait speed[82].

DISCUSSION

Summary of Findings

Thirty-three unique instruments for measuring the level of multimorbidity with low risk of bias were identified and categorised according to the classification by Sarfati.[8] The most commonly used instrument was 'Disease Count'. It was also the only instrument that was associated with the three essential outcomes from the core outcomes set of multimorbidity research (COSmm)[10] i.e., quality of life, mental health, and mortality.

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Comparison with previous research

Although the most common instrument identified in this systematic review was similar to that of Huntley et al.,[12] several instruments including Duke Severity of Illness Checklist (DUSOI) and Functional Comorbidity Index (FCI) identified in their paper were not found in this systematic review. The possible reasons for not finding these instruments in this review could be due to the lack of interest in the instrument by the research community in recent years (last publication using DUSOI was in 2004[108]), or the exclusion of studies specifying an index condition.

Advantages and disadvantages of selected instruments

Disease count

The advantage of using 'disease count' is its simplicity and the ease of data ascertainment with minimal resources required. However, using 'disease count' does not consider the severity of each condition where the complexity of multimorbidity may not be properly addressed.[109] The other disadvantage noted was the lack of transparency in the operational definition of multimorbidity, especially regarding the list of conditions considered for multimorbidity, and the cut-points used. Despite its simplicity, the level of multimorbidity measured using 'disease count' was the only instrument that was found to be associated with the three essential core outcomes (quality of life, mental health, and mortality).

Weighted Indices

The common weighted indices identified in this systematic review were Charlson Comorbidity Index (CCI), Elders Risk Assessment (ERA), Elixhauser Index (EI) and Multimorbidity-weighted Index (MWI). These weighted indices were often utilised in

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prognostic models to build complex multivariable regression models in which the weights were calculated from hazard ratios, odds ratios, or regression coefficients.[110]

The advantage of these weighted indices is that the weights allow the adaptation of an index to a specific outcome. An investigator could recalibrate the correct weight by creating a prognostic model to produce a contextualised instrument for a different setting. Prognostic models can provide clinically relevant risk stratification and help to allocate resources.[111] The disadvantage of such indices is that calculated weights are greatly influenced by the population, outcomes used, and the instrument's original conception and purpose, hampering the ability to compare across studies.

Case-Mix

The Adjusted Clinical Group (ACG) system has a good track record in the United States and several other countries, especially for measuring the outcomes of healthcare utilisation. However, the instrument is proprietary, and the exact algorithm of the instrument is not open to the public and may not be suitable in certain settings. The Clinical Risk Group (CRG) system has a good track record in Spain. It measures the severity of each condition and its algorithm is fully transparent. The common disadvantage of both systems is the financial costs involved in obtaining the license.

Pharmaceutical-based instruments

Medication-based indices include versions of the Chronic Disease Score,[102] which later became known as the RxRisk,[112] and its adaptation for use in the veteran population, the RxRisk-V.[113] Like the disease count, its main advantage is the ease of use with

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minimal resources required. However, many studies were not transparent regarding which type of drugs were included.

Data sources

Data sources used by these instruments included medical record information, patient self-report, clinical judgment, and large administrative databases. Each data source has its inherent advantages and disadvantages. For patient self-report, patients with cognitive impairment may under-report symptoms and may be seen less frequently by their physicians, resulting in an under-recognition or under-treatment of conditions.[114] It has also been shown that health administrative data based on billing system underestimated the prevalence of many chronic conditions.[115]

The available data in a particular setting may strongly influence the ultimate instrument chosen for multimorbidity research. As there is currently no consensus on the gold standard for sources of data, it is difficult to assess which data source was superior from this review.

Outcomes

The findings from the different studies showed that the association between the explanatory variables and reported outcomes were in the same direction and did not conflict with each other (Table 3). There were 17 multimorbidity outcomes identified by a Delphi process involving a panel of international experts in multimorbidity intervention studies.[10] However, only ten out of the seventeen outcomes were reported in the 69 studies identified in this systematic review with low risk of bias using the modified Newcastle Ottawa Scale. The most common outcome that was investigated was health care use. The seven missing outcomes belong to 'patient-reported impact and behaviours' and 'consultation-related' outcome groups, most likely

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indicating that there is a dearth of multimorbidity studies looking at these two groups of outcomes measures.

Clinical Implications

Although there was no restriction on the type of study design in our search strategy, we only found observational studies in our search. The lack of interventional studies suggested that either there was a lack of interventional studies on multimorbidity or that interventional studies did not use any instruments to measure the level of multimorbidity.

Ideally, a single instrument measuring the level of multimorbidity should be able to predict a variety of relevant outcomes. However, Byles et al.[116] reported that a single instrument could not be used to predict different outcomes, in different patient groups and settings, unless different weights were assigned to these factors in calculating a score. Such multiple-scoring instruments may be the way forward for validation of prognostic models for different outcomes and different populations with established multimorbidity instruments. For example, depending on the outcome, study population and setting, the choice of conditions included in the multiple-scoring instrument should include those with a high prevalence in that study population and the weights should be determined by their significant impact (i.e. outcome) on the affected population.

For pragmatic reasons, the final selection of the conditions to be included in such a multiple-scoring instrument may still have to take into account the availability of relevant and reliable data. A certain degree of reductionism will also have to be accepted because a single instrument will not be able to encompass all the nuances of the different interactions of chronic conditions on an individual living in his/her unique milieu. We recommend that researchers perform

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validation studies using the instruments listed in this systematic review to adjust the weights according to the specific outcome of interest for the study population relevant to their setting.

Strengths and limitations of the study

The main strengths of this systematic review were the involvement of a health science librarian in our search strategy, a published protocol, adherence to the protocol without changes during the systematic review process,[117] and the critical appraisal of all the primary studies with a risk-of-bias assessment tool.

The systematic review had several limitations. We excluded grey literature and included only articles that were published in the English language. We also did not contact authors directly for a suggestion of studies, nor identified a list of instruments from the preliminary search and then performed an additional search using the same databases.[118] However, given the consistency of our results as shown in Table 3 where the associated outcomes were all in the same direction, it is unlikely that missed studies would significantly alter the main findings of our review. This systematic review did not review the validity and reliability of all the instruments as it was beyond the scope of the intended work. We have, however, included the references of the original papers or validation studies in Table 2 for each of the instrument where available. Finally, this review specifically aimed to look at the association of the level of multimorbidity as the main independent variable and excluded the level of multimorbidity as a mediating, confounding or effect-modifying variable. We believe that the strict criteria would reduce the number of studies included but will clearly exclude studies that do not differentiate the subtle differences between comorbidity and multimorbidity.[14 119]

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CONCLUSIONS

In this systematic review, we found 33 instruments for measuring the level of multimorbidity in community-dwelling individuals that predict or explore the association of multimorbidity with at least one specified outcome. Disease count and weighted indices like the Charlson Comorbidity Index (CCI), the ERA, and Elixhauser Index (EI) were commonly used for measuring the level of multimorbidity. Other approaches to measuring the level of multimorbidity included case-mix or pharmaceutical-based instruments.

We found continuing interest in measuring the level of multimorbidity with disease count and drug count. There has also been a rise in the development of novel weighted indices using prognostic models or validation of existing well-established instruments like the CCI over the last few years. There is currently an absence of a gold standard for where to obtain chronic disease information. The most suitable instrument will depend on the specified outcome of interest, the study population, and the type of data and resources available.

Finally, there is still much work to improve on the body of knowledge of multimorbidity when most investigators in the last decade measured multimorbidity without including some of the important outcome measures of multimorbidity. We also suggest that a clear description of the instruments is required in the publication of multimorbidity studies to counter the frequent lack of information currently seen so as to contribute to robust multimorbidity research in future.

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ESL, BLR, MF and MS contributed to the design of the work. ESL, together with a health science librarian, created the search strategy and conducted the search. ESL, HLK, EH, TSH, and FYW screened papers, extracted and analysed data. ESL drafted the paper with inputs from BLR, MF and MS, and all authors reviewed drafts and approved the final version.

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Figure 1 – PRISMA Flow Diagram

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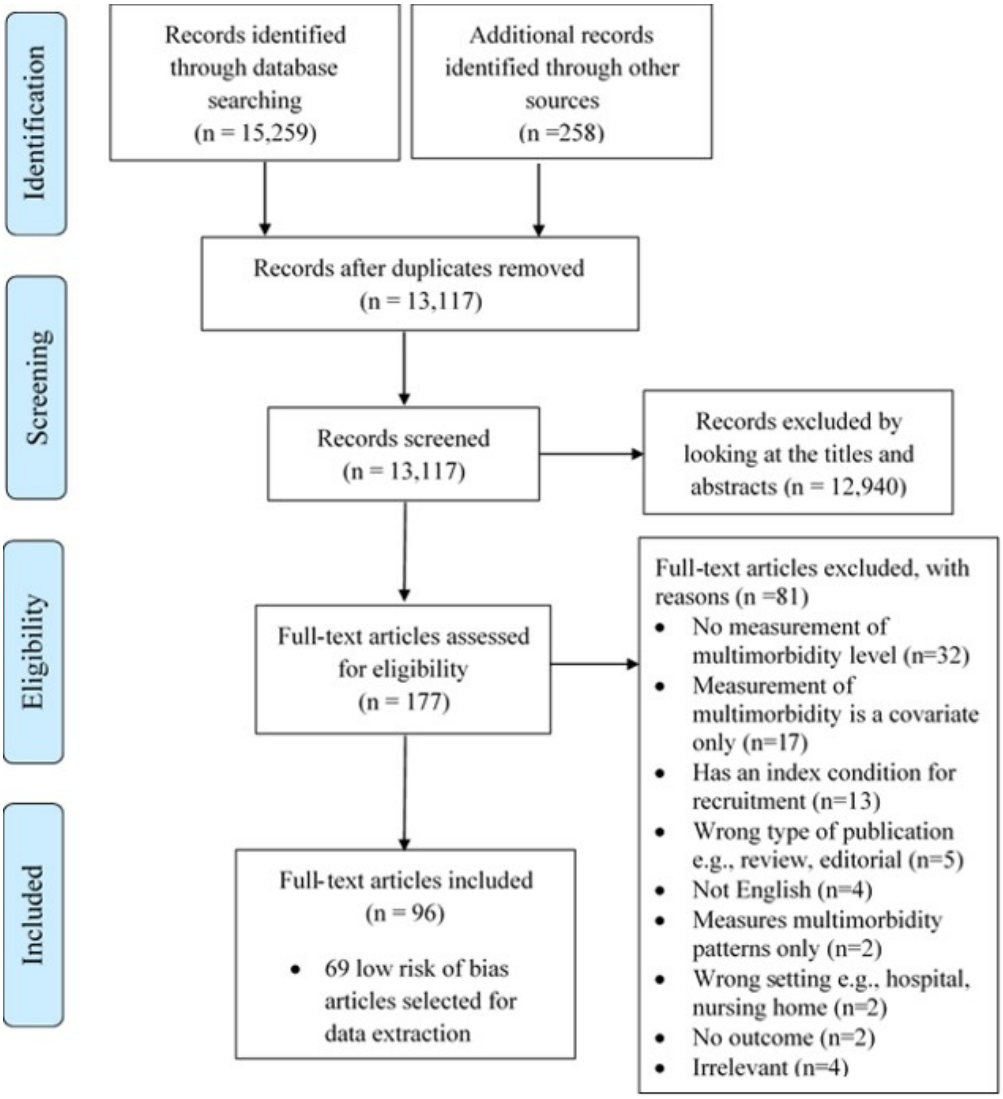


Figure 1 - PRISMA flow diagram

171x186mm (94 x 94 DPI)

Appendix 1. Search Strategy

Table 1. Medline Search 1946 to October 23, 2020

#	Searches	Results
1	exp comorbidity/ or multiple chronic conditions/	111191
2	(multimorbid* or multi-morbid* or comorbid* or co-morbid*).ab,ti,kw.	180089
3	((multiple or coexist* or co-exist* or concurrent* or simultaneous*) adj2 (disease* or illness* or diagnos* or condition* or morbid* or disorder*)).ab,ti,kw.	52139
4	1 or 2 or 3	292316
5	primary health care/ or "continuity of patient care"/ or exp general practice/ or ambulatory care/ or physicians, family/ or physicians, primary care/ or community health services/ or general practitioners/	249484
6	((ambulatory or community or general or family or primary) adj2 (care or health* or practi* or physician* or medicine or center* or centre* or facilit* or clinic*)).ab,ti,kw.	391013
7	5 or 6	508037
8	epidemiologic measurements/ or risk assessment/ or "Outcome Assessment (Health Care)"/ or patient reported outcome measures/ or health status indicators/ or "severity of illness index"/ or sickness impact profile/ or diagnosis-related groups/ or case mix/	594279
9	((health status or risk or outcome* or sickness impact) adj2 (appraisal* or assessment* or evaluation* or measur* or index* or indic* or profile*)).ab,ti,kw.	398319
10	((severity or burden) adj2 (illness* or diseas* or disorder* or condition* or diagnos*)).ab,ti,kw.	85595
11	(charlson comorbidity index or charlson index or charlson score or elixhauser* or cumulative illness rating scale* or CIRS or adjusted clinical group* or ACG* or disease count or Duke severity* or DUSOI or casemix or case-mix).ab,ti,kw.	21547
12	8 or 9 or 10 or 11	939031
13	4 and 7 and 12	4862
14	limit 13 to (english language and yr="2010 - Current")	2827

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Table 2. Embase Classic+Embase 1947 to October 23, 2020

#	Searches	Results
1	comorbidity/ or multiple chronic conditions/	265602
2	(multimorbid* or multi-morbid* or comorbid* or co-morbid*).ab,ti,kw.	329397
3	((multiple or coexist* or co-exist* or concurrent* or simultaneous*) adj2 (disease* or illness* or diagnos* or condition* or morbid* or disorder*)).ab,ti,kw.	81506
4	1 or 2 or 3	493137
5	primary health care/ or patient care/ or primary medical care/ or general practice/ or general practitioner/ or ambulatory care/ or family medicine/ or community care/ or community health services/	664384
6	((ambulatory or community or general or family or primary) adj2 (care or health* or practi* or physician* or medicine or center* or centre* or facilit* or clinic*)).ab,ti,kw.	539525
7	5 or 6	943743
8	risk assessment/ or outcome assessment/ or patient reported outcome/ or health status indicator/ or disease activity score/ or global disease burden/ or organ dysfunction score/ or "severity of illness index"/ or sickness impact profile/ or general health status assessment/ or disease severity/ or diagnosis related group/ or charlson comorbidity index/ or comorbidity assessment/ or elixhauser comorbidity index/ or case mix/	1590115
9	((health status or risk or outcome* or sickness impact) adj2 (appraisal* or assessment* or evaluation* or measur* or index* or indic* or profile*)).ab,ti,kw.	547083
10	((severity or burden) adj2 (illness* or diseas* or disorder* or condition* or diagnos*)).ab,ti,kw.	138554
11	(charlson comorbidity index or charlson index or charlson score or elixhauser* or cumulative illness rating scale* or CIRS or adjusted clinical group* or ACG* or disease count or Duke severity* or DUSOI or casemix or case-mix).ab,ti,kw.	42616
12	8 or 9 or 10 or 11	2022565
13	4 and 7 and 12	12245
14	limit 13 to (english language and yr="2010 - Current")	8896

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Table 3. CINAHL – since inception to October 23, 2020

#	Searches	Results
1	(MH “comorbidity”)	38970
2	(multimorbid* or multi-morbid* or comorbid* or co-morbid*)	66442
3	(multiple or coexist* or co-exist* or concurrent* or simultaneous*) N2 (disease* or illness* or diagnos* or condition* or morbid* or disorder*)	12822
4	1 or 2 or 3	77266
5	((MH “primary health care”) or (MH “family practice”) or (MH “ambulatory care") or (MH “ambulatory care facilities") or (MM “community health services”) or (MM “community health centers”) or (MH “physicians, family”) or (MH “continuity of patient care"))	100224
6	((ambulatory or community or general or family or primary) N2 (care or health* or practi* or physician* or medicine or center* or centre* or facilit* or clinic*))	332346
7	5 or 6	340014
8	((MH “risk assessment”) or (MH "Outcome Assessment") or (MH “patient-reported outcomes) or (MH “health status indicators”) or (MH "severity of illness indices") or (MH “sickness impact profile”) or (MH “diagnosis-related groups”) or (MH “case mix”))	102350
9	((health status or risk or outcome* or sickness impact) N2 (appraisal* or assessment* or evaluation* or measur* or index* or indic* or profile*))	340802
10	(severity or burden) N2 (illness* or diseas* or disorder* or condition* or diagnos*)	68742
11	(charlson comorbidity index or charlson index or charlson score or elixhauser* or cumulative illness rating scale* or CIRS or adjusted clinical group* or ACG* or disease count or Duke severity* or DUSOI or casemix or case-mix)	6729
12	8 or 9 or 10 or 11	396387
13	4 and 7 and 12	4729
14	limit 13 to (english language and yr="2010 - Current")	3536

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Appendix 2. Coding Description for the Modified Newcastle-Ottawa Scale for Cohort Studies

Category	Description
Selection	
1) Representativeness of the sample	This item assesses the representativeness of sample in the community, not from some general population. a) Truly representative***** (e.g., everyone from the database) b) Somewhat representative***** (with at least 2 criteria but selection method was convincing due to random sampling) c) Selected group (e.g., only certain socio-economic groups or areas) d) No description of sampling strategy
2) Ascertainment of multimorbidity	This item assesses the method by which multimorbidity was confirmed. a) Secure record***** (e.g., GP questionnaire) b) Structured interview***** (e.g., interviewer-administered questionnaire) c) Written self-report (e.g., mailed survey, if items are unable to be confirmed by objective measure) d) No description / Other
3) Demonstration that outcome of interest was not present at start of study	A statement of no history of disease earns a star. In the case of mortality studies, outcome of interest is still the presence of a disease, rather than death. a) Yes***** b) No
Comparability	
1) Study controls for age and sex	Covariates must be adjusted for in the analysis. Statements of no differences between groups or that differences were not statistically significant are not sufficient for establishing comparability. Note: If the relative risk for the exposure of interest is adjusted for the covariates listed, then the groups will be considered to be comparable on each variable used. a) Yes***** b) No
2) Study controls for other factors	a) Yes***** b) No
Outcome	
1) Statistical test	Statistical test(s) must be clearly described and appropriate before assessing the other items under the Outcome category. a) Clearly described and appropriate b) Not described, incomplete or inappropriate
2) Assessment of outcome	This item assesses the method by which the outcome of interest was confirmed. For some outcomes, reference to the medical record is sufficient to satisfy the requirement for confirmation. This may not be adequate for other outcomes where reference to specific tests or measures would be required. a) Independent or blind assessment***** (e.g., interviewer-administered questionnaire) b) Record linkage***** (e.g., identified through ICD codes on database records) c) Self-report (i.e., no reference to original medical records) d) No description / Other

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3)	Was follow-up long enough for outcomes to occur?	An acceptable length of time was at least 1 year of follow-up. a) Yes***** b) No
4)	Adequacy of follow-up of cohorts	This item assesses the follow-up of the sample to ensure that losses are not related to the outcome. a) Complete follow-up - all subject accounted for***** b) Subjects lost to follow-up unlikely to introduce bias***** (Number lost ≤20% or description of those lost suggested no different from those followed.) c) Follow-up rate less than 80% and no description of those lost d) No statement

Thresholds for converting the modified Newcastle-Ottawa scales to Good, Fair, and Poor Quality:

- Good Quality - 2-3 stars in Selection category (Representativeness of Sample item must be fulfilled) AND 1-2 stars in Comparability category AND 2-3 stars in Outcome category.
- Fair Quality - 1-2 stars in Selection category AND 1-2 stars in Comparability category AND 2-3 stars in Outcome category.
- Poor Quality - 0 star in Selection category OR 0 stars in Comparability category OR 0-2 stars in Outcome category.

Wells GA, Shea B, O'Connell D, Peterson J, Welch V, Losos M, Tugwell P. The Newcastle-Ottawa Scale (NOS) for assessing the quality of nonrandomized studies in meta-analysis. Available from: http://www.ohri.ca/programs/clinical_epidemiology/oxford.asp

Modifications from original NOS include:

- 'Statistical test' was added to the Outcome category
- 'Representativeness of sample' item under Selection category and 'Statistical test' item under Outcome category must both be fulfilled for study to be considered Good Quality.
- 'Selection of the non-exposed cohort' item was removed from Selection category as most studies did not describe a non-exposed cohort, and this review sought to compare the different levels of multimorbidity within the group.
- 'Representativeness of the exposed cohort' item under Selection category was renamed 'Representativeness of the sample' since the 'non-exposed cohort' was removed above.
- 'Ascertainment of exposure' under the Selection category was renamed as 'Ascertainment of multimorbidity' to specify that multimorbidity is the exposure in this review.
- 'Study controls for age and sex' and 'Study controls for other factors' were revised to be items under Comparability category
- The thresholds for converting the scales to quality were amended accordingly due to the above modifications.

Consensus:

The study team decided that a period of one-year was a reasonable period of follow-up under 'Was follow-up long enough for outcomes to occur?' item under Outcome category.

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Appendix 3. Coding Description for the Modified Newcastle-Ottawa Scale for Cross-sectional Studies

Category	Description
Selection	
1) Representativeness of the sample	This item assesses the representativeness of sample in the specified population, not from some general population. e) Truly representative***** (e.g., everyone from the database, random sampling) f) Somewhat representative***** (with at least 2 criteria but selection method was convincing due to random sampling) g) Selected group (e.g., only certain socio-economic groups or areas) h) No description of sampling strategy
2) Ascertainment of multimorbidity	This item assesses the method by which multimorbidity was confirmed. e) Secure record***** (e.g., Clinical records, GP questionnaire) f) Structured interview***** (e.g., interviewer-administered questionnaire) g) Written self-report (e.g., mailed survey, if items are unable to be confirmed by objective measure) h) No description / Other
3) Sample Size	If there is no description, a reported sample size of 800 and above is satisfactory. c) Justified and satisfactory***** d) Not justified
4) Non-respondents	Acceptable response rates for surveys through various methods† • In-person: 57% • Mail: 50% • Average: 33% • Email: 30% • Internet: 29% • Telephone: 18% • In-app: 13% a) Comparability between respondents and non-respondents characteristics is established, and the response rate is satisfactory.***** b) The response rate is unsatisfactory, or the comparability between respondents and non-respondents is unsatisfactory. c) No description of the response rate or the characteristics of the responders and the non-responders
Comparability	
3) Study controls for age and sex	Confounders must be adjusted for in the analysis. Statements of no differences between groups or that differences were not statistically significant are not sufficient for establishing comparability. Note: If the relative risk for the exposure of interest is adjusted for the confounders listed, then the groups will be considered to be comparable on each variable used in the adjustment. c) Yes***** d) No
4) Study controls for other factors	c) Yes***** d) No

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Outcome	
5) Statistical test	Statistical test(s) must be clearly described and appropriate before assessing the other items under the Outcome category. c) Clearly described and appropriate d) Not described, incomplete or inappropriate
6) Assessment of outcome	This item assesses the method by which the outcome of interest was confirmed. For some outcomes, reference to the medical record is sufficient to satisfy the requirement for confirmation. This may not be adequate for other outcomes where reference to specific tests or measures would be required. e) Independent or blind assessment***** f) Record linkage***** (e.g., identified through ICD codes on database records) g) Self-report (i.e., no reference to original medical records) h) No description / Other

Thresholds for converting the modified Newcastle-Ottawa scales to Good, Fair, and Poor Quality:

- Good Quality – 3-4 stars in Selection category (Representativeness of Sample item must be fulfilled) AND 1-2 stars in Comparability category AND 1 star in Outcome category.
- Fair Quality - 2 stars in Selection category AND 1-2 stars in Comparability category AND 1 star in Outcome category.
- Poor Quality - 0 star in Selection category OR 0 stars in Comparability category OR 0 stars in Outcome category.

†Lindermann N. What's the average survey response rate? [2018 benchmark]. SurveyAnyplace. Available from: https://surveyanyplace.com/average-survey-response-rate/.

Alshabanat A, Zafari Z, Albanyan O, Dairi M, FitzGerald JM. Asthma and COPD Overlap Syndrome (ACOS): a systematic review and meta-analysis. PLoS ONE. 2015;10(9):e0136065. Doi:10.1371/journal.pone.0136065.

Modifications from original NOS include:

- 'Ascertainment of exposure' item under Selection category was renamed as 'Ascertainment of multimorbidity' to specify that multimorbidity is the exposure in this review.
- The ratings for the 'Ascertainment of exposure' item under Selection category were revised to 'secure record, structured interview, written self-report, no description/other' to align with the Modified NOS for cohort studies in this review.
- 'Study controls for age and sex' and 'Study controls for other factors' items were revised to be items under Comparability category.
- 'Was follow-up long enough for outcomes to occur' item under Outcome category was renamed as 'Statistical test'. This item must be fulfilled for the study to be considered as Good Quality.
- The thresholds for converting the scales to quality were amended accordingly due to the above modifications.

Consensus:

- The study team decided on a list of acceptable response rates for various survey methods for the 'Non-respondents' item under Selection category.
- The study team decided that a sample size of 800 was reasonable under 'Sample size' item under Selection category.

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Appendix 4. Summary of risk of bias (RoB) appraisal of included cohort articles using the modified Newcastle-Ottawa Quality Assessment Scale (44 articles)

	Barile et al. (2013)[20]	Biehl et al. (2016)[23]	Boeckxstaens et al. (2015b)[87]	Brilleman et al. (2014)[24]	Brilleman and Salisbury(2013)[25]	Carey et al. (2013)[26]	Charlson et al. (2014)[103]	Chapman et al. (2015)[27]	Crane et al. (2010)[31]	Crooks et al. (2016)[32]	Crooks et al. (2015)[33]	Formiga et al. (2013)[35]	Formiga et al. (2011a)[36]	Formiga et al. (2016)[38]	Fraccaro et al. (2016)[39]	Haas et al. (2013)[43]	Hwang et al. (2015)[44]	Jennings et al. (2015)[104]	Jia and Lubetkin (2017)[105]	Jia et al. (2018)[106]	Lemke et al. (2012)[49]	Monterde et al. (2020)[55]	Md Yusof et al. (2010)[88]	Pavne et al. (2013)[59]	Pavne et al. (2014)[58]	Pavne et al. (2020)[60]	Ouail et al. (2011)[61]	Reves et al. (2014)[64]	Salisburv et al. (2011)[66]	Saver et al. (2014)[107]	Stanlev and Sarfati (2017)[69]	St John et al. (2014)[70]	St John et al. (2019)[90]	Streit et al. (2014)[71]	Takahashi et al. (2016)[73]	Takahashi et al. (2012)[74]	Tvack et al. (2016)[108]	Wallace et al. (2016a)[78]	Wallace et al. (2016b)[79]	Wei et al. (2019a)[109]	Wei et al. (2019b)[81]	Wei et al. (2020a)[82]	Wei et al. (2020b)[110]	Wei and Mukamal (2018)[83]
Selection Criteria																																										
Representativeness of the sample	*	*	-	*	*	*	-	*	*	*	*	*	*	*	*	*	*	*	-	*	*	*	*	-	*	*	*	*	*	*	*	*	*	*	*	-	-	*	*	*	-	*	*	*
Ascertainment of Multimorbidity	-	*	*	*	*	*	*	-	*	*	*	*	*	*	*	*	*	*	-	*	-	*	*	-	*	*	*	*	*	*	*	*	*	*	*	-	-	*	*	*	-	*	*	-
Demonstration that outcome of interest was not present at the start of the study	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*
Comparability Criteria																																											
Study controls for age and sex	*	-	*	*	*	*	*	*	-	*	*	-	*	-	*	*	*	*	*	-	*	*	-	*	*	*	*	-	*	*	*	*	*	*	*	*	*	*	-	*	-	-		
Study controls for others	*	*	-	*	*	*	*	*	*	-	*	*	*	*	*	*	*	*	-	-	*	*	*	*	*	-	*	*	*	*	-	*	*	*	*	*	*	*	*	*	-	*		
Outcome Criteria																																											
Statistical test†	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	X	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Assessment of outcome	-	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*
Follow-up was long enough for outcomes to occur	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*
Adequacy of follow-up of cohorts	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	-	*	*	*	*	*	-	*	*	*	-	*	*	*	*	-	-	*	*
Overall RoB Judgement[a]	G	G	F	G	G	G	G	G	G	G	G	G	G	G	G	G	G	G	F	P	P	G	G	F	G	G	G	G	G	G	G	G	G	G	G	G	G	F	G	G	G	F	G	P	G

Note. G = Good; F = Fair; P = Poor.

†Statistical test must be clearly described and appropriate before assessing the other items under the Outcome category. [a] Good rating is given when there are 2 to 3 stars in Selection category (Representativeness of Sample must be fulfilled) AND 1 to 2 stars in Comparability category AND 2 to 3 stars in Outcome category; Fair rating is given when there are 1 star in Selection category AND 1 to 2 stars in Comparability category AND 2 to 3 stars in Outcome category; Poor rating is given when there are 0 star in Selection category OR 0 star in Comparability category OR 0 to 2 stars in Outcome category.

"*" Study satisfies the criteria; "-" Study did not satisfy the criteria; "✓" Statistical test is clearly described and appropriate; "X" Statistical test is not described, incomplete or inappropriate.

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Appendix 5. Summary of risk of bias (RoB) appraisal of included cross-sectional articles using the modified Newcastle-Ottawa Quality Assessment Scale (52 articles)

	Agbeyangaya et al (2013)[18]	Bahler et al (2015)[19]	Barile et al (2012)[21]	Barnett et al (2012)[22]	Boerkystaens et al (2015a)[86]	Caballer-Tarazona et al	Chen et al (2011)[29]	Chen et al (2018)[112]	Chu (2018)[30]	Clynes et al (2020)[113]	DiNapoli et al (2017)[34]	Formiga et al (2011b)[37]	Galenkamp et al (2011)[40]	Garin et al (2014)[41]	Glynn et al (2011)[42]	Gunn et al (2012)[114]	Hanmer et al (2010)[115]	Hu et al (2017)[116]	Isaacs et al (2014)[117]	Jindai et al (2016)[45]	Kim et al (2012)[118]	Kojima et al (2011)[89]	Kristensen et al (2014)[46]	Lapi et al (2015)[47]	Lawson et al (2013)[48]	Li et al (2016)[119]	Lorrinzi (2016)[50]	Marinko et al (2019)[51]	Marengoni et al (2011)[52]	McDaid et al (2013)[53]	Milla-Perspner (2019)[54]	Mnoah et al (2012)[120]	Muiica-Mota et al. (2015)[11]	Nanssens et al (2011)[121]	Østergaard and Foldager	Pati et al (2019)[57]	Palladino et al (2019)[56]	Peters et al (2018)[122]	Ramstad et al (2014)[62]	Reinke et al (2019)[63]	Renne and Gabbans (2018)[123]	Ryu et al (2015)[65]	Shadmi et al (2011)[68]	Sibley et al (2014)[124]	Sullivan et al (2012)[72]	Thalde-Lopez et al (2016)[75]	van den Bussche et al (2014)[76]	van Oostrom et al (2014)[77]	Vos et al (2013)[125]	Wei et al (2018)[80]	Wikman et al (2011)[84]	Wister et al (2015)[85]
Selection																																																					
Representativeness of the sample	*	*	*	*	-	*	*	-	*	-	*	*	*	*	*	*	*	-	*	-	*	-	*	*	*	*	*	*	*	*	*	*	-	-	*	*	*	*	*	-	*	*	*	*	*	*	*	*	*	*	*	*
Ascertainment of Multimorbidity	*	*	-	*	*	*	*	-	*	*	*	*	*	*	*	*	-	-	*	*	*	*	*	*	*	-	*	*	*	*	*	*	*	-	*	*	*	-	-	*	*	*	*	*	*	*	*	*	*	*	*	*
Sample size	*	*	*	*	-	*	*	*	*	-	*	-	*	*	*	*	*	-	*	*	*	-	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	-	*	*	*
Non-respondents	-	*	*	*	-	*	*	*	*	-	*	*	-	-	*	-	-	*	*	*	*	-	*	*	*	-	*	-	-	-	-	*	-	*	-	*	*	-	-	*	*	*	-	*	*	*	*	*	*	*	*	-
Comparability																																																					
Study controls for age and sex	*	*	*	*	*	-	*	*	*	-	*	-	*	*	*	*	*	*	-	-	-	*	*	*	*	-	*	*	*	*	*	-	*	-	-	*	*	*	*	*	*	*	-	*	*	*	*	*	*	*	*	*
Study controls for others	*	*	*	*	*	-	*	*	*	*	*	*	-	*	*	*	*	*	-	*	-	*	*	*	*	-	*	*	*	*	*	-	*	-	-	*	*	*	*	*	*	*	-	*	*	*	*	*	*	*	*	-
Outcome																																																					
Statistical test†	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	X	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Assessment of outcome	*	*	✓	*	*	*	*	-	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	-	*	*	*	*	*	*	*	*	*
Overall RoB Judgement[a]	G	G	G	G	P	P	G	P	G	P	G	G	G	G	G	F	F	F	P	G	P	P	G	G	G	P	G	G	G	G	G	P	F	P	P	G	G	F	G	G	P	G	G	P	G	G	P	G	G	G

Note. G = Good; F = Fair; P = Poor.

†Statistical test must be clearly described and appropriate before assessing the other items under the Outcome category. [a] Good rating is given when there are 3 to 4 stars in Selection category (Representativeness of Sample must be fulfilled) AND 1 to 2 stars in Comparability category AND 1 star in Outcome category; Fair rating is given when there are 2 stars in Selection category AND 1 to 2 stars in Comparability category AND 1 star in Outcome category; Poor rating is given when there are 0 to 1 star in Selection category OR 0 star in Comparability category OR 0 star in Outcome category.

"*" Study satisfies the criteria; "-" Study did not satisfy the criteria; "✓" Statistical test is clearly described and appropriate; "X" Statistical test is not described, incomplete or inappropriate

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	3
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	5
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	6
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	6
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	6-7
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	6-8
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	Appendix 1
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	7
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	7
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	8
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	7
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	NA
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	NA

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	NA
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	NA
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	Figure 1
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	9-12
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	Appendix 4,5
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	13-22
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	NA
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	NA
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	NA
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	22-27
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	27
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	28
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	29

From: Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(7): e1000097. doi:10.1371/journal.pmed1000097

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A systematic review on the instruments used for measuring the association of the level of multimorbidity and clinically important outcomes

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TITLE PAGE

A systematic review on the instruments used for measuring the association of the level of multimorbidity and clinically important outcomes

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ABSTRACT

Objectives: There are multiple instruments for measuring multimorbidity. The main objective of this systematic review was to provide a list of instruments that measure the association of a specific outcome with different levels of multimorbidity as the main independent variable in community-dwelling individuals. The secondary objective was to provide details of the requirements, strengths and limitations of these instruments, and the chosen outcomes.

Methods: A systematic review was conducted according to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses guidelines (PROSPERO registration number: CRD42018105297). We searched MEDLINE, Embase and CINAHL electronic databases and manually searched the Journal of Comorbidity published in English language between 1st January 2010 and 23rd October 2020 inclusive. Studies also had to select adult patients from primary care or the general population, and had at least one specified outcome variable. Two authors screened the titles, abstracts and full texts independently. Disagreements were resolved with a third author. The modified Newcastle Ottawa Scale was used for quality assessment.

Results: Ninety-six studies were identified with 69 of them rated to have a low risk of bias. In total, 33 unique instruments were described. Disease count and weighted indices like the Charlson Comorbidity Index were commonly used. Other approaches included case-mix or pharmaceutical-based instruments. Disease count was the common instrument used for measuring all three essential core outcomes of multimorbidity research: mortality, mental health and quality of life. There was a rise in the development of novel weighted indices by using prognostic models. The data obtained for measuring multimorbidity were from various sources including medical records, patient self-reports, and large administrative databases.

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Conclusions: We listed the details of 33 instruments for measuring the level of multimorbidity as a resource for investigators interested in the measurement of multimorbidity for its association or prediction of a specific outcome.

ARTICLE SUMMARY

Strengths and Limitations of this study

- This review builds upon Huntley et al.'s 2012 review article and provides an updated, comprehensive list of instruments that measure levels of multimorbidity in community-dwelling individuals
- Thorough literature search of three major electronic databases with the involvement of a health science librarian
- Reporting of search as per Preferred Reporting Items for Systematic Reviews and Meta-Analysis guidelines
- Exclusion of non-English language articles and grey literature

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MAIN TEXT

BACKGROUND

Multimorbidity is defined as the co-occurrence of two or more chronic medical conditions in an individual.[1] It is a growing public health challenge and accounts for most of the expenditures in the healthcare system.[2] The complex interactions of several co-existing diseases have profound implications on individuals[3 4] and their healthcare providers.[5 6]

There are multiple instruments for measuring multimorbidity and many of them do not usually specify the severity of individual conditions.[7] No gold standard multimorbidity measurement instrument exists and there is also no agreed categorisation of the available instruments. Sarfati[8 9] classified the various measurement instruments into four broad approaches. They are: (1) by simple counts of individual conditions (i.e., disease count), (2) by organ or system-based approaches, (3) by weighting conditions and combining them into indices, and (4) by other miscellaneous approaches. Most of these measurements are used to measure the prevalence or patterns of multimorbidity. However, they can also be used to predict an outcome or to evaluate an intervention for a desired outcome. A set of core outcomes of multimorbidity (COSmm) was proposed after consulting a panel of international experts in multimorbidity intervention studies using a Delphi process.[10] Core outcome sets (COS) represent the minimum that should be measured and reported in all clinical trials of multimorbidity.[11]

Huntley et al.[12] published a systematic review in 2012 describing the instruments used to measure the morbidity burden in primary care and the general population. They found 17 different instruments from 194 articles. The most widely used instruments and those with the most significant evidence of validity were the Charlson Comorbidity Index, disease counts,

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and the Adjusted Clinical Groups (ACG) system.[12] However, this review was conducted in 2009 and the multimorbidity research has increased exponentially since then.

The present review was to build upon the review article by Huntley et al.[12] in order to provide a current and comprehensive list of instruments that measure levels of multimorbidity for community-dwelling individuals. We used the term 'level of multimorbidity' to refer to the combined effects of multiple conditions on an individual. The main objective of this review was to list instruments for measuring the levels of multimorbidity. We specifically look for studies that measure the association of a clinically important outcome with different levels of multimorbidity as the main independent variable in community-dwelling individuals. Our second objective was to provide details of the requirements, strengths and limitations of these instruments, and the chosen outcomes in the studies so that clinicians and researchers can select or develop instruments that match their needs for predicting a specific outcome.

METHODS

A protocol for this systematic review (CRD42018105297) was published online on PROSPERO.[13] We searched MEDLINE, EMBASE, CINAHL and also manually searched the Journal of Comorbidity for potential studies. The medical subject headings and keywords used for the search are shown in Appendix 1.

We selected studies that included: (1) adult patients from primary care or the general population as the majority of patients with multimorbidity are managed by primary care physicians[14]; (2) at least one specified outcome variable; and (3) published full-text articles from 1st January 2010 to 23 October 2020. Studies were excluded if they (1) selected patients from the hospital or nursing home only or patient data were drawn solely from the hospital or the nursing home;

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or (2) selected patients with an index condition; or (3) used level of multimorbidity as a covariate and not the main independent variable; or (4) were not written in English. We did not include a specific definition of multimorbidity instrument because, given a lack of consensus in the literature on the use of this term, we wanted to include a diverse range of studies on the above topic.

One reviewer (ESL) conducted a preliminary screen of titles and abstracts to exclude articles that were irrelevant. Abstracts of the remaining articles were screened independently by two reviewers (ESL & EH) according to the eligibility criteria. Disagreements were resolved through discussion until a consensus was reached. The full-text articles were then retrieved for the agreed list and independently assessed according to the eligibility criteria by the same reviewers. Disagreements were resolved through discussion with a third reviewer (TSH) until a consensus was reached. After agreement on the list of articles, the reference lists of included articles were hand-searched for additional eligible articles. We reported multimorbidity instruments that were described in all selected articles.

The risk of bias of the study design of selected articles was next appraised independently by three reviewers (ESL, EH and TSH) using the modified Newcastle-Ottawa Scale (NOS)[15 16]. Each article was assessed under the three broad categories: a) Selection; b) Comparability; and c) Outcome (Appendices 2 & 3).

We contacted the authors, as needed, for additional information or clarification up to three times spaced one week apart. We contacted 25 authors and 19 of them replied. Any disagreements on the risk of bias were resolved among the three reviewers through regular meetings. HLK and FYW were responsible for tracking and updating the final outcome of the risk of bias assessment.

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RESULTS

The number of included studies was 96, of which 69 were assessed to have low risk of bias. A summary of the PRISMA flow chart is depicted in Figure 1. Forty-eight studies selected participants from the general population and the other 48 studies selected participants from primary care. Most of the studies in this systematic review were from Europe and North America with very few Asian studies. There were 44 cohort studies of which 36 were assessed to have low risk of bias, and 52 cross-sectional studies of which 33 were assessed to have low risk of bias. We found 33 unique instruments from the 96 studies. The instruments were categorised according to Sarfati[8 9] into: 1) simple counts of individual conditions; 2) organ or system-based approaches; 3) conditions that have been weighted and combined into indices; and 4) other approaches. A total of 150 outcomes were reported from all the studies. No studies were excluded for an outcome that was not deemed to be clinically important. Appendices 4 & 5 summarise the risk of bias assessment of each study. Table 1 provides a summary of the study design, population source, age group, multimorbidity measurements, outcome measures, and risk of bias assessment of all the studies.

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Table 1. Summary of included studies

Author (Year)	Study Design	Population source	Age	Multimorbidity Measurement	Outcomes measured	Risk of bias
Agborsangaya (2013)[17]	CS	GP	≥ 18	DC	HRQoL	Good
Bahler (2015)[18]	CS	GP	≥ 65	DC - ATC classification system	Total number of consultations	Good
Barile (2013)[19]	Cohort	GP	≥ 65	DC	ADL limitations, Physically unhealthy days, Mentally unhealthy days	Good
Barile (2012)[20]	CS	GP	≥ 65	DC	Physical HRQoL, Mental HRQoL	Good
Barnett (2012)[21]	CS	PC	≥ 0	DC	Presence of mental health disorder	Good
Biehl (2016)[22]	Cohort	PC	≥ 65	ERA, CCI	Presence of critical illness	Good
Boeckxstaens (2015a)[23]	CS	PC	≥ 80	DC, CCI, CIRS	Disability (measured by ADL), Frailty (5 components)	Poor
Boeckxstaens (2015b)[24]	Cohort	PC	≥ 80	DC, mCCI, CIRS	Mortality at 3 years, Hospitalisation at 3 years, Functional Decline at 19 months (ADL, Physical, Mental decline)	Fair
Brilleman (2014)[25]	Cohort	PC	≥ 18	QOF count, CCI, EDC count, ACG, RUB	Primary healthcare cost The EDC performed best followed by the QOF and ACG.	Good
Brilleman & Salisbury (2013)[26]	Cohort	PC	≥ 18	QOF count, CCI, EDC count, ACG, RUB, Prescribed drugs count	Mortality: The CCI was the best performing measure followed by the number of prescribed drugs. Number of primary care consultations (3-years period): The number of prescribed drugs had the greatest predictive validity followed by the ACG based measures (ACG, EDC count, and RUB).	Good
Caballer-Tarazona (2019)[27]	CS	GP	≥ 0	CRG	Expenditure of integrated health care (hospital, primary health care (PHC) and pharmaceutical prescription)	Poor
Carey (2013)[28]	Cohort	PC	≥ 60	Standard QOF, Extended QOF, CCI (Khan)	Mortality (1-year period) The standard QOF score outperformed the CCI (Khan). The extended QOF score produced only a modest improvement in overall model performance.	Good
Chapman (2015)[29]	Cohort	GP	≥ 18	CCI, CCI-PSR	Mortality (5, 10, 15, 20, 25-years period) The CCI-PSR showed substantially better discrimination than the CCI.	Good
Charlson (2014)[30]	Cohort	GP	≥ 0	CCI	Health care cost, Utilisation of services	Good
Chen (2011)[31]	CS	GP	≥ 18	DC	General Health, Mental Distress, Physical Distress, Activity limitations	Good
Chen (2018)[32]	CS	GP	≥ 45	DC	Health service utilisation	Poor
Chu (2018)[33]	CS	PC	≥ 40	DC, CIRS	Health care utilization	Good
Clynes (2020)[34]	CS	GP	(born in 1931-1939)	DC	Physical functioning	Poor
Crane (2010)[35]	Cohort	PC	≥ 60	ERA	No. of hospital visits, ED visits, Hospital admissions, Days hospitalised (1-year period)	Good
Crooks (2016)[36]	Cohort	PC	20 to 100	Co-morbidity linked score, CCI, EI	Mortality (1-year period) The linked score had significantly improved discrimination and fit compared to the CCI and the Elixhauser Index	Good
Crooks (2015)[37]	Cohort	PC	≥ 20	CCI (Read), CCI (ICD-10), CCI (Read & ICD-10)	All-cause mortality (1-5 years) There was no large difference in the discrimination of the model for whichever codes that were used to derive the CCI.	Good

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Table 1. Summary of included studies (Continued)

Author (Year)	Study Design	Population source	Age	Multimorbidity Measurement	Outcomes measured	Risk of bias
DiNapoli (2017)[38]	CS	PC	≥ 50	Organ systems with chronic disease	Presence of depressive or anxiety disorder	Good
Formiga (2013)[39]	Cohort	PC	85	CCI	Mortality (3-years period)	Good
Formiga (2011a)[40]	Cohort	GP	90 to 99	CCI	Mortality (5-years period)	Good
Formiga (2011b)[41]	CS	PC	85	CCI	Successful aging	Good
Formiga (2016)[42]	Cohort	PC	85	CCI	Mortality (5-years period)	Good
Fraccaro (2016)[43]	Cohort	PC	≥ 18	CCI (Khan)	Mortality (1, 5, 10-years period), Mortality (3, 6, 12-months period)	Good
Galenkamp (2011)[44]	CS	GP	57 to 98	DC	SRH	Good
Garin (2014)[45]	CS	GP	≥ 50	DC	QOL, Disability	Good
Glynn (2011)[46]	CS	PC	> 50	DC	Primary care consultations, Hospital outpatient visits, Hospital admissions, Healthcare cost (all 1-year period)	Good
Gunn (2012)[47]	CS	PC	18 to 76	DC	Depressive symptoms (CES-D score)	Fair
Haas (2013)[48]	Cohort	PC	≥ 18	ACG, Minnesota Health Care Home Tiering, HCC, ERA, CCC, CCI, Hybrid Model	Hospitalisation, ED visits, Readmission within 30 days, Healthcare expenditure (all 1-year period) The ACG model outperformed the other 5 models in all outcomes.	Good
Hanmer (2010)[49]	CS	GP	22 to 106	Additive Model, Minimum Model, Multiplicative Model	Health Utility (SF-6D)	Fair
Hu (2017)[50]	CS	PC	≥65	Age-adjusted CCI	Frequency of family physician visits	Fair
Hwang (2015)[51]	Cohort	GP	≥ 0	ACE-27, ACE-27 count	Healthcare expenditure The model, using year 1 data to determine if an individual would be classified into the persistent high-user group for the following 3 years, indicates a very high level of accuracy in predicting membership in a high-user group.	Good
Isaacs (2014)[52]	CS	PC	18 to 101	DC	Prescription Costs	Poor
Jennings (2015)[53]	Cohort	PC	≥ 75	DC	Count of fall-related injuries in the 24 months after the date of screening	Fair
Jia (2018)[54]	Cohort	GP	≥ 65	DC	Quality-adjusted life years (QALY)	Poor
Jia & Lebetkin (2017)[55]	Cohort	GP	≥ 65	DC	Quality-adjusted life years (QALY)	Poor
Jindai (2016)[56]	CS	GP	≥ 65	DC	Functional limitations (ADL, IADL, Leisure and social activities, Lower-extremity mobility, General physical activities)	Good
Kim (2012)[57]	CS	GP	≥ 65	DC	Quality of life (EQ5D)	Poor
Kojima (2011)[58]	CS	PC	≥ 65	DC	Fall tendency	Poor
Kristensen (2014)[59]	CS	PC	> 0	RUB	Fee-for-services expenditures	Good

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Table 1. Summary of included studies (Continued)

Author (Year)	Study Design	Population source	Age	Multimorbidity Measurement	Outcomes measured	Risk of bias
Lapi (2015)[60]	CS	PC	≥ 15	HSMI	Total mean healthcare cost per year The HSMI explained 50.17% of the variation in costs	Good
Lawson (2013)[61]	CS	GP	≥ 20	DC	Preference-weighted HRQoL	Good
Lemke (2012)[62]	Cohort	GP	≥ 0	CCI, ACG	Inpatient hospitalisations ACG-based predictive model was superior to CCI model.	Good
Li (2016)[63]	CS	GP	16 to 68	DC	Health-related Quality of life	Poor
Loprinzi (2016)[64]	CS	GP	60-85	DC	Cognitive function	Good
Macinko (2019)[65]	CS	GP	≥ 18	DC (categorical 2 and 3 or more) (Self-reported)	Primary care experience (self-reported)	Good
Marengoni (2011)[66]	CS	GP	≥ 75 (baseline) ≥ 77 (follow-up)	DC	Disability	Good
McDaid (2013)[67]	CS	GP	≥ 50	DC	Disability, QoL, SRH	Good
Md Yusof (2010)[68]	Cohort	GP	64 to 85	CCI,	Mortality over 7 years	Fair
Milla-Perseguer (2019)[69]	CS	PC	≥ 18	CRG	Health-related quality of life (HRQL) - EQ-5D-3L	Good
Monterde (2020)[70]	Cohort	GP	≥ 18	Adjusted Morbidity Group (GMA), CCI, DC, CRG	Use of healthcare resources	Good
Muggah (2012)[71]	CS	GP	≥ 20	DC	Primary health care use	Poor
Mujica-Mota (2015)[72]	CS	PC	≥ 18	DC	Health-related quality of life (EQ5D)	Fair
Naessens (2011)[73]	CS	GP	18 to 64	DC	Healthcare cost	Poor
Ostergaard & Foldager (2011)[74]	CS	PC	≥ 18	DC	Major Depressive Episode (measured by DSQ)	Poor
Palladino (2019)[75]	CS	GP	≥ 50	DC	Primary care use, Reduced functional capacity, Self-perceived health, Hospital admissions, Quality of life	Good
Pati (2019)[76]	CS	PC	≥ 18	Severity burden score (21 conditions)	Health-related quality of life (SF-12)	Good
Payne (2013)[77]	Cohort	PC	≥ 20	DC	Unplanned hospital admission, Potentially preventable admission (all 1-year period)	Good
Payne (2014)[78]	Cohort	PC	≥ 20	DC	Unplanned hospital admissions (1-year period)	Good
Payne (2020)[79]	Cohort	PC	≥ 20	CCI, DC (37 read codes), Cambridge Multimorbidity Score	Mortality, Unplanned inpatient hospital admission, primary care consultations	Good
Peters (2018)[80]	CS	PC	18 to 101	DC, DBIS	Quality of life	Fair

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Table 1. Summary of included studies (Continued)

Author (Year)	Study Design	Population source	Age	Multimorbidity Measurement	Outcomes measured	Risk of bias
Quail (2011)[81]	Cohort	GP	≥ 20	DC, CCI (Quan), Elixhauser (Quan), No. of different dispended drugs, CDS	Mortality (1-year period): Elixhauser (Quan) performed best followed by CCI. One or more hospitalisations; Two or more hospitalisations: DC was the best performing measure	Good
Ranstad (2014)[82]	CS	GP	≥ 0	RUB	Registered active listing in primary care and all healthcare	Good
Reinke (2019)[83]	CS	PC	30-94	DC	Symptom burden (MSAS-SF), quality of life (Veterans RAND 12)	Good
Renne & Gobbens (2018)[84]	CS	PC	≥ 70	DC	Quality of life	Poor
Reyes (2014)[85]	Cohort	PC (Men)	≥ 65	CCI	Hip fractures	Good
Ryu (2015)[86]	CS	PC	≥ 18	DC	Deficits of perceived general health, Depressive symptoms	Good
Salisbury (2011)[87]	Cohort	PC	≥ 18	QOF count, EDC count	Primary Care consultation rates, Continuity of care (all 3-year period)	Good
Saver (2014)[88]	Cohort	GP	≥ 65	CCI (Romano) + Hypertension	Acute ACSH, Chronic ACSH	Good
Shadmi (2011)[89]	CS	GP	≥ 18	ADG, CCI	No. of primary care physician visits, specialist visits, hospitalisation ADG explained the largest percent of variance or in health care resource use	Good
Sibley (2014)[90]	CS	GP	≥ 65	DC	Self-reported falls in the last 12 months	Poor
Stanley & Sarfati (2017)[91]	Cohort	PC	≥ 18	M3 Index, CCI, Elixhauser (van Walraven)	Mortality, Overnight hospitalisation (all 1-year period) M3 Index outperformed both CCI and Elixhauser (van Walraven)	Good
St John (2014)[92]	Cohort	GP	≥ 65	DC (0-36 conditions)	Mortality in 5 years	Good
St John (2019)[93]	Cohort	GP	≥ 65	DC	Functional Impairment in 5 years	Good
Streit (2014)[94]	Cohort	PC	50 to 80	CCI, DC	Quality of cardiovascular preventive care, Quality of preventive care	Good
Sullivan (2012)[95]	CS	GP	≥ 18	DC	Preference-based HRQoL	Good
Takahashi (2011)[96]	Cohort	PC	> 60	ERA	Mortality, Nursing home placement (all 2-year period)	Good
Takahashi (2016)[97]	Cohort	PC	≥ 18	Minnesota Tiering (ACG), Enhanced model	Hospitalisation / ED visits The enhanced model is better	Good
Tyack (2016)[98]	Cohort	PC	≥ 18	DC	Health-related quality of life	Fair
Ubalde-Lopez (2016)[99]	CS	GP	F (mean): 35.9, M (mean): 37.9	MDMS	Sickness absence episodes taken in last 2 years	Good
van den Bussche (2011)[100]	CS	PC	≥ 65	DC	Frequency of contacts with physicians, Number of different ambulatory physicians contacted (all 1-year period)	Good

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Table 1. Summary of included studies (Continued)

Author (Year)	Study Design	Population source	Age	Multimorbidity Measurement	Outcomes measured	Risk of bias
van Oostrom (2014)[101]	CS	PC	≥ 55	DC	No. of contacts with general practice, Medications prescribed, Referrals	Good
Vos (2013)[102]	CS	PC	70 to 74	DC	Self-rated health (SF-36)	Poor
Wallace (2016a)[103]	Cohort	PC	≥ 70	Pra tool, Modified Pra tool	Emergency hospital admission (1-year period) Both models demonstrated poor model discrimination	Good
Wallace (2016b)[104]	Cohort	PC	≥ 70	DC, Barnett conditions DC, CCI, Prescribed drugs count, RxRisk-V	Emergency admission, Functional decline (all 2-year period) All measures demonstrated poor discrimination	Good
Wei (2018)[105]	CS	GP	≥ 51	MWI	Subjective physical functioning, Grip strength, Gait speed, Cognitive performance, ADL limitations, IADL limitations	Good
Wei (2019a)[106]	Cohort	GP	≥ 51	MWI	Physical Functioning - SF-36, Mortality	Good
Wei (2019b)[107]	Cohort	GP	≥ 51	MWI	Suicide mortality, health-related quality of life	Fair
Wei (2020a)[108]	Cohort	GP	≥ 51	MWI	Cognitive functioning	Good
Wei (2020b)[109]	Cohort	GP	≥ 51	MWI-ICD, DC, CCI, Elixhauser, health-related quality of life comorbidity index	Mortality, future physical functioning	Poor
Wei & Mukamal (2018)[110]	Cohort	GP	≥ 36	MWI, DC, CCI	Mortality (10-years period), Future physical functioning MWI performed best in predicting mortality as compared to DC and CCI	Good
Wikman (2011) [111]	CS	GP	≥ 50	DC	QoL, Affective well-being	Good
Wister (2015) [112]	CS	GP	≥ 65	MM additive scale, MM weighted by HUI3, MM weighted by ADL Scale, MM weighted by HUI3 betas	Life satisfaction, Perceived health status	Good

Note. ACE = Adult Comorbidity Evaluation; ACG = Adjusted Clinical Groups; ACSH = Ambulatory Care Sensitive Hospitalisation; ADG = Aggregated Diagnosis Groups; ADL = Activities of Daily Living; CCC = Chronic Condition Count; CCI = Charlson Comorbidity Index; CCI-PSR = Charlson Comorbidity Index-Psychosocial Risk; CDS = Chronic Disease Score; CIRS = Cumulative illness Rating Scale; CRG = Clinical Risk Groups; CS = Cross-Sectional; DBIS = Disease Burden Impact Scale; DC = Disease Count (Unweighted); ED = Emergency Department; EDC = Expanded Diagnosis Clusters; EI = Elixhauser Index; ERA = Elder Risk Assessment; GP = General Population; HCC = Hierarchical Condition Categories; HRQoL = Health-related Quality of Life; HSMI = Health Search Morbidity Index; HUI3 = Health Utility Index; IADL = Instrumental Activities of Daily Living; ICD-10 = International Classification of Diseases, Tenth Revision; M3 Index = Multimorbidity Measure Index; mCCI = modified Charlson Comorbidity Index; MDMS = Multidimensional Multimorbidity Score; MM = Multimorbidity; MWI = Multimorbidity-Weighted Index; PC = Primary Care; Pra tool = Probability of repeated admission risk prediction tool; QOF = Quality and Outcomes Framework; QoL = Quality of Life; RUB = Resource Utilisation Band; RxRisk-V = A Veterans Association adapted pharmacy-based case-mix instrument; SRH = Self-Rated Health

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Table 2 summarises the 33 instruments that were identified from all the studies. Table 3 provides a summary of multimorbidity instruments and their associations with the outcomes measured from all the included studies.

Simple counts of individual conditions

Disease count was based on the total number of all the conditions an individual had, usually from a pre-specified list of chronic conditions. It was used in 59 out of the 96 studies (61.5%). Disease count was reported to be associated with activity limitations, continuity of care, disability, healthcare cost, healthcare utilisation, medications, mental disorders, mortality, general health, physical function, quality of life, and self-rated health (Table 3).

Organ or system-based approaches

There were two instruments in this category. They were Cumulative Illness Rating Scale (CIRS)[23 24] and Organ Systems with chronic disease count (Organ-CDC).[38]

Weighted Indices

There were 17 unique weighted instruments found in the included studies. The original Charlson Comorbidity Index (CCI) with its different modifications was the most frequently used instrument and was used in 29 studies. The CCI was based on disease count, but the 17 conditions were weighted originally based on their impact on one-year mortality.[30] The final score was derived by the summation of all the weighted conditions. There were many

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Table 2. Description of instruments used for measurement of multimorbidity and the data sources and resources required

Category	Instrument	System/Condition based	Weightage; Scoring Method	Data Sources and resources required
A: Count of Individual Conditions				
A-1	DC	Condition (7-147)	Unweighted; Condition count	ATC list of conditions, Elixhauser list of conditions, EMR, GP records, Health service database, Hospital discharge abstract, Insurance claims or Questionnaires - telephone, face-to-face, mailed surveys. Participant involvement required
A-2	CCC[73]	Condition (6)	Unweighted; Based on AHRQ's clinical classification software & no. of conditions for each category	EMR
B: Organ or System-based Approaches				
B-3	Organ systems with CDC	Organ system (17)	Unweighted sum of organ systems	EMR
B-4	CIRS[113 114]	Body systems (13)	1-5 (based on severity of the condition); different weightage for diseases	EMR
C: Weighted Indices				
C-5	ACE[115]	Condition (27)	1-3 (based on severity of most severe condition); Highest score of single item	Insurance Claims' database
C-6	Cambridge MM Score[79]	Condition (20)	Weighted based on three different outcomes – primary care consultation, unplanned admission & mortality	EMR linked to mortality, hospital admission and socioeconomic deprivation
C-7	CCI[30]	Condition (19)	1-6 (based on impact on 1-year mortality (RR) – original); Sum of weighted conditions	Administrative database, EMR, Medical chart review, orInterviews or postal questionnaire where participant involvement is required
C-8	CLS[36]	Condition (98)	Based on impact for mortality (hazard ratio); Sum of beta coefficients of each category	Linked patients' records of all primary care events, hospital admissions and causes of death.
C-9	DBIS[116 117]	Conditions (25-28)	Weighted according to the degree in which each condition interferes with daily activities	Patient involvement in the questionnaire is required
C-10	EI (original & modified)[118]	Condition (21-31)	Based on impact on in-hospital mortality; Summing of beta coefficients	Insurance Claims' or Medical services database
C-11	ERA[35]	Condition (6-9)	Weighted (based on impact on future hospitalisation); Sum of weighted regression coefficients	EMR and administrative database
C-12	HCC[119]	Condition (70)	Based on Medicare capitation payments for health expenditure; More severe manifestations of a condition dominating (and zeroing out the effect of) less serious ones. Other diseases are summed additively.	EMR & HCC software licensing and fees
C-13	M3 Index[91]	Condition (55)	Weighted based on 1-year mortality; Summing of beta coefficients	Linked patients' records
C-14	MDMS[99]	Condition (7 chronic conditions, 2 health behaviours for 1st dimension & 5 symptoms for 2nd dimension)	Weighted but not based on any specific outcome; Sum of the value for the weighted absolute contributions of each of the dimensions.	Standardised medical evaluation (Interviewer-administered); Participant involvement is required
C-15	MM weighted by ADL Scale[120]	Condition (19)	Weighted based on OARS functional status scale measuring ADL; Sum of weighted conditions	Face-to-face or telephone interviews where participant involvement is required

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Table 2. Description of instruments used for measurement of multimorbidity and the data sources and resources required (Continued)

Category	Instrument	System/Condition based	Weightage; Scoring Method	Data Sources and resources required
C-16	MM weighted by HUI[121]	Condition (19)	Weighted based on correlation with health utility index; Sum of weighted conditions	Face-to-face or telephone interviews where participant involvement is required
C-17	MM weighted by HUI betas[121]	Condition (19)	Weighted based on correlation with health utility index and adjusted for age and sex; Summing of beta coefficients	Face-to-face or telephone interviews where participant involvement is required
C-18	MWI[122]	Condition (81)	Weighted based on impact on SF-36 physical functioning scale; Sum of weights	Interviewer-administered or mail questionnaire where participant involvement is required
C-19	QOF standard (weighted)[28]	Condition (14)	0-6, based on impact on 1-year mortality (RR); Sum of weighted conditions	EMR
C-20	QOF extended (weighted)[28]	Condition (9)	1-3, based on impact on 1-year mortality (RR); Sum of weighted conditions	EMR
C-21	Severity Burden Score[116]	Condition (21)	Sum of weights of diseases by the level of interference for each condition	Interviewer-administered structured questionnaire by nurses where participant involvement is required
D: Other Approaches (D1 = Case Mix, D2 = Pharmaceutical-based)				
D1-22	ACG[123 124]	Condition (93 mutually exclusive ACGs. Some are modified to 68 ACGs)	Incorporated into ACGs based on impact on resource use (proprietary); Variable	EMR & ACG software licensing and fees
D1-23	ADG[123 124]	Condition (32 groups)	Based on duration, severity, diagnostic certainty, aetiology, and need for specialty care; Variable	EMR & ACG software licensing and fees
D1-24	CRG[125]	NA; Diagnostic categories derived from organ systems or clinical category (37)	Pre-formulated based on the 3M clinical risk groups and consists of 9 core health ranks	EMR - inpatient and outpatient & 3M Clinical Risk Grouping software v1.6 and service fees
D1-25	Adjusted Morbidity Groups (GMA)[126]	NA; mutually exclusive categories (31)	Based on multimorbidity and levels of patient complexity	Registry data
D1-26	HM[48]	Condition (NS)	Only MN tier 4 + MN tier 3 with ERA > 10; Variable	EMR, HCC software licensing, fees & administrative data
D1-27	HSMI[60]	Condition (73 chronic and acute conditions)	Based upon yearly health care costs directly derived from primary care setting; Sum of regression coefficients (range from -0.06 to 1.04)	EMR
D1-28	Minnesota Tiering[123 124]	Condition (NS)	Grouping patients into 'complexity tiers' based on the number of major condition categories; Condition Count	EMR or administrative data & MN Tiering software licensing and fees
D1-29	Resource Utilisation Band[123 124]	Condition (6 mutually exclusive bands)	Based upon ACG algorithm on impact on resource use (proprietary); Variable	EMR & ACG software licensing and fees
D2-30	CDS[127]	Condition (17)	Weighted 1-5; Sum of weights based on pharmacological database	Prescription drug database
D2-31	Drug Count	NA; Variable. Some may be based on pharmacologic-therapeutic classification system	Weighted; Medication count	Self-reported questionnaire where participant involvement is required
D2-32	Modified Pra tool using RxRisk-V[103 128 129]	NA; Pra tool + RxRisk-V	Weighted due to RxRisk-V; 4 categories	GP medical record + linked pharmacy claims database

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Table 2. Description of instruments used for measurement of multimorbidity and the data sources and resources required (Continued)

Category	Instrument	System/Condition based	Weightage; Scoring Method	Data Sources and resources required
D2-33	RxRisk-V[129]	NA; WHO-ATC classification system	Weighted according to the diagnostic group of drugs to predict future health care costs; Sum of weights	GP medical record + linked pharmacy claims database

Note. ACE = Adult Comorbidity Evaluation; ACG = Adjusted Clinical Groups; ADG = Aggregated Diagnosis Groups; ADL = Activities of Daily Living; ATC = Anatomical Therapeutic Chemical; CCC = Chronic Condition Count; CCI = Charlson Comorbidity Index; mCCI = modified Charlson Comorbidity Index; CLS = Comorbidity Linked Score; CDC= Chronic Disease Count; CDS = Chronic Disease Score; CGI-S = Clinical Global Impression-Severity Scale; CIRS = Cumulative Illness Rating Scale; CRG = Clinical Risk Groups; DBIS = Disease Burden Impact Scale; DC = Disease Count; EDC = Expanded Diagnosis Clusters; EI = Elixhauser Index; EMR = Electronic Medical Records; ERA = Elder Risk Assessment; GP = General Practitioner; HCC = Hierarchical Condition Categories; HM = Hybrid Model (MN Tier + ERA); HUI = Health Utility Index; HSMI = Health Search Morbidity Index; mCCI = Modified Charlson Comorbidity Index; MDMS = Multidimensional Multimorbidity Score; MM = Multimorbidity; M3 Index = Multimorbidity Measure Index; MWI = Multimorbidity-Weighted Index; OARS = Older Americans Resources and Services; Pra tool = Probability of repeated admission risk prediction tool; QOF = Quality and Outcomes Framework; RR = Risk Ratio; RxRisk-V = A Veterans Association adapted pharmacy-based case-mix instrument; SF-36 = 36-item Short Form Survey

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Table 3. Summary of multimorbidity instruments and their associations with outcomes measured from all the included studies

Multimorbidity Measures	Association between Outcomes and Multimorbidity	
	Evidence of an association	No Evidence of an association
A = Count of Individual Conditions		
DC (Many different groupings ranging from 7[44 58 130] to 147[46] conditions and some are further categorised[48])	ADL limitations[19], Activity limitations[31], Affective well-being[111], Cognitive Function[64], Continuity of care (3-years)[87], Deficits of perceived general health[86], Depression[74], Depressive symptoms[86 47], Disability[23 45 66 67], Emergency hospital admission (2-years)[104], Fall-related injuries[53], Fall risk[58], Frequency of contacts with physicians (1-year)[100], Functional capacity[75], Functional decline (2-years)[104], Functional Impairment[93], Functional Limitations[56], Future physical functioning[110], General health[31], Healthcare costs[51 73], Health-related quality of life[49 57 63 72 98], Hospitalisation (3-years)[24], Hospital admissions (1-year)[77 78 81], Hospital outpatient visits (1-year)[97], Hospitalisation/Emergency Department Visits[97], Life satisfaction[112], Mental distress[31], Mortality (1-year)[81], (3-years)[26 24], (5 years)[92], (10-years)[110], No. of contacts with general practice (1-year)[101], No. of medications prescribed (1-year)[101], No. of mentally unhealthy days[19 20], No. of physically unhealthy days[19 20], No. of different ambulatory physicians contacted (1-year)[100], No. of primary care consultations (1-year)[26], (3-years)[26], No. of referrals (1-year)[101], Outpatient/Inpatient service use[32], Physical distress[31], Physical function[34], Prescription costs[52], Perceived health status[112], Presence of mental health disorder[46], Primary care consultations (1-year period)[87], (3-years)[87], Primary care experience – Self-reported[65], Primary healthcare cost[25], Primary health care use[71], Potentially preventable unplanned admission (1-year period)[77], Quality-adjusted life years[54 55], Quality of life[75 83 80 84], Self-rated health[102], Self-reported falls (12-month)[90], Symptom burden[83], Self-rated Health[44 67], Self-perceived health[75], Total number of consultation[18], total health care costs[18]	Functional decline[24], Quality of cardiovascular preventive care[94], Quality of preventive care [94]
CCC	Healthcare costs[48], Hospital admissions (1-year)[48], No. of emergency department visits (1-year)[48], Readmission within 30 days (1-year)[48]	
B = Organ or System-Based Approaches		
Organ systems with CDC	Presence of depressive or anxiety disorder[38]	
CIRS	Disability [23], Frailty[23], Health care utilisation[33], Hospitalisation (3-years)[24], Mortality[24]	Functional decline[24]
C = Weighted Indices		
ACE	Healthcare expenditure[51]	
Cambridge MM Score	Mortality[79], Primary care consultation[79], Unplanned admission[79]	

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Table 3. Summary of multimorbidity instruments and their associations with outcomes measured from all the included studies (Continued)

Multimorbidity Measures	Association between Outcomes and Multimorbidity	
	Evidence of an association	No evidence of an association
CCI	Ambulatory care sensitive hospitalisations (acute & chronic)[88], Disability [23], Emergency department visits (1-year)[48], Emergency hospital admission (2-years)[103], Frailty[23], Functional decline (2-years)[103], Future physical functioning[110], Healthcare expenditure[48], Hip fractures[85], Hospitalisation (1-year)[44 48 62 81 91], Hospitalisation (3-years)[24], Mortality (1-year)[28 43 81 91], Mortality (3-years)[24], (5-years)[29 43], (10-years)[29 43], (15, 20, 25-years)[29], No. of primary care consultations (3-years)[26], No. of primary care physician visits (1-year) [89], No. of specialist visits (1-year)[89], Potentially preventable unplanned admission (1-year)[78], Presence of critical illness[22], Primary healthcare cost[25], Mortality (1-year)[36 37], (3-years)[26 39], (5-years)[37 40 42], (7-years)[68], (10-years)[110], Readmission within 30 days (1-year)[48], Successful aging[41]	Functional decline[24], Primary care visits[50], Quality of cardiovascular preventive care[94], Quality of preventive care[94]
CLS	Mortality (1-year)[36]	
DBIS	Quality of life[80]	
EI (Original & Modified)	Hospitalisation (1-year)[81 91], Mortality (1-year)[36 81 91]	
ERA	Healthcare expenditure [48], Mortality (2-years)[130], No. of days hospitalised (1-year)[35], No. of emergency department visits (1-year)[35 48], No. of hospital admissions (1-year)[35 48], No. of hospital visits (1-year)[35 48], Nursing home placement (2-years)[130], Presence of critical illness[22], Readmission within 30 days (1-year)[48]	
HCC	Hospitalisation (1-year)[48], ED visits (1-year)[48], Readmission within 30 days (1-year)[48], Healthcare expenditure (1-year)[48]	
M3 Index	Hospitalisation (1-year)[91], Mortality (1-year)[91]	
MDMS	Sickness absence episodes taken in 2 years (male)[99]	Sickness absence episodes taken in 2 years (female)[99]
MM weighted by ADL Scale	Life satisfaction[112], Perceived health status[112]	
MM weighted by HUI	Life satisfaction[112], Perceived health status[112]	
MM weighted by HUI betas	Life satisfaction[112], Perceived health status[112]	
MWI	ADL limitations[105], IADL limitations[105], Mortality (10-years)[110], Cognitive performance[105], Future physical functioning[110 109], Grip strength[105], Health-related quality of life[107], Mortality[109], Subjective physical functioning[105], Suicide mortality[107]	Gait speed[110]
QOF (Standard)	Mortality (1-year) [28]	
QOF (Extended)	Mortality (1-year) [28]	
Severity Burden Score	Mental component score (SF-12)[76]	

Table 3. Summary of multimorbidity instruments and their associations with outcomes measured from all the included studies (Continued)

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Multimorbidity Measures	Association between Outcomes and Multimorbidity	
	Evidence of an association	No evidence of an association
D = Other Approaches (D-1 = Case-Mix, D2 = Pharmaceutical-based)		
ACG	Hospitalisation (1-year)[62], Mortality (3-years)[26], No. of primary care consultations (3-years)[26], Primary healthcare cost[25], Readmission within 30 days (1-year)[48]	
ADG	Hospitalisation (1-year)[89], No. of primary care physician visits (1-year)[89], No. of specialist visits (1-year)[89]	
CRG	Healthcare expenditure [27], HRQoL using EQ-5D-3L[69]	
Adjusted Morbidity Groups (GMA)	Use of healthcare resources[70]	
HM	Emergency department visits (1-year)[48], Healthcare expenditure[48], Hospitalisation (1-year)[48], Readmission within 30 days (1-year)[48]	
HSMI	Healthcare cost (primary care)[60]	
Minnesota Tiering	Emergency department visits (1-year)[48 97], Healthcare expenditure[48], Hospitalisation (1-year)[48 97], Readmission within 30 days (1-year)[48]	
Resource Utilisation Band	Fee-for-service expenditures[59], Primary healthcare cost[25], Mortality (3-years)[26], Number of primary care consultations (3-years)[26], Registered active listing in primary care[82], Registered active listing in all healthcare[82]	
CDS	Hospitalisation (1-year)[81], Mortality (1-year) [81]	
Drug Count	Emergency hospital admission (2-years)[104], Functional decline (2-years)[104], Hospitalisation (1-year)[81], Mortality (1-year)[81], (3-years)[26], No. of primary care consultations (3-years)[26]	
Pra tool Modified using RxRisk-V	Emergency hospital admission (1-year)[103]	
RxRisk-V	Emergency hospital admission (2-years)[104], Functional decline (2-years)[104]	

Note. ACE-27 = Adult Comorbidity Evaluation; ACG = Adjusted Clinical Groups; ADG = Aggregated Diagnosis Groups; ADL = Activities of Daily Living; CCC = Chronic Condition Count; CCI = Charlson Comorbidity Index; CDC= Chronic Disease Count; CDS = Chronic Disease Score; CIRS = Cumulative Illness Rating Scale; CLS = Comorbidity Linked Score; CRG = Clinical Risk Groups; DBIS = Disease Burden Impact Scale; DC = Disease Count; Elixhauser Index = EI; ERA = Elder Risk Assessment; HCC = Hierarchical Condition Categories; HM = Hybrid Model (MN Tier + ERA); HRQoL = Health-Related Quality of Life; HSMI = Health Search Morbidity Index; HUI = Health Utility Index; MM = Multimorbidity; MDMS = Multidimensional Multimorbidity Score; M3 Index = Multimorbidity Measure (M3) Index; MWI = Multimorbidity-Weighted Index; Pra tool = Probability of repeated admission risk prediction tool; QOF = Quality and Outcomes Framework; RxRisk-V = A Veterans Association adapted pharmacy-based case-mix instrument; SF-12 = Short Form-12

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variations and modifications of the score including the addition of psychosocial factors. The CCI instrument was found to be associated with multiple outcomes other than one-year mortality.

Most of the other weighted index instruments were novel, like the Multimorbidity-Weighted Index (MWI), in which the investigators built multivariable prognostic models from a set of potential predictor conditions and weighted the conditions based on an outcome of clinical interest. The most common outcomes chosen were mortality and physical function. Other outcomes included health expenditure,[48] health utility index,[112] and severity of the most severe condition.[51] The Multidimensional Multimorbidity Score (MDMS)[99] was unique as it was weighted based on health behaviours and patient symptoms and not based on any specific outcome.

Other Approaches to measuring multimorbidity

Other approaches included case-mix and pharmaceutical-based instruments. For case-mix approach, the Adjusted Clinical Group (ACG) and Resource Utilisation Band (RUB) were the most commonly used instruments.[131] Most of the case-mix instruments required proprietary software licenses from the United States of America and obtained data from electronic medical records or administrative data. The Clinical Risk Groups instrument was similar but took into account the severity of individual conditions.[125]

The second group of instruments in this category was related to pharmaceutical data. The most frequent type was the unweighted drug count. The other three (Chronic disease score, A Veterans Association adapted pharmacy-based case-mix instrument like RxRisk-V, and modified Probability of repeated admission risk prediction tool using RxRisk-V) were all

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weighted indices. Except for the drug count that was based on a self-report questionnaire, the rest required a prescription drug database to obtain the data.

Outcomes

We classified the 150 outcomes into 17 categories as reported in the core outcomes set of multimorbidity research (COSmm).[10] The most commonly reported outcomes were health care use (n=45), mortality (n=18), health-related quality of life (n=18), and physical function (n=13). The different studies unanimously showed that higher levels of multimorbidity was associated with higher health care use and mortality, lower health-related quality of life, and poorer physical function. Seven outcomes in the COSmm were not found in all the 96 studies. These were treatment burden, self-management behaviour, self-efficacy, adherence, communications, shared decision making and prioritisation. There were 19 outcomes that were not described in the COSmm. These included cognitive function, risk of suicide, frailty and falls. The outcomes not found to have any association with the instruments for measuring the level of multimorbidity were preventive care[94], sickness absence episodes (female)[99], and gait speed[110].

DISCUSSION

Summary of Findings

Thirty-three unique instruments for measuring the level of multimorbidity were identified and categorised according to the classification by Sarfati.[8] The most commonly used instrument was 'Disease Count'. It was also the only instrument that was associated with the three essential outcomes from the core outcomes set of multimorbidity research (COSmm)[10] i.e., quality of life, mental health, and mortality.

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Comparison with previous research

Although the most common instrument identified in this systematic review was similar to that of Huntley et al.,[12] several instruments including Duke Severity of Illness Checklist (DUSOI) and Functional Comorbidity Index (FCI) identified in their article were not found in this systematic review. The possible reasons for not finding these instruments in this review could be due to the lack of interest in the instrument by the research community in recent years (last publication using DUSOI was in 2004[132]), or the exclusion of studies specifying an index condition.

Advantages and disadvantages of selected instruments

Disease count

The advantage of using 'disease count' is its simplicity and the ease of data ascertainment with minimal resources required. However, using 'disease count' does not consider the severity of each condition where the complexity of multimorbidity may not be properly addressed.[133] The other disadvantage noted was the lack of transparency in the operational definition of multimorbidity, especially regarding the list of conditions considered for multimorbidity, and the cut-points used. Despite its simplicity, the level of multimorbidity measured using 'disease count' was the only instrument that was found to be associated with the three essential core outcomes (quality of life, mental health, and mortality).

Weighted Indices

The common weighted indices identified in this systematic review were Charlson Comorbidity Index (CCI), Elders Risk Assessment (ERA), Elixhauser Index (EI) and

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Multimorbidity-weighted Index (MWI). These weighted indices were often utilised in prognostic models to build complex multivariable regression models in which the weights were calculated from hazard ratios, odds ratios, or regression coefficients.[134]

The advantage of these weighted indices is that the weights allow the adaptation of an index to a specific outcome. An investigator could recalibrate the correct weight by creating a prognostic model to produce a contextualised instrument for a different setting. Prognostic models can provide clinically relevant risk stratification and help to allocate resources.[135] The disadvantage of such indices is that calculated weights are greatly influenced by the population, outcomes used, and the instrument's original conception and purpose, hampering the ability to compare across studies.

Case-Mix

The Adjusted Clinical Group (ACG) system has a good track record in the United States and several other countries, especially for measuring the outcomes of healthcare utilisation. However, the instrument is proprietary, and the exact algorithm of the instrument is not open to the public and may not be suitable in certain settings. The Clinical Risk Group (CRG) system has a good track record in Spain. It measures the severity of each condition and its algorithm is fully transparent. The common disadvantage of both systems is the financial costs involved in obtaining the license.

Pharmaceutical-based instruments

Medication-based indices include versions of the Chronic Disease Score,[127] which later became known as the RxRisk,[136] and its adaptation for use in the veteran population, the RxRisk-V.[137] Like the disease count, its main advantage is the ease of use with

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minimal resources required. However, many studies were not transparent regarding which type of drugs were included.

Data sources

Data sources used by these instruments included medical record information, patient self-report, clinical judgment, and large administrative databases. Each data source has its inherent advantages and disadvantages. For patient self-report, patients with cognitive impairment may under-report symptoms and may be seen less frequently by their physicians, resulting in an under-recognition or under-treatment of conditions.[138] It has also been shown that health administrative data based on billing system underestimated the prevalence of many chronic conditions.[139]

The available data in a particular setting may strongly influence the ultimate instrument chosen for multimorbidity research. As there is currently no consensus on the gold standard for sources of data, it is difficult to assess which data source was superior from this review.

Outcomes

There were 17 multimorbidity outcomes identified by a Delphi process involving a panel of international experts in multimorbidity intervention studies.[10] However, only ten out of the seventeen outcomes were reported in the 96 studies identified in this systematic review. The most common outcome that was investigated was health care use. The seven missing outcomes belong to 'patient-reported impact and behaviours' and 'consultation-related' outcome groups, most likely indicating that there is a dearth of multimorbidity studies looking at these two groups of outcomes measures.

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Clinical Implications

Ideally, a single instrument measuring the level of multimorbidity should be able to predict a variety of relevant outcomes. However, Byles et al.[140] reported that a single instrument could not be used to predict different outcomes, in different patient groups and settings, unless different weights were assigned to these factors in calculating a score. Such multiple-scoring instruments may be the way forward for validation of prognostic models for different outcomes and different populations with established multimorbidity instruments. For example, depending on the outcome, study population and setting, the choice of conditions included in the multiple-scoring instrument should include those with a high prevalence in that study population and the weights should be determined by their significant impact (i.e. outcome) on the affected population.

For pragmatic reasons, the final selection of the conditions to be included in such a multiple-scoring instrument may still have to take into account the availability of relevant and reliable data. A certain degree of reductionism will also have to be accepted because a single instrument will not be able to encompass all the nuances of the different interactions of chronic conditions on an individual living in his/her unique milieu. We recommend that researchers perform validation studies using the instruments listed in this systematic review to adjust the weights according to the specific outcome of interest for the study population relevant to their setting.

Strengths and limitations of the study

The main strengths of this systematic review were the involvement of a health science librarian in our search strategy, a published protocol, adherence to the protocol without major changes during the systematic review process,[141] and the critical appraisal of all the primary studies with a risk of bias assessment tool.

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The systematic review had several limitations. We excluded grey literature and included only studies that were published in the English language. We also did not contact authors directly for a suggestion of studies, nor identified a list of instruments from the preliminary search and then performed an additional search using the same databases.[142] Additionally, this systematic review did not review the validity and reliability of all the instruments as it was beyond the scope of the intended work. We have, however, included the references of the original articles or validation studies in Table 2 for each of the instrument where available. Finally, this review specifically aimed to look at the association of the level of multimorbidity as the main independent variable and excluded the level of multimorbidity as a mediating, confounding or effect-modifying variable. This strict criterion excluded 17 studies (Figure 1) as a result. Excluding these 17 studies did not alter the findings as the instruments used in all the 17 studies were Disease Count (n=9), Cumulative Illness Rating Scale (CIRS, n=3), Charlson Comorbdity Index (CCI, n=3), Elixhauser Index (EI, n=1) and Aggregated Diagnosis Groups (ADG, n=1) where no new instruments were identified.

CONCLUSIONS

In this systematic review, we found 33 instruments for measuring the level of multimorbidity in community-dwelling individuals that predict or explore the association of multimorbidity with at least one specified outcome. Disease count and weighted indices like the Charlson Comorbidity Index (CCI), the ERA, and Elixhauser Index (EI) were commonly used for measuring the level of multimorbidity. Other approaches to measuring the level of multimorbidity included case-mix or pharmaceutical-based instruments.

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We found continuing interest in measuring the level of multimorbidity with disease count and drug count. There has also been a rise in the development of novel weighted indices using prognostic models or validation of existing well-established instruments like the CCI over the last few years. There is currently an absence of a gold standard for where to obtain chronic disease information. The most suitable instrument will depend on the specified outcome of interest, the study population, and the type of data and resources available.

Finally, there is still much work to improve on the body of knowledge of multimorbidity when most investigators in the last decade measured multimorbidity without including some of the important outcome measures of multimorbidity. We also suggest that a clear description of the instruments is required in the publication of multimorbidity studies to counter the frequent lack of information currently seen so as to contribute to robust multimorbidity research in future.

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All authors made substantial contribution to the design of the work. BLR, MF and MS mentored and provided guidance to the review. ESL, together with a health science librarian, created the search strategy and conducted the search. ESL, EH, TSH, HLK and FYW screened articles, extracted and analysed data. ESL drafted the manuscript with valuable inputs from BLR, MF and MS and all authors reviewed drafts and approved the final version.

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Patient and public involvement: This research was done without patient involvement. Patients were not invited to comment on the study design and were not consulted to develop patient relevant outcomes or interpret the results. Patients were not invited to contribute to the writing or editing of this document for readability or accuracy.

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Figure 1 – PRISMA Flow Diagram

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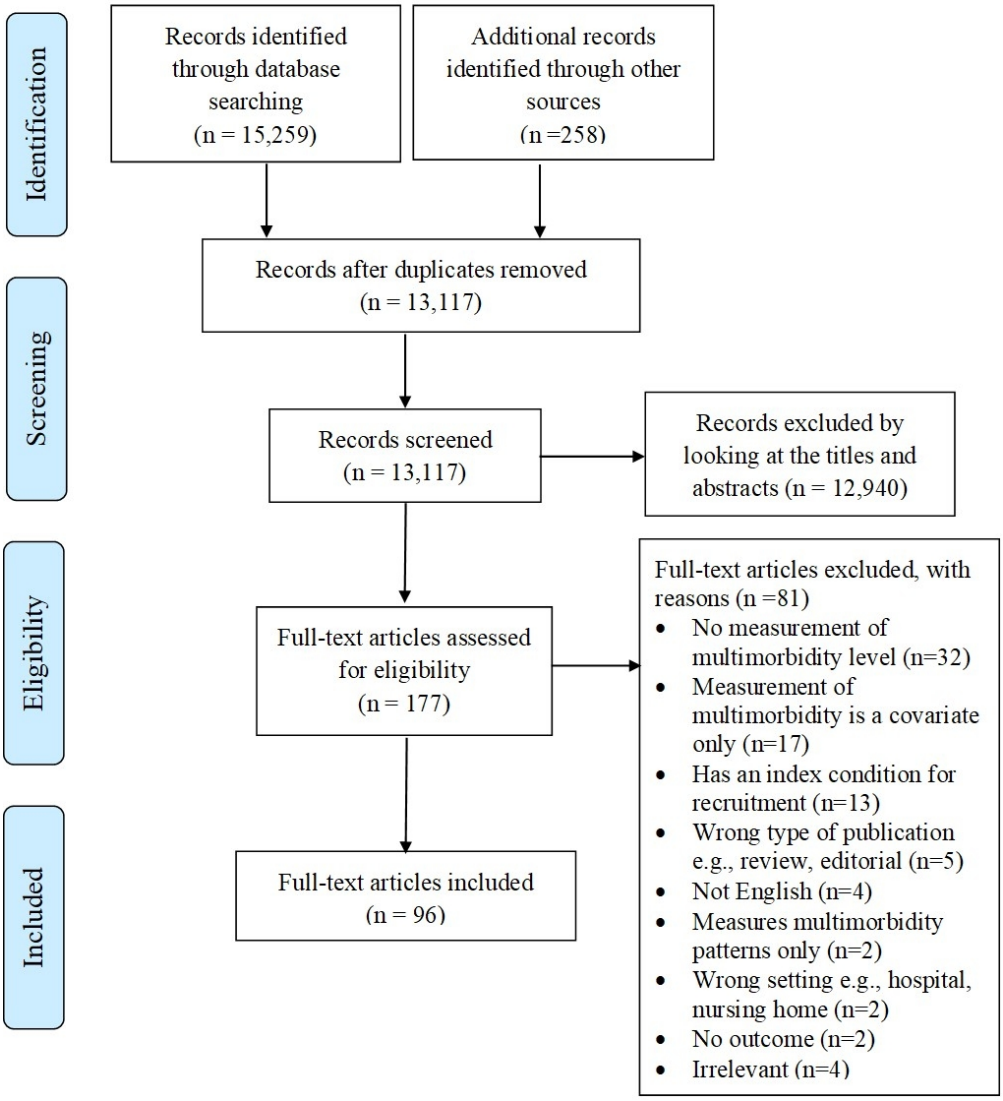


Figure 1 - PRISMA Flow Diagram

172x197mm (150 x 150 DPI)

Appendix 1. Search Strategy

Table 1. Medline Search 1946 to October 23, 2020

#	Searches	Results
1	exp comorbidity/ or multiple chronic conditions/	111191
2	(multimorbid* or multi-morbid* or comorbid* or co-morbid*).ab,ti,kw.	180089
3	((multiple or coexist* or co-exist* or concurrent* or simultaneous*) adj2 (disease* or illness* or diagnos* or condition* or morbid* or disorder*)).ab,ti,kw.	52139
4	1 or 2 or 3	292316
5	primary health care/ or "continuity of patient care"/ or exp general practice/ or ambulatory care/ or physicians, family/ or physicians, primary care/ or community health services/ or general practitioners/	249484
6	((ambulatory or community or general or family or primary) adj2 (care or health* or practi* or physician* or medicine or center* or centre* or facilit* or clinic*)).ab,ti,kw.	391013
7	5 or 6	508037
8	epidemiologic measurements/ or risk assessment/ or "Outcome Assessment (Health Care)"/ or patient reported outcome measures/ or health status indicators/ or "severity of illness index"/ or sickness impact profile/ or diagnosis-related groups/ or case mix/	594279
9	((health status or risk or outcome* or sickness impact) adj2 (appraisal* or assessment* or evaluation* or measur* or index* or indic* or profile*)).ab,ti,kw.	398319
10	((severity or burden) adj2 (illness* or diseas* or disorder* or condition* or diagnos*)).ab,ti,kw.	85595
11	(charlson comorbidity index or charlson index or charlson score or elixhauser* or cumulative illness rating scale* or CIRS or adjusted clinical group* or ACG* or disease count or Duke severity* or DUSOI or casemix or case-mix).ab,ti,kw.	21547
12	8 or 9 or 10 or 11	939031
13	4 and 7 and 12	4862
14	limit 13 to (english language and yr="2010 - Current")	2827

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Table 2. Embase Classic+Embase 1947 to October 23, 2020

#	Searches	Results
1	comorbidity/ or multiple chronic conditions/	265602
2	(multimorbid* or multi-morbid* or comorbid* or co-morbid*).ab,ti,kw.	329397
3	((multiple or coexist* or co-exist* or concurrent* or simultaneous*) adj2 (disease* or illness* or diagnos* or condition* or morbid* or disorder*)).ab,ti,kw.	81506
4	1 or 2 or 3	493137
5	primary health care/ or patient care/ or primary medical care/ or general practice/ or general practitioner/ or ambulatory care/ or family medicine/ or community care/ or community health services/	664384
6	((ambulatory or community or general or family or primary) adj2 (care or health* or practi* or physician* or medicine or center* or centre* or facilit* or clinic*)).ab,ti,kw.	539525
7	5 or 6	943743
8	risk assessment/ or outcome assessment/ or patient reported outcome/ or health status indicator/ or disease activity score/ or global disease burden/ or organ dysfunction score/ or "severity of illness index"/ or sickness impact profile/ or general health status assessment/ or disease severity/ or diagnosis related group/ or charlson comorbidity index/ or comorbidity assessment/ or elixhauser comorbidity index/ or case mix/	1590115
9	((health status or risk or outcome* or sickness impact) adj2 (appraisal* or assessment* or evaluation* or measur* or index* or indic* or profile*)).ab,ti,kw.	547083
10	((severity or burden) adj2 (illness* or diseas* or disorder* or condition* or diagnos*)).ab,ti,kw.	138554
11	(charlson comorbidity index or charlson index or charlson score or elixhauser* or cumulative illness rating scale* or CIRS or adjusted clinical group* or ACG* or disease count or Duke severity* or DUSOI or casemix or case-mix).ab,ti,kw.	42616
12	8 or 9 or 10 or 11	2022565
13	4 and 7 and 12	12245
14	limit 13 to (english language and yr="2010 - Current")	8896

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Table 3. CINAHL – since inception to October 23, 2020

#	Searches	Results
1	(MH “comorbidity”)	38970
2	(multimorbid* or multi-morbid* or comorbid* or co-morbid*)	66442
3	(multiple or coexist* or co-exist* or concurrent* or simultaneous*) N2 (disease* or illness* or diagnos* or condition* or morbid* or disorder*)	12822
4	1 or 2 or 3	77266
5	((MH “primary health care”) or (MH “family practice”) or (MH “ambulatory care") or (MH “ambulatory care facilities") or (MM “community health services”) or (MM “community health centers”) or (MH “physicians, family”) or (MH “continuity of patient care"))	100224
6	((ambulatory or community or general or family or primary) N2 (care or health* or practi* or physician* or medicine or center* or centre* or facilit* or clinic*))	332346
7	5 or 6	340014
8	((MH “risk assessment”) or (MH "Outcome Assessment") or (MH “patient-reported outcomes) or (MH “health status indicators”) or (MH "severity of illness indices") or (MH “sickness impact profile”) or (MH “diagnosis-related groups”) or (MH “case mix”))	102350
9	((health status or risk or outcome* or sickness impact) N2 (appraisal* or assessment* or evaluation* or measur* or index* or indic* or profile*))	340802
10	(severity or burden) N2 (illness* or diseas* or disorder* or condition* or diagnos*)	68742
11	(charlson comorbidity index or charlson index or charlson score or elixhauser* or cumulative illness rating scale* or CIRS or adjusted clinical group* or ACG* or disease count or Duke severity* or DUSOI or casemix or case-mix)	6729
12	8 or 9 or 10 or 11	396387
13	4 and 7 and 12	4729
14	limit 13 to (english language and yr="2010 - Current")	3536

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Appendix 2. Coding Description for the Modified Newcastle-Ottawa Scale for Cohort Studies

Category	Description
Selection	
1) Representativeness of the sample	This item assesses the representativeness of sample in the community, not from some general population. a) Truly representative***** (e.g., everyone from the database) b) Somewhat representative***** (with at least 2 criteria but selection method was convincing due to random sampling) c) Selected group (e.g., only certain socio-economic groups or areas) d) No description of sampling strategy
2) Ascertainment of multimorbidity	This item assesses the method by which multimorbidity was confirmed. a) Secure record***** (e.g., GP questionnaire) b) Structured interview***** (e.g., interviewer-administered questionnaire) c) Written self-report (e.g., mailed survey, if items are unable to be confirmed by objective measure) d) No description / Other
3) Demonstration that outcome of interest was not present at start of study	A statement of no history of disease earns a star. In the case of mortality studies, outcome of interest is still the presence of a disease, rather than death. a) Yes***** b) No
Comparability	
1) Study controls for age and sex	Covariates must be adjusted for in the analysis. Statements of no differences between groups or that differences were not statistically significant are not sufficient for establishing comparability. Note: If the relative risk for the exposure of interest is adjusted for the covariates listed, then the groups will be considered to be comparable on each variable used. a) Yes***** b) No
2) Study controls for other factors	a) Yes***** b) No
Outcome	
1) Statistical test	Statistical test(s) must be clearly described and appropriate before assessing the other items under the Outcome category. a) Clearly described and appropriate b) Not described, incomplete or inappropriate
2) Assessment of outcome	This item assesses the method by which the outcome of interest was confirmed. For some outcomes, reference to the medical record is sufficient to satisfy the requirement for confirmation. This may not be adequate for other outcomes where reference to specific tests or measures would be required. a) Independent or blind assessment***** (e.g., interviewer-administered questionnaire) b) Record linkage***** (e.g., identified through ICD codes on database records) c) Self-report (i.e., no reference to original medical records) d) No description / Other

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3)	Was follow-up long enough for outcomes to occur?	An acceptable length of time was at least 1 year of follow-up. a) Yes***** b) No
4)	Adequacy of follow-up of cohorts	This item assesses the follow-up of the sample to ensure that losses are not related to the outcome. a) Complete follow-up - all subject accounted for***** b) Subjects lost to follow-up unlikely to introduce bias***** (Number lost ≤20% or description of those lost suggested no different from those followed.) c) Follow-up rate less than 80% and no description of those lost d) No statement

Thresholds for converting the modified Newcastle-Ottawa scales to Good, Fair, and Poor Quality:

- Good Quality - 2-3 stars in Selection category (Representativeness of Sample item must be fulfilled) AND 1-2 stars in Comparability category AND 2-3 stars in Outcome category.
- Fair Quality - 1 star in Selection category AND 1-2 stars in Comparability category AND 2-3 stars in Outcome category.
- Poor Quality - 0 star in Selection category OR 0 star in Comparability category OR 0-2 stars in Outcome category.

Wells GA, Shea B, O'Connell D, Peterson J, Welch V, Losos M, Tugwell P. The Newcastle-Ottawa Scale (NOS) for assessing the quality of nonrandomized studies in meta-analysis. Available from: http://www.ohri.ca/programs/clinical_epidemiology/oxford.asp

Modifications from original NOS include:

- 'Statistical test' was added to the Outcome category
- 'Representativeness of sample' item under Selection category and 'Statistical test' item under Outcome category must both be fulfilled for study to be considered Good Quality.
- 'Selection of the non-exposed cohort' item was removed from Selection category as most studies did not describe a non-exposed cohort, and this review sought to compare the different levels of multimorbidity within the group.
- 'Representativeness of the exposed cohort' item under Selection category was renamed 'Representativeness of the sample' since the 'non-exposed cohort' was removed above.
- 'Ascertainment of exposure' under the Selection category was renamed as 'Ascertainment of multimorbidity' to specify that multimorbidity is the exposure in this review.
- 'Study controls for age and sex' and 'Study controls for other factors' were revised to be items under Comparability category
- The thresholds for converting the scales to quality were amended accordingly due to the above modifications.

Consensus:

- The study team decided that a period of one-year was a reasonable period of follow-up under 'Was follow-up long enough for outcomes to occur?' item under Outcome category.

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Appendix 3. Coding Description for the Modified Newcastle-Ottawa Scale for Cross-sectional Studies

Category	Description
Selection	
1) Representativeness of the sample	This item assesses the representativeness of sample in the specified population, not from some general population. a) Truly representative***** (e.g., everyone from the database, random sampling) b) Somewhat representative***** (with at least 2 criteria but selection method was convincing due to random sampling) c) Selected group (e.g., only certain socio-economic groups or areas) d) No description of sampling strategy
2) Ascertainment of multimorbidity	This item assesses the method by which multimorbidity was confirmed. a) Secure record***** (e.g., Clinical records, GP questionnaire) b) Structured interview***** (e.g., interviewer-administered questionnaire) c) Written self-report (e.g., mailed survey, if items are unable to be confirmed by objective measure) d) No description / Other
3) Sample Size	If there is no description, a reported sample size of 800 and above is satisfactory. a) Justified and satisfactory***** b) Not justified
4) Non-respondents	Acceptable response rates for surveys through various methods† • In-person: 57% • Mail: 50% • Average: 33% • Email: 30% • Internet: 29% • Telephone: 18% • In-app: 13% a) Comparability between respondents and non-respondents characteristics is established, and the response rate is satisfactory.***** b) The response rate is unsatisfactory, or the comparability between respondents and non-respondents is unsatisfactory. c) No description of the response rate or the characteristics of the responders and the non-responders
Comparability	
1) Study controls for age and sex	Confounders must be adjusted for in the analysis. Statements of no differences between groups or that differences were not statistically significant are not sufficient for establishing comparability. Note: If the relative risk for the exposure of interest is adjusted for the confounders listed, then the groups will be considered to be comparable on each variable used in the adjustment. a) Yes***** b) No
2) Study controls for other factors	a) Yes***** b) No

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Outcome	
1) Statistical test	Statistical test(s) must be clearly described and appropriate before assessing the other items under the Outcome category. a) Clearly described and appropriate b) Not described, incomplete or inappropriate
2) Assessment of outcome	This item assesses the method by which the outcome of interest was confirmed. For some outcomes, reference to the medical record is sufficient to satisfy the requirement for confirmation. This may not be adequate for other outcomes where reference to specific tests or measures would be required. a) Independent or blind assessment***** b) Record linkage***** (e.g., identified through ICD codes on database records) c) Self-report (i.e., no reference to original medical records) d) No description / Other

Thresholds for converting the modified Newcastle-Ottawa scales to Good, Fair, and Poor Quality:

- Good Quality – 3-4 stars in Selection category (Representativeness of Sample item must be fulfilled) AND 1-2 stars in Comparability category AND 1 star in Outcome category.
- Fair Quality - 2 stars in Selection category AND 1-2 stars in Comparability category AND 1 star in Outcome category.
- Poor Quality – 0-1 star in Selection category OR 0 star in Comparability category OR 0 star in Outcome category.

†Lindermann N. What's the average survey response rate? [2018 benchmark]. SurveyAnyplace. Available from: https://surveyanyplace.com/average-survey-response-rate/.

Alshabanat A, Zafari Z, Albanyan O, Dairi M, FitzGerald JM. Asthma and COPD Overlap Syndrome (ACOS): a systematic review and meta-analysis. PLoS ONE. 2015;10(9):e0136065. Doi:10.1371/journal.pone.0136065.

Modifications from original NOS include:

- 'Ascertainment of exposure' item under Selection category was renamed as 'Ascertainment of multimorbidity' to specify that multimorbidity is the exposure in this review.
- The ratings for the 'Ascertainment of exposure' item under Selection category were revised to 'secure record, structured interview, written self-report, no description/other' to align with the Modified NOS for cohort studies in this review.
- 'Study controls for age and sex' and 'Study controls for other factors' items were revised to be items under Comparability category.
- 'Was follow-up long enough for outcomes to occur' item under Outcome category was renamed as 'Statistical test'. This item must be fulfilled for the study to be considered as Good Quality.
- The thresholds for converting the scales to quality were amended accordingly due to the above modifications.

Consensus:

- The study team decided on a list of acceptable response rates for various survey methods for the 'Non-respondents' item under Selection category.
- The study team decided that a sample size of 800 was reasonable under 'Sample size' item under Selection category.

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Appendix 4. Summary of risk of bias (RoB) appraisal of included cohort articles using the modified Newcastle-Ottawa Quality Assessment Scale (44 articles)

	Barile et al. (2013)[19]	Biehl et al. (2016)[22]	Boeckxstaens et al. (2015b)[24]	Brilleman et al. (2014)[25]	Brilleman and Salisbury(2013)[26]	Carey et al. (2013)[28]	Chapman et al. (2015)[29]	Charlson et al. (2014)[30]	Crane et al. (2010)[35]	Crooks et al. (2016)[36]	Crooks et al. (2015)[37]	Formiga et al. (2013)[39]	Formiga et al. (2011a)[40]	Formiga et al. (2016)[42]	Fraccaro et al. (2016)[43]	Haas et al. (2013)[48]	Hwang et al. (2015)[51]	Jennings et al. (2015)[53]	Jia et al. (2018)[54]	Jia and Lubetkin (2017)[55]	Lemke et al. (2012)[62]	Md Yusof et al. (2010)[68]	Monterde et al. (2020)[70]	Pavne et al. (2013)[77]	Pavne et al. (2014)[78]	Pavne et al. (2020)[79]	Ouail et al. (2011)[81]	Reves et al. (2014)[85]	Salisbury et al. (2011)[87]	Saver et al. (2014)[88]	Stanley and Sarfati (2017)[91]	St John et al. (2014)[92]	St John et al. (2019)[93]	Streit et al. (2014)[94]	Takahashi et al. (2011)[96]	Takahashi et al. (2016)[97]	Tvack et al. (2016)[98]	Wallace et al. (2016a)[103]	Wallace et al. (2016b)[104]	Wei et al. (2019a)[106]	Wei et al. (2019b)[107]	Wei et al. (2020a)[108]	Wei et al. (2020b)[109]	Wei and Mukamal (2018)[110]
Selection Criteria																																												
Representativeness of the sample	*	*	-	*	*	*	*	-	*	*	*	*	*	*	*	*	*	*	-	*	*	*	-	*	*	*	*	*	*	*	*	*	*	*	*	*	-	*	*	*	-	*	*	*
Ascertainment of Multimorbidity	-	*	*	*	*	*	*	-	*	*	*	*	*	*	*	*	*	*	*	-	*	*	-	*	*	*	*	*	*	*	*	*	*	*	*	*	-	*	*	*	*	*	*	-
Demonstration that outcome of interest was not present at the start of the study	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*
Comparability Criteria																																												
Study controls for age and sex	*	-	*	*	*	*	*	*	-	*	*	-	*	-	*	*	*	-	*	*	-	*	*	*	*	*	*	*	-	*	*	*	*	*	*	*	-	*	-	-	*	-	-	-
Study controls for others	*	*	-	*	*	*	*	*	*	-	*	*	*	*	*	*	*	*	-	-	*	*	*	*	*	-	*	*	*	*	-	*	*	*	*	*	*	*	-	*	-	*	-	*
Outcome Criteria																																												
Statistical test†	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	X	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Assessment of outcome	-	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*
Follow-up was long enough for outcomes to occur	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*
Adequacy of follow-up of cohorts	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	-	*	*	*	*	*	-	*	*	*	-	*	*	*	-	-	*	*
Overall RoB Judgement[a]	G	G	F	G	G	G	G	G	G	G	G	G	G	G	G	G	G	G	F	P	P	G	F	G	G	G	G	G	G	G	G	G	G	G	G	G	F	G	G	G	F	G	P	G

Note. G = Good; F = Fair; P = Poor.

†Statistical test must be clearly described and appropriate before assessing the other items under the Outcome category. [a] Good rating is given when there are 2 to 3 stars in Selection category (Representativeness of Sample must be fulfilled) AND 1 to 2 stars in Comparability category AND 2 to 3 stars in Outcome category; Fair rating is given when there are 1 star in Selection category AND 1 to 2 stars in Comparability category AND 2 to 3 stars in Outcome category; Poor rating is given when there are 0 star in Selection category OR 0 star in Comparability category OR 0 to 2 stars in Outcome category.

"*" Study satisfies the criteria; "-" Study did not satisfy the criteria; "✓" Statistical test is clearly described and appropriate; "X" Statistical test is not described, incomplete or inappropriate.

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Appendix 5. Summary of risk of bias (RoB) appraisal of included cross-sectional articles using the modified Newcastle-Ottawa Quality Assessment Scale (52 articles)

	Agborsangaya et al. (2013)[17]	Bahler et al. (2015)[18]	Barile et al. (2012)[20]	Barnett et al. (2012)[21]	Boeckxstaens et al. (2015a)[23]	Caballer-Tarazona et al. (2019)[27]	Chen et al. (2011)[31]	Chen et al. (2018)[32]	Chu (2018)[33]	Clynes et al. (2020)[34]	DiNapoli et al. (2017)[38]	Formiga et al. (2011b)[41]	Galenkamp et al. (2011)[44]	Garin et al. (2014)[45]	Glynn et al. (2011)[46]	Gunn et al. (2012)[47]	Hanmer et al. (2010)[49]	Hu et al. (2017)[50]	Isaacs et al. (2014)[52]	Jindai et al. (2016)[56]	Kim et al. (2012)[57]	Koiima et al. (2011)[58]	Kristensen et al. (2014)[59]	Lapi et al. (2015)[60]	Lawson et al. (2013)[61]	Li et al. (2016)[63]	Lonrinzi (2016)[64]	Macinko et al. (2019)[65]	Marengoni et al. (2011)[66]	McDaid et al. (2013)[67]	Milla-Perseguer (2019)[69]	Musgah et al. (2012)[71]	Mujica-Mota et al. (2015)[72]	Naessens et al. (2011)[73]	Østergaard and Foldager (2011)[74]	Palladino et al. (2019)[75]	Pati et al. (2019)[76]	Peters et al. (2018)[80]	Ranstad et al. (2014)[82]	Reinke et al. (2019)[83]	Renne and Gobbens (2018)[84]	Ryu et al. (2015)[86]	Shadmi et al. (2011)[89]	Siblev et al. (2014)[90]	Sullivan et al. (2012)[95]	Ubalde-Lopez et al. (2016)[99]	van den Bussche et al. (2014)[100]	van Oostrom et al. (2014)[101]	Vos et al. (2013)[102]	Wei et al. (2018)[105]	Wikman et al. (2011)[111]	Wister et al. (2015)[112]
Selection																																																					
Representativeness of the sample	*	*	*	*	-	*	*	-	*	-	*	*	*	*	*	*	*	-	*	-	*	-	*	*	*	*	*	*	*	*	*	*	-	-	*	*	*	*	-	*	*	*	*	*	-	*	*	*	*	*	*	*
Ascertainment of Multimorbidity	*	*	-	*	*	*	*	-	*	*	*	*	*	*	*	*	-	-	*	*	*	*	*	*	*	*	-	*	*	*	*	*	*	-	*	*	*	-	*	*	-	-	*	*	*	*	*	*	*	*
Sample size	*	*	*	*	-	*	*	*	*	-	*	-	*	*	*	*	*	*	-	*	*	*	-	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	-	*	*	*	*	*	*	*	-	*	*	*
Non-respondents	-	*	*	*	-	*	*	*	*	-	*	*	-	-	*	-	-	*	*	*	*	-	*	*	*	-	*	-	-	-	-	*	-	*	-	*	*	-	*	*	-	-	*	*	*	*	*	*	*	-
Comparability																																																				
Study controls for age and sex	*	*	*	*	*	-	*	*	*	-	*	-	*	*	*	*	*	*	-	-	-	*	*	*	*	-	*	*	*	*	*	-	*	-	-	*	*	*	*	*	*	-	*	*	*	*	*	-	*	*	*	*
Study controls for others	*	*	*	*	*	-	*	*	*	*	*	*	-	*	*	*	*	*	-	*	-	*	*	*	*	-	*	*	*	*	*	-	*	-	-	*	*	*	*	*	-	*	*	*	*	*	*	-	*	*	*	-
Outcome																																																				
Statistical test†	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	X	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Assessment of outcome	*	*	✓	*	*	*	*	-	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	-	*	*	*	*	*	*	*	*	*
Overall RoB Judgement[a]	G	G	G	G	P	P	G	P	G	P	G	G	G	G	G	F	F	F	P	G	P	P	G	G	G	P	G	G	G	G	G	P	F	P	P	G	G	F	G	G	P	G	G	P	G	G	G	G	P	G	G	G

Note. G = Good; F = Fair; P = Poor.

[†]Statistical test must be clearly described and appropriate before assessing the other items under the Outcome category. [a] Good rating is given when there are 3 to 4 stars in Selection category (Representativeness of Sample must be fulfilled) AND 1 to 2 stars in Comparability category AND 1 star in Outcome category; Fair rating is given when there are 2 stars in Selection category AND 1 to 2 stars in Comparability category AND 1 star in Outcome category; Poor rating is given when there are 0 to 1 star in Selection category OR 0 star in Comparability category OR 0 star in Outcome category.

"*" Study satisfies the criteria; "-" Study did not satisfy the criteria; "✓" Statistical test is clearly described and appropriate; "X" Statistical test is not described, incomplete or inappropriate

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	3
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	5
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	6
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	6
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	6-7
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	6-7
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	Appendix 1
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	6-7
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	7
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	7
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	7, Appendix 2,3
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	NA
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	NA

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	NA
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	NA
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	Figure 1
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	8-13
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	Appendix 4,5
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	14-22
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	NA
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	NA
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	NA
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	22-26
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	27
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	27-28
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	29

From: Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(7): e1000097. doi:10.1371/journal.pmed1000097

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A systematic review on the instruments used for measuring the association of the level of multimorbidity and clinically important outcomes

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Secondary Subject Heading:	Evidence based practice
Keywords:	PRIMARY CARE, Protocols & guidelines < HEALTH SERVICES ADMINISTRATION & MANAGEMENT, GENERAL MEDICINE (see Internal Medicine)

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TITLE PAGE

A systematic review on the instruments used for measuring the association of the level of multimorbidity and clinically important outcomes

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ABSTRACT

Objectives: There are multiple instruments for measuring multimorbidity. The main objective of this systematic review was to provide a list of instruments that are suitable for use in studies aiming to measure the association of a specific outcome with different levels of multimorbidity as the main independent variable in community-dwelling individuals. The secondary objective was to provide details of the requirements, strengths and limitations of these instruments, and the chosen outcomes.

Methods: We conducted the review according to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses guidelines (PROSPERO registration number: CRD42018105297). We searched MEDLINE, Embase and CINAHL electronic databases published in English and manually searched the Journal of Comorbidity between 1st January 2010 and 23rd October 2020 inclusive. Studies also had to select adult patients from primary care or general population, and had at least one specified outcome variable. Two authors screened the titles, abstracts and full texts independently. Disagreements were resolved with a third author. The modified Newcastle Ottawa Scale was used for quality assessment.

Results: Ninety-six studies were identified with 69 of them rated to have a low risk of bias. In total, 33 unique instruments were described. Disease count and weighted indices like Charlson Comorbidity Index were commonly used. Other approaches included pharmaceutical-based instruments. Disease count was the common instrument used for measuring all three essential core outcomes of multimorbidity research: mortality, mental health and quality of life. There was a rise in the development of novel weighted indices by using prognostic models. The data obtained for measuring multimorbidity were from sources including medical records, patient self-reports, and large administrative databases.

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Conclusions: We listed the details of 33 instruments for measuring the level of multimorbidity as a resource for investigators interested in the measurement of multimorbidity for its association with or prediction of a specific outcome.

ARTICLE SUMMARY

Strengths and Limitations of this study

Strengths

- This review builds upon Huntley et al.'s 2012 review article and provides an updated, comprehensive list of instruments that measure levels of multimorbidity in community-dwelling individuals.
- There was a thorough literature search of three major electronic databases with the involvement of a health science librarian.
- The review is reported based on Preferred Reporting Items for Systematic Reviews and Meta-Analysis guidelines.

Limitation

- This review excludes non-English language articles and grey literature.

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MAIN TEXT

BACKGROUND

Multimorbidity is defined as the co-occurrence of two or more chronic medical conditions in an individual.[1] It is a growing public health challenge and accounts for most of the expenditures in the healthcare system.[2] The complex interactions of several co-existing diseases have profound implications on individuals[3 4] and their healthcare providers.[5 6]

There are multiple instruments for measuring multimorbidity and many of them do not usually specify the severity of individual conditions.[7] No gold standard multimorbidity measurement instrument exists and there is also no agreed categorisation of the available instruments. Sarfati[8 9] classified the various measurement instruments into four broad approaches. They are: (1) by simple counts of individual conditions (i.e., disease count), (2) by organ or system-based approaches, (3) by weighting conditions and combining them into indices, and (4) by other miscellaneous approaches. Most of these measurements are used to measure the prevalence or patterns of multimorbidity. However, they can also be used to predict an outcome or to evaluate an intervention for a desired outcome. A set of core outcomes of multimorbidity (COSmm) was proposed after consulting a panel of international experts in multimorbidity intervention studies using a Delphi process.[10] Core outcome sets (COS) represent the minimum that should be measured and reported in all clinical trials of multimorbidity.[11]

Huntley et al.[12] published a systematic review in 2012 describing the instruments used to measure the morbidity burden in primary care and the general population. They found 17 different instruments from 194 articles. The most widely used instruments and those with the most significant evidence of validity were the Charlson Comorbidity Index, disease counts,

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and the Adjusted Clinical Groups (ACG) system.[12] However, this review was conducted in 2009 and the multimorbidity research has increased exponentially since then.

The present review was to build upon the review article by Huntley et al.[12] in order to provide a current and comprehensive list of instruments that measure levels of multimorbidity for community-dwelling individuals. We used the term 'level of multimorbidity' to refer to the combined effects of multiple conditions on an individual. The main objective of this review was to list instruments for measuring the levels of multimorbidity. We specifically look for studies that measure the association of a clinically important outcome with different levels of multimorbidity as the main independent variable in community-dwelling individuals. Our second objective was to provide details of the requirements, strengths and limitations of these instruments, and the chosen outcomes in the studies so that clinicians and researchers can select or develop instruments that match their needs for predicting a specific outcome.

METHODS

A protocol for this systematic review (CRD42018105297) was published online on PROSPERO.[13] We searched MEDLINE, EMBASE, CINAHL and also manually searched the Journal of Comorbidity for potential studies. The medical subject headings and keywords used for the search are shown in Appendix 1.

We selected studies that included: (1) adult patients from primary care or the general population as the majority of patients with multimorbidity are managed by primary care physicians[14]; (2) at least one specified outcome variable; and (3) published full-text articles from 1st January 2010 to 23 October 2020. Studies were excluded if they (1) selected patients from the hospital or nursing home only or patient data were drawn solely from the hospital or the nursing home;

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or (2) selected patients with an index condition; or (3) used level of multimorbidity as a covariate and not the main independent variable; or (4) were not written in English. We did not include a specific definition of multimorbidity instrument because, given a lack of consensus in the literature on the use of this term, we wanted to include a diverse range of studies on the above topic.

One reviewer (ESL) conducted a preliminary screen of titles and abstracts to exclude articles that were irrelevant. Abstracts of the remaining articles were screened independently by two reviewers (ESL & EH) according to the eligibility criteria. Disagreements were resolved through discussion until a consensus was reached. The full-text articles were then retrieved for the agreed list and independently assessed according to the eligibility criteria by the same reviewers. Disagreements were resolved through discussion with a third reviewer (TSH) until a consensus was reached. After agreement on the list of articles, the reference lists of included articles were hand-searched for additional eligible articles. We reported multimorbidity instruments that were described in all selected articles.

The risk of bias of the study design of selected articles was next appraised independently by three reviewers (ESL, EH and TSH) using the modified Newcastle-Ottawa Scale (NOS)[15 16]. Each article was assessed under the three broad categories: a) Selection; b) Comparability; and c) Outcome (Appendices 2 & 3).

We contacted the authors, as needed, for additional information or clarification up to three times spaced one week apart. We contacted 25 authors and 19 of them replied. Any disagreements on the risk of bias were resolved among the three reviewers through regular meetings. HLK and FYW were responsible for tracking and updating the final outcome of the risk of bias assessment.

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Patient and public involvement: This research was done without patient involvement. Patients were not invited to comment on the study design and were not consulted to develop patient relevant outcomes or interpret the results. Patients were not invited to contribute to the writing or editing of this document for readability or accuracy.

RESULTS

The number of included studies was 96, of which 69 were assessed to have low risk of bias. A summary of the PRISMA flow chart is depicted in Figure 1. Forty-eight studies selected participants from the general population and the other 48 studies selected participants from primary care. Most of the studies in this systematic review were from Europe and North America with very few Asian studies. There were 44 cohort studies of which 36 were assessed to have low risk of bias, and 52 cross-sectional studies of which 33 were assessed to have low risk of bias. We found 33 unique instruments from the 96 studies. The instruments were categorised according to Sarfati[8 9] into: 1) simple counts of individual conditions; 2) organ or system-based approaches; 3) conditions that have been weighted and combined into indices; and 4) other approaches. A total of 150 outcomes were reported from all the studies. No studies were excluded for an outcome that was not deemed to be clinically important. Appendices 4 & 5 summarise the risk of bias assessment of each study. Table 1 provides a summary of the study design, population source, age group, multimorbidity measurements, outcome measures, and risk of bias assessment of all the studies.

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Table 1. Summary of included studies

Author (Year)	Study Design	Population source	Age	Multimorbidity Measurement	Outcomes measured	Risk of bias
Agborsangaya (2013)[17]	CS	GP	≥ 18	DC	HRQoL	Good
Bahler (2015)[18]	CS	GP	≥ 65	DC - ATC classification system	Total number of consultations	Good
Barile (2013)[19]	Cohort	GP	≥ 65	DC	ADL limitations, Physically unhealthy days, Mentally unhealthy days	Good
Barile (2012)[20]	CS	GP	≥ 65	DC	Physical HRQoL, Mental HRQoL	Good
Barnett (2012)[21]	CS	PC	≥ 0	DC	Presence of mental health disorder	Good
Biehl (2016)[22]	Cohort	PC	≥ 65	ERA, CCI	Presence of critical illness	Good
Boeckxstaens (2015a)[23]	CS	PC	≥ 80	DC, CCI, CIRS	Disability (measured by ADL), Frailty (5 components)	Poor
Boeckxstaens (2015b)[24]	Cohort	PC	≥ 80	DC, mCCI, CIRS	Mortality at 3 years, Hospitalisation at 3 years, Functional Decline at 19 months (ADL, Physical, Mental decline)	Fair
Brilleman (2014)[25]	Cohort	PC	≥ 18	QOF count, CCI, EDC count, ACG, RUB	Primary healthcare cost The EDC performed best followed by the QOF and ACG.	Good
Brilleman & Salisbury (2013)[26]	Cohort	PC	≥ 18	QOF count, CCI, EDC count, ACG, RUB, Prescribed drugs count	Mortality: The CCI was the best performing measure followed by the number of prescribed drugs. Number of primary care consultations (3-years period): The number of prescribed drugs had the greatest predictive validity followed by the ACG based measures (ACG, EDC count, and RUB).	Good
Caballer-Tarazona (2019)[27]	CS	GP	≥ 0	CRG	Expenditure of integrated health care (hospital, primary health care (PHC) and pharmaceutical prescription)	Poor
Carey (2013)[28]	Cohort	PC	≥ 60	Standard QOF, Extended QOF, CCI (Khan)	Mortality (1-year period) The standard QOF score outperformed the CCI (Khan). The extended QOF score produced only a modest improvement in overall model performance.	Good
Chapman (2015)[29]	Cohort	GP	≥ 18	CCI, CCI-PSR	Mortality (5, 10, 15, 20, 25-years period) The CCI-PSR showed substantially better discrimination than the CCI.	Good
Charlson (2014)[30]	Cohort	GP	≥ 0	CCI	Health care cost, Utilisation of services	Good
Chen (2011)[31]	CS	GP	≥ 18	DC	General Health, Mental Distress, Physical Distress, Activity limitations	Good
Chen (2018)[32]	CS	GP	≥ 45	DC	Health service utilisation	Poor
Chu (2018)[33]	CS	PC	≥ 40	DC, CIRS	Health care utilization	Good
Clynes (2020)[34]	CS	GP	(born in 1931-1939)	DC	Physical functioning	Poor
Crane (2010)[35]	Cohort	PC	≥ 60	ERA	No. of hospital visits, ED visits, Hospital admissions, Days hospitalised (1-year period)	Good
Crooks (2016)[36]	Cohort	PC	20 to 100	Co-morbidity linked score, CCI, EI	Mortality (1-year period) The linked score had significantly improved discrimination and fit compared to the CCI and the Elixhauser Index	Good
Crooks (2015)[37]	Cohort	PC	≥ 20	CCI (Read), CCI (ICD-10), CCI (Read & ICD-10)	All-cause mortality (1-5 years) There was no large difference in the discrimination of the model for whichever codes that were used to derive the CCI.	Good

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Table 1. Summary of included studies (Continued)

Author (Year)	Study Design	Population source	Age	Multimorbidity Measurement	Outcomes measured	Risk of bias
DiNapoli (2017)[38]	CS	PC	≥ 50	Organ systems with chronic disease	Presence of depressive or anxiety disorder	Good
Formiga (2013)[39]	Cohort	PC	85	CCI	Mortality (3-years period)	Good
Formiga (2011a)[40]	Cohort	GP	90 to 99	CCI	Mortality (5-years period)	Good
Formiga (2011b)[41]	CS	PC	85	CCI	Successful aging	Good
Formiga (2016)[42]	Cohort	PC	85	CCI	Mortality (5-years period)	Good
Fraccaro (2016)[43]	Cohort	PC	≥ 18	CCI (Khan)	Mortality (1, 5, 10-years period), Mortality (3, 6, 12-months period)	Good
Galenkamp (2011)[44]	CS	GP	57 to 98	DC	SRH	Good
Garin (2014)[45]	CS	GP	≥ 50	DC	QOL, Disability	Good
Glynn (2011)[46]	CS	PC	> 50	DC	Primary care consultations, Hospital outpatient visits, Hospital admissions, Healthcare cost (all 1-year period)	Good
Gunn (2012)[47]	CS	PC	18 to 76	DC	Depressive symptoms (CES-D score)	Fair
Haas (2013)[48]	Cohort	PC	≥ 18	ACG, Minnesota Health Care Home Tiering, HCC, ERA, CCC, CCI, Hybrid Model	Hospitalisation, ED visits, Readmission within 30 days, Healthcare expenditure (all 1-year period) The ACG model outperformed the other 5 models in all outcomes.	Good
Hanmer (2010)[49]	CS	GP	22 to 106	Additive Model, Minimum Model, Multiplicative Model	Health Utility (SF-6D)	Fair
Hu (2017)[50]	CS	PC	≥65	Age-adjusted CCI	Frequency of family physician visits	Fair
Hwang (2015)[51]	Cohort	GP	≥ 0	ACE-27, ACE-27 count	Healthcare expenditure The model, using year 1 data to determine if an individual would be classified into the persistent high-user group for the following 3 years, indicates a very high level of accuracy in predicting membership in a high-user group.	Good
Isaacs (2014)[52]	CS	PC	18 to 101	DC	Prescription Costs	Poor
Jennings (2015)[53]	Cohort	PC	≥ 75	DC	Count of fall-related injuries in the 24 months after the date of screening	Fair
Jia (2018)[54]	Cohort	GP	≥ 65	DC	Quality-adjusted life years (QALY)	Poor
Jia & Lebetkin (2017)[55]	Cohort	GP	≥ 65	DC	Quality-adjusted life years (QALY)	Poor
Jindai (2016)[56]	CS	GP	≥ 65	DC	Functional limitations (ADL, IADL, Leisure and social activities, Lower-extremity mobility, General physical activities)	Good
Kim (2012)[57]	CS	GP	≥ 65	DC	Quality of life (EQ5D)	Poor
Kojima (2011)[58]	CS	PC	≥ 65	DC	Fall tendency	Poor
Kristensen (2014)[59]	CS	PC	> 0	RUB	Fee-for-services expenditures	Good

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Table 1. Summary of included studies (Continued)

Author (Year)	Study Design	Population source	Age	Multimorbidity Measurement	Outcomes measured	Risk of bias
Lapi (2015)[60]	CS	PC	≥ 15	HSMI	Total mean healthcare cost per year The HSMI explained 50.17% of the variation in costs	Good
Lawson (2013)[61]	CS	GP	≥ 20	DC	Preference-weighted HRQoL	Good
Lemke (2012)[62]	Cohort	GP	≥ 0	CCI, ACG	Inpatient hospitalisations ACG-based predictive model was superior to CCI model.	Good
Li (2016)[63]	CS	GP	16 to 68	DC	Health-related Quality of life	Poor
Loprinzi (2016)[64]	CS	GP	60-85	DC	Cognitive function	Good
Macinko (2019)[65]	CS	GP	≥ 18	DC (categorical 2 and 3 or more) (Self-reported)	Primary care experience (self-reported)	Good
Marengoni (2011)[66]	CS	GP	≥ 75 (baseline) ≥ 77 (follow-up)	DC	Disability	Good
McDaid (2013)[67]	CS	GP	≥ 50	DC	Disability, QoL, SRH	Good
Md Yusof (2010)[68]	Cohort	GP	64 to 85	CCI,	Mortality over 7 years	Fair
Milla-Perseguer (2019)[69]	CS	PC	≥ 18	CRG	Health-related quality of life (HRQL) - EQ-5D-3L	Good
Monterde (2020)[70]	Cohort	GP	≥ 18	Adjusted Morbidity Group (GMA), CCI, DC, CRG	Use of healthcare resources	Good
Muggah (2012)[71]	CS	GP	≥ 20	DC	Primary health care use	Poor
Mujica-Mota (2015)[72]	CS	PC	≥ 18	DC	Health-related quality of life (EQ5D)	Fair
Naessens (2011)[73]	CS	GP	18 to 64	DC	Healthcare cost	Poor
Ostergaard & Foldager (2011)[74]	CS	PC	≥ 18	DC	Major Depressive Episode (measured by DSQ)	Poor
Palladino (2019)[75]	CS	GP	≥ 50	DC	Primary care use, Reduced functional capacity, Self-perceived health, Hospital admissions, Quality of life	Good
Pati (2019)[76]	CS	PC	≥ 18	Severity burden score (21 conditions)	Health-related quality of life (SF-12)	Good
Payne (2013)[77]	Cohort	PC	≥ 20	DC	Unplanned hospital admission, Potentially preventable admission (all 1-year period)	Good
Payne (2014)[78]	Cohort	PC	≥ 20	DC	Unplanned hospital admissions (1-year period)	Good
Payne (2020)[79]	Cohort	PC	≥ 20	CCI, DC (37 read codes), Cambridge Multimorbidity Score	Mortality, Unplanned inpatient hospital admission, primary care consultations	Good
Peters (2018)[80]	CS	PC	18 to 101	DC, DBIS	Quality of life	Fair

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Table 1. Summary of included studies (Continued)

Author (Year)	Study Design	Population source	Age	Multimorbidity Measurement	Outcomes measured	Risk of bias
Quail (2011)[81]	Cohort	GP	≥ 20	DC, CCI (Quan), Elixhauser (Quan), No. of different dispended drugs, CDS	Mortality (1-year period): Elixhauser (Quan) performed best followed by CCI. One or more hospitalisations; Two or more hospitalisations: DC was the best performing measure	Good
Ranstad (2014)[82]	CS	GP	≥ 0	RUB	Registered active listing in primary care and all healthcare	Good
Reinke (2019)[83]	CS	PC	30-94	DC	Symptom burden (MSAS-SF), quality of life (Veterans RAND 12)	Good
Renne & Gobbens (2018)[84]	CS	PC	≥ 70	DC	Quality of life	Poor
Reyes (2014)[85]	Cohort	PC (Men)	≥ 65	CCI	Hip fractures	Good
Ryu (2015)[86]	CS	PC	≥ 18	DC	Deficits of perceived general health, Depressive symptoms	Good
Salisbury (2011)[87]	Cohort	PC	≥ 18	QOF count, EDC count	Primary Care consultation rates, Continuity of care (all 3-year period)	Good
Saver (2014)[88]	Cohort	GP	≥ 65	CCI (Romano) + Hypertension	Acute ACSH, Chronic ACSH	Good
Shadmi (2011)[89]	CS	GP	≥ 18	ADG, CCI	No. of primary care physician visits, specialist visits, hospitalisation ADG explained the largest percent of variance or in health care resource use	Good
Sibley (2014)[90]	CS	GP	≥ 65	DC	Self-reported falls in the last 12 months	Poor
Stanley & Sarfati (2017)[91]	Cohort	PC	≥ 18	M3 Index, CCI, Elixhauser (van Walraven)	Mortality, Overnight hospitalisation (all 1-year period) M3 Index outperformed both CCI and Elixhauser (van Walraven)	Good
St John (2014)[92]	Cohort	GP	≥ 65	DC (0-36 conditions)	Mortality in 5 years	Good
St John (2019)[93]	Cohort	GP	≥ 65	DC	Functional Impairment in 5 years	Good
Streit (2014)[94]	Cohort	PC	50 to 80	CCI, DC	Quality of cardiovascular preventive care, Quality of preventive care	Good
Sullivan (2012)[95]	CS	GP	≥ 18	DC	Preference-based HRQoL	Good
Takahashi (2011)[96]	Cohort	PC	> 60	ERA	Mortality, Nursing home placement (all 2-year period)	Good
Takahashi (2016)[97]	Cohort	PC	≥ 18	Minnesota Tiering (ACG), Enhanced model	Hospitalisation / ED visits The enhanced model is better	Good
Tyack (2016)[98]	Cohort	PC	≥ 18	DC	Health-related quality of life	Fair
Ubalde-Lopez (2016)[99]	CS	GP	F (mean): 35.9, M (mean): 37.9	MDMS	Sickness absence episodes taken in last 2 years	Good
van den Bussche (2011)[100]	CS	PC	≥ 65	DC	Frequency of contacts with physicians, Number of different ambulatory physicians contacted (all 1-year period)	Good

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Table 1. Summary of included studies (Continued)

Author (Year)	Study Design	Population source	Age	Multimorbidity Measurement	Outcomes measured	Risk of bias
van Oostrom (2014)[101]	CS	PC	≥ 55	DC	No. of contacts with general practice, Medications prescribed, Referrals	Good
Vos (2013)[102]	CS	PC	70 to 74	DC	Self-rated health (SF-36)	Poor
Wallace (2016a)[103]	Cohort	PC	≥ 70	Pra tool, Modified Pra tool	Emergency hospital admission (1-year period) Both models demonstrated poor model discrimination	Good
Wallace (2016b)[104]	Cohort	PC	≥ 70	DC, Barnett conditions DC, CCI, Prescribed drugs count, RxRisk-V	Emergency admission, Functional decline (all 2-year period) All measures demonstrated poor discrimination	Good
Wei (2018)[105]	CS	GP	≥ 51	MWI	Subjective physical functioning, Grip strength, Gait speed, Cognitive performance, ADL limitations, IADL limitations	Good
Wei (2019a)[106]	Cohort	GP	≥ 51	MWI	Physical Functioning - SF-36, Mortality	Good
Wei (2019b)[107]	Cohort	GP	≥ 51	MWI	Suicide mortality, health-related quality of life	Fair
Wei (2020a)[108]	Cohort	GP	≥ 51	MWI	Cognitive functioning	Good
Wei (2020b)[109]	Cohort	GP	≥ 51	MWI-ICD, DC, CCI, Elixhauser, health-related quality of life comorbidity index	Mortality, future physical functioning	Poor
Wei & Mukamal (2018)[110]	Cohort	GP	≥ 36	MWI, DC, CCI	Mortality (10-years period), Future physical functioning MWI performed best in predicting mortality as compared to DC and CCI	Good
Wikman (2011) [111]	CS	GP	≥ 50	DC	QoL, Affective well-being	Good
Wister (2015) [112]	CS	GP	≥ 65	MM additive scale, MM weighted by HUI3, MM weighted by ADL Scale, MM weighted by HUI3 betas	Life satisfaction, Perceived health status	Good

Note. ACE = Adult Comorbidity Evaluation; ACG = Adjusted Clinical Groups; ACSH = Ambulatory Care Sensitive Hospitalisation; ADG = Aggregated Diagnosis Groups; ADL = Activities of Daily Living; CCC = Chronic Condition Count; CCI = Charlson Comorbidity Index; CCI-PSR = Charlson Comorbidity Index-Psychosocial Risk; CDS = Chronic Disease Score; CIRS = Cumulative illness Rating Scale; CRG = Clinical Risk Groups; CS = Cross-Sectional; DBIS = Disease Burden Impact Scale; DC = Disease Count (Unweighted); ED = Emergency Department; EDC = Expanded Diagnosis Clusters; EI = Elixhauser Index; ERA = Elder Risk Assessment; GP = General Population; HCC = Hierarchical Condition Categories; HRQoL = Health-related Quality of Life; HSMI = Health Search Morbidity Index; HUI3 = Health Utility Index; IADL = Instrumental Activities of Daily Living; ICD-10 = International Classification of Diseases, Tenth Revision; M3 Index = Multimorbidity Measure Index; mCCI = modified Charlson Comorbidity Index; MDMS = Multidimensional Multimorbidity Score; MM = Multimorbidity; MWI = Multimorbidity-Weighted Index; PC = Primary Care; Pra tool = Probability of repeated admission risk prediction tool; QOF = Quality and Outcomes Framework; QoL = Quality of Life; RUB = Resource Utilisation Band; RxRisk-V = A Veterans Association adapted pharmacy-based case-mix instrument; SRH = Self-Rated Health

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Table 2 summarises the 33 instruments that were identified from all the studies. Table 3 provides a summary of multimorbidity instruments and their associations with the outcomes measured from all the included studies.

Simple counts of individual conditions

Disease count was based on the total number of all the conditions an individual had, usually from a pre-specified list of chronic conditions. It was used in 59 out of the 96 studies (61.5%). Disease count was reported to be associated with activity limitations, continuity of care, disability, healthcare cost, healthcare utilisation, medications, mental disorders, mortality, general health, physical function, quality of life, and self-rated health (Table 3).

Organ or system-based approaches

There were two instruments in this category. They were Cumulative Illness Rating Scale (CIRS)[23 24] and Organ Systems with chronic disease count (Organ-CDC).[38]

Weighted Indices

There were 17 unique weighted instruments found in the included studies. The original Charlson Comorbidity Index (CCI) with its different modifications was the most frequently used instrument and was used in 29 studies. The CCI was based on disease count, but the 17 conditions were weighted originally based on their impact on one-year mortality.[30] The final score was derived by the summation of all the weighted conditions. There were many

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Table 2. Description of instruments used for measurement of multimorbidity and the data sources and resources required

Category	Instrument	System/Condition based	Weightage; Scoring Method	Data Sources and resources required
A: Count of Individual Conditions				
A-1	DC	Condition (7-147)	Unweighted; Condition count	ATC list of conditions, Elixhauser list of conditions, EMR, GP records, Health service database, Hospital discharge abstract, Insurance claims or Questionnaires - telephone, face-to-face, mailed surveys. Participant involvement required
A-2	CCC[73]	Condition (6)	Unweighted; Based on AHRQ's clinical classification software & no. of conditions for each category	EMR
B: Organ or System-based Approaches				
B-3	Organ systems with CDC	Organ system (17)	Unweighted sum of organ systems	EMR
B-4	CIRS[113 114]	Body systems (13)	1-5 (based on severity of the condition); different weightage for diseases	EMR
C: Weighted Indices				
C-5	ACE[115]	Condition (27)	1-3 (based on severity of most severe condition); Highest score of single item	Insurance Claims' database
C-6	Cambridge MM Score[79]	Condition (20)	Weighted based on three different outcomes – primary care consultation, unplanned admission & mortality	EMR linked to mortality, hospital admission and socioeconomic deprivation
C-7	CCI[30]	Condition (19)	1-6 (based on impact on 1-year mortality (RR) – original); Sum of weighted conditions	Administrative database, EMR, Medical chart review, orInterviews or postal questionnaire where participant involvement is required
C-8	CLS[36]	Condition (98)	Based on impact for mortality (hazard ratio); Sum of beta coefficients of each category	Linked patients' records of all primary care events, hospital admissions and causes of death.
C-9	DBIS[116 117]	Conditions (25-28)	Weighted according to the degree in which each condition interferes with daily activities	Patient involvement in the questionnaire is required
C-10	EI (original & modified)[118]	Condition (21-31)	Based on impact on in-hospital mortality; Summing of beta coefficients	Insurance Claims' or Medical services database
C-11	ERA[35]	Condition (6-9)	Weighted (based on impact on future hospitalisation); Sum of weighted regression coefficients	EMR and administrative database
C-12	HCC[119]	Condition (70)	Based on Medicare capitation payments for health expenditure; More severe manifestations of a condition dominating (and zeroing out the effect of) less serious ones. Other diseases are summed additively.	EMR & HCC software licensing and fees
C-13	M3 Index[91]	Condition (55)	Weighted based on 1-year mortality; Summing of beta coefficients	Linked patients' records
C-14	MDMS[99]	Condition (7 chronic conditions, 2 health behaviours for 1st dimension & 5 symptoms for 2nd dimension)	Weighted but not based on any specific outcome; Sum of the value for the weighted absolute contributions of each of the dimensions.	Standardised medical evaluation (Interviewer-administered); Participant involvement is required
C-15	MM weighted by ADL Scale[120]	Condition (19)	Weighted based on OARS functional status scale measuring ADL; Sum of weighted conditions	Face-to-face or telephone interviews where participant involvement is required

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Table 2. Description of instruments used for measurement of multimorbidity and the data sources and resources required (Continued)

Category	Instrument	System/Condition based	Weightage; Scoring Method	Data Sources and resources required
C-16	MM weighted by HUI[121]	Condition (19)	Weighted based on correlation with health utility index; Sum of weighted conditions	Face-to-face or telephone interviews where participant involvement is required
C-17	MM weighted by HUI betas[121]	Condition (19)	Weighted based on correlation with health utility index and adjusted for age and sex; Summing of beta coefficients	Face-to-face or telephone interviews where participant involvement is required
C-18	MWI[122]	Condition (81)	Weighted based on impact on SF-36 physical functioning scale; Sum of weights	Interviewer-administered or mail questionnaire where participant involvement is required
C-19	QOF standard (weighted)[28]	Condition (14)	0-6, based on impact on 1-year mortality (RR); Sum of weighted conditions	EMR
C-20	QOF extended (weighted)[28]	Condition (9)	1-3, based on impact on 1-year mortality (RR); Sum of weighted conditions	EMR
C-21	Severity Burden Score[116]	Condition (21)	Sum of weights of diseases by the level of interference for each condition	Interviewer-administered structured questionnaire by nurses where participant involvement is required
D: Other Approaches (D1 = Case Mix, D2 = Pharmaceutical-based)				
D1-22	ACG[123 124]	Condition (93 mutually exclusive ACGs. Some are modified to 68 ACGs)	Incorporated into ACGs based on impact on resource use (proprietary); Variable	EMR & ACG software licensing and fees
D1-23	ADG[123 124]	Condition (32 groups)	Based on duration, severity, diagnostic certainty, aetiology, and need for specialty care; Variable	EMR & ACG software licensing and fees
D1-24	CRG[125]	NA; Diagnostic categories derived from organ systems or clinical category (37)	Pre-formulated based on the 3M clinical risk groups and consists of 9 core health ranks	EMR - inpatient and outpatient & 3M Clinical Risk Grouping software v1.6 and service fees
D1-25	Adjusted Morbidity Groups (GMA)[126]	NA; mutually exclusive categories (31)	Based on multimorbidity and levels of patient complexity	Registry data
D1-26	HM[48]	Condition (NS)	Only MN tier 4 + MN tier 3 with ERA > 10; Variable	EMR, HCC software licensing, fees & administrative data
D1-27	HSMI[60]	Condition (73 chronic and acute conditions)	Based upon yearly health care costs directly derived from primary care setting; Sum of regression coefficients (range from -0.06 to 1.04)	EMR
D1-28	Minnesota Tiering[123 124]	Condition (NS)	Grouping patients into 'complexity tiers' based on the number of major condition categories; Condition Count	EMR or administrative data & MN Tiering software licensing and fees
D1-29	Resource Utilisation Band[123 124]	Condition (6 mutually exclusive bands)	Based upon ACG algorithm on impact on resource use (proprietary); Variable	EMR & ACG software licensing and fees
D2-30	CDS[127]	Condition (17)	Weighted 1-5; Sum of weights based on pharmacological database	Prescription drug database
D2-31	Drug Count	NA; Variable. Some may be based on pharmacologic-therapeutic classification system	Weighted; Medication count	Self-reported questionnaire where participant involvement is required
D2-32	Modified Pra tool using RxRisk-V[103 128 129]	NA; Pra tool + RxRisk-V	Weighted due to RxRisk-V; 4 categories	GP medical record + linked pharmacy claims database

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Table 2. Description of instruments used for measurement of multimorbidity and the data sources and resources required (Continued)

Category	Instrument	System/Condition based	Weightage; Scoring Method	Data Sources and resources required
D2-33	RxRisk-V[129]	NA; WHO-ATC classification system	Weighted according to the diagnostic group of drugs to predict future health care costs; Sum of weights	GP medical record + linked pharmacy claims database

Note. ACE = Adult Comorbidity Evaluation; ACG = Adjusted Clinical Groups; ADG = Aggregated Diagnosis Groups; ADL = Activities of Daily Living; ATC = Anatomical Therapeutic Chemical; CCC = Chronic Condition Count; CCI = Charlson Comorbidity Index; mCCI = modified Charlson Comorbidity Index; CLS = Comorbidity Linked Score; CDC= Chronic Disease Count; CDS = Chronic Disease Score; CGI-S = Clinical Global Impression-Severity Scale; CIRS = Cumulative Illness Rating Scale; CRG = Clinical Risk Groups; DBIS = Disease Burden Impact Scale; DC = Disease Count; EDC = Expanded Diagnosis Clusters; EI = Elixhauser Index; EMR = Electronic Medical Records; ERA = Elder Risk Assessment; GP = General Practitioner; HCC = Hierarchical Condition Categories; HM = Hybrid Model (MN Tier + ERA); HUI = Health Utility Index; HSMI = Health Search Morbidity Index; mCCI = Modified Charlson Comorbidity Index; MDMS = Multidimensional Multimorbidity Score; MM = Multimorbidity; M3 Index = Multimorbidity Measure Index; MWI = Multimorbidity-Weighted Index; OARS = Older Americans Resources and Services; Pra tool = Probability of repeated admission risk prediction tool; QOF = Quality and Outcomes Framework; RR = Risk Ratio; RxRisk-V = A Veterans Association adapted pharmacy-based case-mix instrument; SF-36 = 36-item Short Form Survey

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Table 3. Summary of multimorbidity instruments and their associations with outcomes measured from all the included studies

Multimorbidity Measures	Association between Outcomes and Multimorbidity	
	Evidence of an association	No Evidence of an association
A = Count of Individual Conditions		
DC (Many different groupings ranging from 7[44 58 130] to 147[46] conditions and some are further categorised[48])	ADL limitations[19], Activity limitations[31], Affective well-being[111], Cognitive Function[64], Continuity of care (3-years)[87], Deficits of perceived general health[86], Depression[74], Depressive symptoms[86 47], Disability[23 45 66 67], Emergency hospital admission (2-years)[104], Fall-related injuries[53], Fall risk[58], Frequency of contacts with physicians (1-year)[100], Functional capacity[75], Functional decline (2-years)[104], Functional Impairment[93], Functional Limitations[56], Future physical functioning[110], General health[31], Healthcare costs[51 73], Health-related quality of life[49 57 63 72 98], Hospitalisation (3-years)[24], Hospital admissions (1-year)[77 78 81], Hospital outpatient visits (1-year)[97], Hospitalisation/Emergency Department Visits[97], Life satisfaction[112], Mental distress[31], Mortality (1-year)[81], (3-years)[26 24], (5 years)[92], (10-years)[110], No. of contacts with general practice (1-year)[101], No. of medications prescribed (1-year)[101], No. of mentally unhealthy days[19 20], No. of physically unhealthy days[19 20], No. of different ambulatory physicians contacted (1-year)[100], No. of primary care consultations (1-year)[26], (3-years)[26], No. of referrals (1-year)[101], Outpatient/Inpatient service use[32], Physical distress[31], Physical function[34], Prescription costs[52], Perceived health status[112], Presence of mental health disorder[46], Primary care consultations (1-year period)[87], (3-years)[87], Primary care experience – Self-reported[65], Primary healthcare cost[25], Primary health care use[71], Potentially preventable unplanned admission (1-year period)[77], Quality-adjusted life years[54 55], Quality of life[75 83 80 84], Self-rated health[102], Self-reported falls (12-month)[90], Symptom burden[83], Self-rated Health[44 67], Self-perceived health[75], Total number of consultation[18], total health care costs[18]	Functional decline[24], Quality of cardiovascular preventive care[94], Quality of preventive care [94]
CCC	Healthcare costs[48], Hospital admissions (1-year)[48], No. of emergency department visits (1-year)[48], Readmission within 30 days (1-year)[48]	
B = Organ or System-Based Approaches		
Organ systems with CDC	Presence of depressive or anxiety disorder[38]	
CIRS	Disability [23], Frailty[23], Health care utilisation[33], Hospitalisation (3-years)[24], Mortality[24]	Functional decline[24]
C = Weighted Indices		
ACE	Healthcare expenditure[51]	
Cambridge MM Score	Mortality[79], Primary care consultation[79], Unplanned admission[79]	

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Table 3. Summary of multimorbidity instruments and their associations with outcomes measured from all the included studies (Continued)

Multimorbidity Measures	Association between Outcomes and Multimorbidity	
	Evidence of an association	No evidence of an association
CCI	Ambulatory care sensitive hospitalisations (acute & chronic)[88], Disability [23], Emergency department visits (1-year)[48], Emergency hospital admission (2-years)[103], Frailty[23], Functional decline (2-years)[103], Future physical functioning[110], Healthcare expenditure[48], Hip fractures[85], Hospitalisation (1-year)[44 48 62 81 91], Hospitalisation (3-years)[24], Mortality (1-year)[28 43 81 91], Mortality (3-years)[24], (5-years)[29 43], (10-years)[29 43], (15, 20, 25-years)[29], No. of primary care consultations (3-years)[26], No. of primary care physician visits (1-year) [89], No. of specialist visits (1-year)[89], Potentially preventable unplanned admission (1-year)[78], Presence of critical illness[22], Primary healthcare cost[25], Mortality (1-year)[36 37], (3-years)[26 39], (5-years)[37 40 42], (7-years)[68], (10-years)[110], Readmission within 30 days (1-year)[48], Successful aging[41]	Functional decline[24], Primary care visits[50], Quality of cardiovascular preventive care[94], Quality of preventive care[94]
CLS	Mortality (1-year)[36]	
DBIS	Quality of life[80]	
EI (Original & Modified)	Hospitalisation (1-year)[81 91], Mortality (1-year)[36 81 91]	
ERA	Healthcare expenditure [48], Mortality (2-years)[130], No. of days hospitalised (1-year)[35], No. of emergency department visits (1-year)[35 48], No. of hospital admissions (1-year)[35 48], No. of hospital visits (1-year)[35 48], Nursing home placement (2-years)[130], Presence of critical illness[22], Readmission within 30 days (1-year)[48]	
HCC	Hospitalisation (1-year)[48], ED visits (1-year)[48], Readmission within 30 days (1-year)[48], Healthcare expenditure (1-year)[48]	
M3 Index	Hospitalisation (1-year)[91], Mortality (1-year)[91]	
MDMS	Sickness absence episodes taken in 2 years (male)[99]	Sickness absence episodes taken in 2 years (female)[99]
MM weighted by ADL Scale	Life satisfaction[112], Perceived health status[112]	
MM weighted by HUI	Life satisfaction[112], Perceived health status[112]	
MM weighted by HUI betas	Life satisfaction[112], Perceived health status[112]	
MWI	ADL limitations[105], IADL limitations[105], Mortality (10-years)[110], Cognitive performance[105], Future physical functioning[110 109], Grip strength[105], Health-related quality of life[107], Mortality[109], Subjective physical functioning[105], Suicide mortality[107]	Gait speed[110]
QOF (Standard)	Mortality (1-year) [28]	
QOF (Extended)	Mortality (1-year) [28]	
Severity Burden Score	Mental component score (SF-12)[76]	

Table 3. Summary of multimorbidity instruments and their associations with outcomes measured from all the included studies (Continued)

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Multimorbidity Measures	Association between Outcomes and Multimorbidity	
	Evidence of an association	No evidence of an association
D = Other Approaches (D-1 = Case-Mix, D2 = Pharmaceutical-based)		
ACG	Hospitalisation (1-year)[62], Mortality (3-years)[26], No. of primary care consultations (3-years)[26], Primary healthcare cost[25], Readmission within 30 days (1-year)[48]	
ADG	Hospitalisation (1-year)[89], No. of primary care physician visits (1-year)[89], No. of specialist visits (1-year)[89]	
CRG	Healthcare expenditure [27], HRQoL using EQ-5D-3L[69]	
Adjusted Morbidity Groups (GMA)	Use of healthcare resources[70]	
HM	Emergency department visits (1-year)[48], Healthcare expenditure[48], Hospitalisation (1-year)[48], Readmission within 30 days (1-year)[48]	
HSMI	Healthcare cost (primary care)[60]	
Minnesota Tiering	Emergency department visits (1-year)[48 97], Healthcare expenditure[48], Hospitalisation (1-year)[48 97], Readmission within 30 days (1-year)[48]	
Resource Utilisation Band	Fee-for-service expenditures[59], Primary healthcare cost[25], Mortality (3-years)[26], Number of primary care consultations (3-years)[26], Registered active listing in primary care[82], Registered active listing in all healthcare[82]	
CDS	Hospitalisation (1-year)[81], Mortality (1-year) [81]	
Drug Count	Emergency hospital admission (2-years)[104], Functional decline (2-years)[104], Hospitalisation (1-year)[81], Mortality (1-year)[81], (3-years)[26], No. of primary care consultations (3-years)[26]	
Pra tool Modified using RxRisk-V	Emergency hospital admission (1-year)[103]	
RxRisk-V	Emergency hospital admission (2-years)[104], Functional decline (2-years)[104]	

Note. ACE-27 = Adult Comorbidity Evaluation; ACG = Adjusted Clinical Groups; ADG = Aggregated Diagnosis Groups; ADL = Activities of Daily Living; CCC = Chronic Condition Count; CCI = Charlson Comorbidity Index; CDC= Chronic Disease Count; CDS = Chronic Disease Score; CIRS = Cumulative Illness Rating Scale; CLS = Comorbidity Linked Score; CRG = Clinical Risk Groups; DBIS = Disease Burden Impact Scale; DC = Disease Count; Elixhauser Index = EI; ERA = Elder Risk Assessment; HCC = Hierarchical Condition Categories; HM = Hybrid Model (MN Tier + ERA); HRQoL = Health-Related Quality of Life; HSMI = Health Search Morbidity Index; HUI = Health Utility Index; MM = Multimorbidity; MDMS = Multidimensional Multimorbidity Score; M3 Index = Multimorbidity Measure (M3) Index; MWI = Multimorbidity-Weighted Index; Pra tool = Probability of repeated admission risk prediction tool; QOF = Quality and Outcomes Framework; RxRisk-V = A Veterans Association adapted pharmacy-based case-mix instrument; SF-12 = Short Form-12

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variations and modifications of the score including the addition of psychosocial factors. The CCI instrument was found to be associated with multiple outcomes other than one-year mortality.

Most of the other weighted index instruments were novel, like the Multimorbidity-Weighted Index (MWI), in which the investigators built multivariable prognostic models from a set of potential predictor conditions and weighted the conditions based on an outcome of clinical interest. The most common outcomes chosen were mortality and physical function. Other outcomes included health expenditure,[48] health utility index,[112] and severity of the most severe condition.[51] The Multidimensional Multimorbidity Score (MDMS)[99] was unique as it was weighted based on health behaviours and patient symptoms and not based on any specific outcome.

Other Approaches to measuring multimorbidity

Other approaches included case-mix and pharmaceutical-based instruments. For case-mix approach, the Adjusted Clinical Group (ACG) and Resource Utilisation Band (RUB) were the most commonly used instruments.[131] Most of the case-mix instruments required proprietary software licenses from the United States of America and obtained data from electronic medical records or administrative data. The Clinical Risk Groups instrument was similar but took into account the severity of individual conditions.[125]

The second group of instruments in this category was related to pharmaceutical data. The most frequent type was the unweighted drug count. The other three (Chronic disease score, A Veterans Association adapted pharmacy-based case-mix instrument like RxRisk-V, and modified Probability of repeated admission risk prediction tool using RxRisk-V) were all

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weighted indices. Except for the drug count that was based on a self-report questionnaire, the rest required a prescription drug database to obtain the data.

Outcomes

We classified the 150 outcomes into 17 categories as reported in the core outcomes set of multimorbidity research (COSmm).[10] The most commonly reported outcomes were health care use (n=45), mortality (n=18), health-related quality of life (n=18), and physical function (n=13). The different studies unanimously showed that higher levels of multimorbidity was associated with higher health care use and mortality, lower health-related quality of life, and poorer physical function. Seven outcomes in the COSmm were not found in all the 96 studies. These were treatment burden, self-management behaviour, self-efficacy, adherence, communications, shared decision making and prioritisation. There were 19 outcomes that were not described in the COSmm. These included cognitive function, risk of suicide, frailty and falls. The outcomes not found to have any association with the instruments for measuring the level of multimorbidity were preventive care[94], sickness absence episodes (female)[99], and gait speed[110].

DISCUSSION

Summary of Findings

Thirty-three unique instruments for measuring the level of multimorbidity were identified and categorised according to the classification by Sarfati.[8] The most commonly used instrument was 'Disease Count'. It was also the only instrument that was associated with the three essential outcomes from the core outcomes set of multimorbidity research (COSmm)[10] i.e., quality of life, mental health, and mortality.

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Comparison with previous research

Although the most common instrument identified in this systematic review was similar to that of Huntley et al.,[12] several instruments including Duke Severity of Illness Checklist (DUSOI) and Functional Comorbidity Index (FCI) identified in their article were not found in this systematic review. The possible reasons for not finding these instruments in this review could be due to the lack of interest in the instrument by the research community in recent years (to our knowledge, the last publication using DUSOI was in 2004[132]), or the exclusion of studies specifying an index condition.

Advantages and disadvantages of selected instruments

Disease count

The advantage of using 'disease count' is its simplicity and the ease of data ascertainment with minimal resources required. However, using 'disease count' does not consider the severity of each condition where the complexity of multimorbidity may not be properly addressed.[133] The other disadvantage noted was the lack of transparency in the operational definition of multimorbidity, especially regarding the list of conditions considered for multimorbidity, and the cut-points used. Despite its simplicity, the level of multimorbidity measured using 'disease count' was the only instrument that was found to be associated with the three essential core outcomes (quality of life, mental health, and mortality).

Weighted Indices

The common weighted indices identified in this systematic review were Charlson Comorbidity Index (CCI), Elders Risk Assessment (ERA), Elixhauser Index (EI) and

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Multimorbidity-weighted Index (MWI). These weighted indices were often utilised in prognostic models to build complex multivariable regression models in which the weights were calculated from hazard ratios, odds ratios, or regression coefficients.[134]

The advantage of these weighted indices is that the weights allow the adaptation of an index to a specific outcome. An investigator could recalibrate the correct weight by creating a prognostic model to produce a contextualised instrument for a different setting. Prognostic models can provide clinically relevant risk stratification and help to allocate resources.[135] The disadvantage of such indices is that calculated weights are greatly influenced by the population, outcomes used, and the instrument's original conception and purpose, hampering the ability to compare across studies.

Case-Mix

The Adjusted Clinical Group (ACG) system has a good track record in the United States and several other countries, especially for measuring the outcomes of healthcare utilisation. However, the instrument is proprietary, and the exact algorithm of the instrument is not open to the public and may not be suitable in certain settings. The Clinical Risk Group (CRG) system has a good track record in Spain. It measures the severity of each condition and its algorithm is fully transparent. The common disadvantage of both systems is the financial costs involved in obtaining the license.

Pharmaceutical-based instruments

Medication-based indices include versions of the Chronic Disease Score,[127] which later became known as the RxRisk,[136] and its adaptation for use in the veteran population, the RxRisk-V.[137] Like the disease count, its main advantage is the ease of use with

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minimal resources required. However, many studies were not transparent regarding which type of drugs were included.

Data sources

Data sources used by these instruments included medical record information, patient self-report, clinical judgment, and large administrative databases. Each data source has its inherent advantages and disadvantages. For patient self-report, patients with cognitive impairment may under-report symptoms and may be seen less frequently by their physicians, resulting in an under-recognition or under-treatment of conditions.[138] It has also been shown that health administrative data based on billing system underestimated the prevalence of many chronic conditions.[139]

The available data in a particular setting may strongly influence the ultimate instrument chosen for multimorbidity research. As there is currently no consensus on the gold standard for sources of data, it is difficult to assess which data source was superior from this review.

Outcomes

There were 17 multimorbidity outcomes identified by a Delphi process involving a panel of international experts in multimorbidity intervention studies.[10] However, only ten out of the seventeen outcomes were reported in the 96 studies identified in this systematic review. The most common outcome that was investigated was health care use. The seven missing outcomes belong to 'patient-reported impact and behaviours' and 'consultation-related' outcome groups, most likely indicating that there is a dearth of multimorbidity studies looking at these two groups of outcomes measures.

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Clinical Implications

Ideally, a single instrument measuring the level of multimorbidity should be able to predict a variety of relevant outcomes. However, Byles et al.[140] reported that a single instrument could not be used to predict different outcomes, in different patient groups and settings, unless different weights were assigned to these factors in calculating a score. Such multiple-scoring instruments may be the way forward for validation of prognostic models for different outcomes and different populations with established multimorbidity instruments. For example, depending on the outcome, study population and setting, the choice of conditions included in the multiple-scoring instrument should include those with a high prevalence in that study population and the weights should be determined by their significant impact (i.e. outcome) on the affected population.

For pragmatic reasons, the final selection of the conditions to be included in such a multiple-scoring instrument may still have to take into account the availability of relevant and reliable data. A certain degree of reductionism will also have to be accepted because a single instrument will not be able to encompass all the nuances of the different interactions of chronic conditions on an individual living in his/her unique milieu. We recommend that researchers perform validation studies using the instruments listed in this systematic review to adjust the weights according to the specific outcome of interest for the study population relevant to their setting.

Strengths and limitations of the study

The main strengths of this systematic review were the involvement of a health science librarian in our search strategy, a published protocol, adherence to the protocol without major changes during the systematic review process,[141] and the critical appraisal of all the primary studies with a risk of bias assessment tool.

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The systematic review had several limitations. We excluded grey literature and included only studies that were published in the English language. We also did not contact authors directly for a suggestion of studies, nor identified a list of instruments from the preliminary search and then performed an additional search using the same databases.[142] Additionally, this systematic review did not review the validity and reliability of all the instruments as it was beyond the scope of the intended work. We have, however, included the references of the original articles or validation studies in Table 2 for each of the instrument where available. Finally, this review specifically aimed to look at the association of the level of multimorbidity as the main independent variable and excluded the level of multimorbidity as a mediating, confounding or effect-modifying variable. This strict criterion excluded 17 studies (Figure 1) as a result. Excluding these 17 studies did not alter the findings as the instruments used in all the 17 studies were Disease Count (n=9), Cumulative Illness Rating Scale (CIRS, n=3), Charlson Comorbdity Index (CCI, n=3), Elixhauser Index (EI, n=1) and Aggregated Diagnosis Groups (ADG, n=1) where no new instruments were identified.

CONCLUSIONS

In this systematic review, we found 33 instruments for measuring the level of multimorbidity in community-dwelling individuals that predict or explore the association of multimorbidity with at least one specified outcome. Disease count and weighted indices like the Charlson Comorbidity Index (CCI), the ERA, and Elixhauser Index (EI) were commonly used for measuring the level of multimorbidity. Other approaches to measuring the level of multimorbidity included case-mix or pharmaceutical-based instruments.

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We found continuing interest in measuring the level of multimorbidity with disease count and drug count. There has also been a rise in the development of novel weighted indices using prognostic models or validation of existing well-established instruments like the CCI over the last few years. There is currently an absence of a gold standard for where to obtain chronic disease information. The most suitable instrument will depend on the specified outcome of interest, the study population, and the type of data and resources available.

Finally, there is still much work to improve on the body of knowledge of multimorbidity when most investigators in the last decade measured multimorbidity without including some of the important outcome measures of multimorbidity. We also suggest that a clear description of the instruments is required in the publication of multimorbidity studies to counter the frequent lack of information currently seen so as to contribute to robust multimorbidity research in future.

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Contributors:

All authors made substantial contribution to the design of the work. BLR, MF and MS mentored and provided guidance to the review. ESL, together with a health science librarian, created the search strategy and conducted the search. ESL, EH, TSH, HLK and FYW screened articles, extracted and analysed data. ESL drafted the manuscript with valuable inputs from BLR, MF and MS and all authors reviewed drafts and approved the final version.

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Figure 1 – PRISMA Flow Diagram

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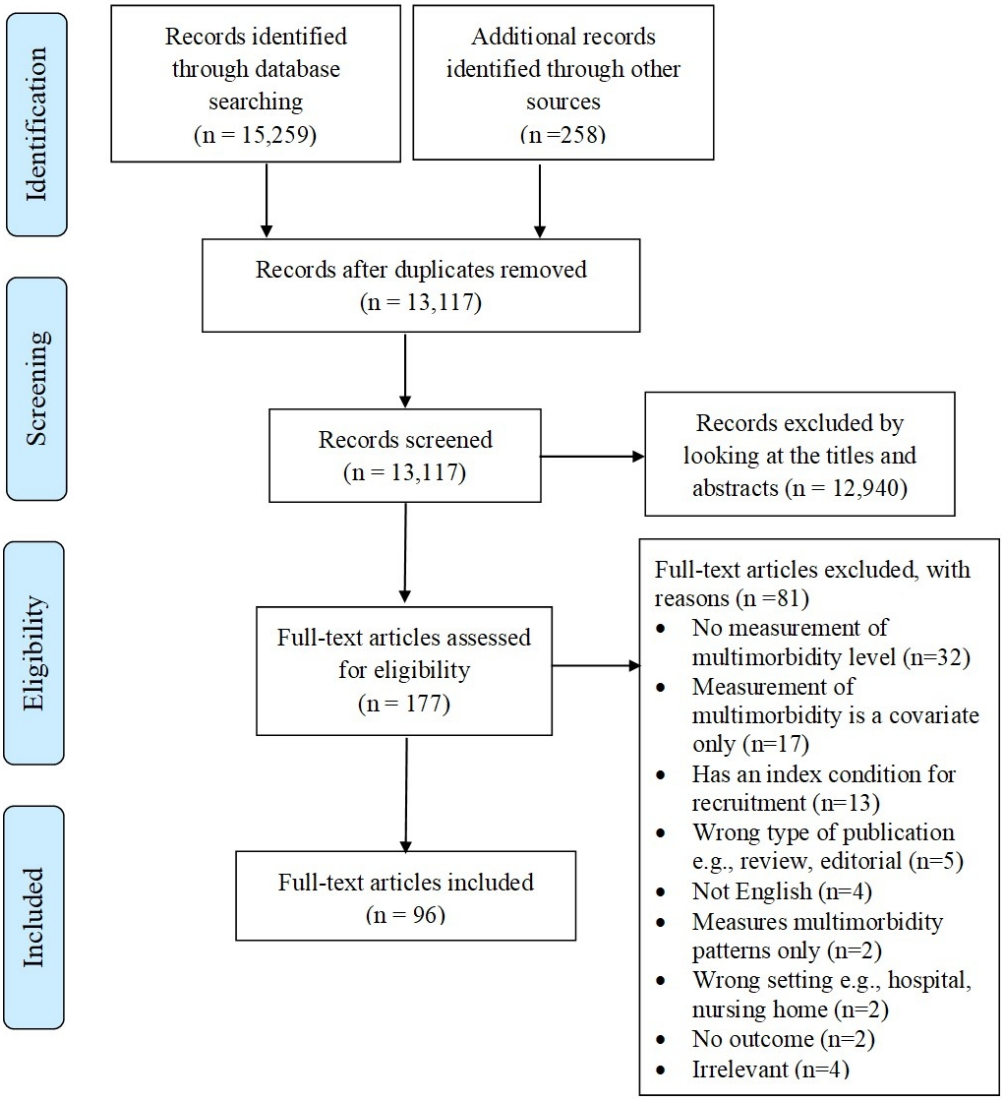


Figure 1 - PRISMA Flow Diagram

172x197mm (150 x 150 DPI)

Appendix 1. Search Strategy

Table 1. Medline Search 1946 to October 23, 2020

#	Searches	Results
1	exp comorbidity/ or multiple chronic conditions/	111191
2	(multimorbid* or multi-morbid* or comorbid* or co-morbid*).ab,ti,kw.	180089
3	((multiple or coexist* or co-exist* or concurrent* or simultaneous*) adj2 (disease* or illness* or diagnos* or condition* or morbid* or disorder*)).ab,ti,kw.	52139
4	1 or 2 or 3	292316
5	primary health care/ or "continuity of patient care"/ or exp general practice/ or ambulatory care/ or physicians, family/ or physicians, primary care/ or community health services/ or general practitioners/	249484
6	((ambulatory or community or general or family or primary) adj2 (care or health* or practi* or physician* or medicine or center* or centre* or facilit* or clinic*)).ab,ti,kw.	391013
7	5 or 6	508037
8	epidemiologic measurements/ or risk assessment/ or "Outcome Assessment (Health Care)"/ or patient reported outcome measures/ or health status indicators/ or "severity of illness index"/ or sickness impact profile/ or diagnosis-related groups/ or case mix/	594279
9	((health status or risk or outcome* or sickness impact) adj2 (appraisal* or assessment* or evaluation* or measur* or index* or indic* or profile*)).ab,ti,kw.	398319
10	((severity or burden) adj2 (illness* or diseas* or disorder* or condition* or diagnos*)).ab,ti,kw.	85595
11	(charlson comorbidity index or charlson index or charlson score or elixhauser* or cumulative illness rating scale* or CIRS or adjusted clinical group* or ACG* or disease count or Duke severity* or DUSOI or casemix or case-mix).ab,ti,kw.	21547
12	8 or 9 or 10 or 11	939031
13	4 and 7 and 12	4862
14	limit 13 to (english language and yr="2010 - Current")	2827

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Table 2. Embase Classic+Embase 1947 to October 23, 2020

#	Searches	Results
1	comorbidity/ or multiple chronic conditions/	265602
2	(multimorbid* or multi-morbid* or comorbid* or co-morbid*).ab,ti,kw.	329397
3	((multiple or coexist* or co-exist* or concurrent* or simultaneous*) adj2 (disease* or illness* or diagnos* or condition* or morbid* or disorder*)).ab,ti,kw.	81506
4	1 or 2 or 3	493137
5	primary health care/ or patient care/ or primary medical care/ or general practice/ or general practitioner/ or ambulatory care/ or family medicine/ or community care/ or community health services/	664384
6	((ambulatory or community or general or family or primary) adj2 (care or health* or practi* or physician* or medicine or center* or centre* or facilit* or clinic*)).ab,ti,kw.	539525
7	5 or 6	943743
8	risk assessment/ or outcome assessment/ or patient reported outcome/ or health status indicator/ or disease activity score/ or global disease burden/ or organ dysfunction score/ or "severity of illness index"/ or sickness impact profile/ or general health status assessment/ or disease severity/ or diagnosis related group/ or charlson comorbidity index/ or comorbidity assessment/ or elixhauser comorbidity index/ or case mix/	1590115
9	((health status or risk or outcome* or sickness impact) adj2 (appraisal* or assessment* or evaluation* or measur* or index* or indic* or profile*)).ab,ti,kw.	547083
10	((severity or burden) adj2 (illness* or diseas* or disorder* or condition* or diagnos*)).ab,ti,kw.	138554
11	(charlson comorbidity index or charlson index or charlson score or elixhauser* or cumulative illness rating scale* or CIRS or adjusted clinical group* or ACG* or disease count or Duke severity* or DUSOI or casemix or case-mix).ab,ti,kw.	42616
12	8 or 9 or 10 or 11	2022565
13	4 and 7 and 12	12245
14	limit 13 to (english language and yr="2010 - Current")	8896

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Table 3. CINAHL – since inception to October 23, 2020

#	Searches	Results
1	(MH “comorbidity”)	38970
2	(multimorbid* or multi-morbid* or comorbid* or co-morbid*)	66442
3	(multiple or coexist* or co-exist* or concurrent* or simultaneous*) N2 (disease* or illness* or diagnos* or condition* or morbid* or disorder*)	12822
4	1 or 2 or 3	77266
5	((MH “primary health care”) or (MH “family practice”) or (MH “ambulatory care") or (MH “ambulatory care facilities") or (MM “community health services”) or (MM “community health centers”) or (MH “physicians, family”) or (MH “continuity of patient care"))	100224
6	((ambulatory or community or general or family or primary) N2 (care or health* or practi* or physician* or medicine or center* or centre* or facilit* or clinic*))	332346
7	5 or 6	340014
8	((MH “risk assessment”) or (MH "Outcome Assessment") or (MH “patient-reported outcomes) or (MH “health status indicators”) or (MH "severity of illness indices") or (MH “sickness impact profile”) or (MH “diagnosis-related groups”) or (MH “case mix”))	102350
9	((health status or risk or outcome* or sickness impact) N2 (appraisal* or assessment* or evaluation* or measur* or index* or indic* or profile*))	340802
10	(severity or burden) N2 (illness* or diseas* or disorder* or condition* or diagnos*)	68742
11	(charlson comorbidity index or charlson index or charlson score or elixhauser* or cumulative illness rating scale* or CIRS or adjusted clinical group* or ACG* or disease count or Duke severity* or DUSOI or casemix or case-mix)	6729
12	8 or 9 or 10 or 11	396387
13	4 and 7 and 12	4729
14	limit 13 to (english language and yr="2010 - Current")	3536

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Appendix 2. Coding Description for the Modified Newcastle-Ottawa Scale for Cohort Studies

Category	Description
Selection	
1) Representativeness of the sample	This item assesses the representativeness of sample in the community, not from some general population. a) Truly representative***** (e.g., everyone from the database) b) Somewhat representative***** (with at least 2 criteria but selection method was convincing due to random sampling) c) Selected group (e.g., only certain socio-economic groups or areas) d) No description of sampling strategy
2) Ascertainment of multimorbidity	This item assesses the method by which multimorbidity was confirmed. a) Secure record***** (e.g., GP questionnaire) b) Structured interview***** (e.g., interviewer-administered questionnaire) c) Written self-report (e.g., mailed survey, if items are unable to be confirmed by objective measure) d) No description / Other
3) Demonstration that outcome of interest was not present at start of study	A statement of no history of disease earns a star. In the case of mortality studies, outcome of interest is still the presence of a disease, rather than death. a) Yes***** b) No
Comparability	
1) Study controls for age and sex	Covariates must be adjusted for in the analysis. Statements of no differences between groups or that differences were not statistically significant are not sufficient for establishing comparability. Note: If the relative risk for the exposure of interest is adjusted for the covariates listed, then the groups will be considered to be comparable on each variable used. a) Yes***** b) No
2) Study controls for other factors	a) Yes***** b) No
Outcome	
1) Statistical test	Statistical test(s) must be clearly described and appropriate before assessing the other items under the Outcome category. a) Clearly described and appropriate b) Not described, incomplete or inappropriate
2) Assessment of outcome	This item assesses the method by which the outcome of interest was confirmed. For some outcomes, reference to the medical record is sufficient to satisfy the requirement for confirmation. This may not be adequate for other outcomes where reference to specific tests or measures would be required. a) Independent or blind assessment***** (e.g., interviewer-administered questionnaire) b) Record linkage***** (e.g., identified through ICD codes on database records) c) Self-report (i.e., no reference to original medical records) d) No description / Other

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3)	Was follow-up long enough for outcomes to occur?	An acceptable length of time was at least 1 year of follow-up. a) Yes***** b) No
4)	Adequacy of follow-up of cohorts	This item assesses the follow-up of the sample to ensure that losses are not related to the outcome. a) Complete follow-up - all subject accounted for***** b) Subjects lost to follow-up unlikely to introduce bias***** (Number lost ≤20% or description of those lost suggested no different from those followed.) c) Follow-up rate less than 80% and no description of those lost d) No statement

Thresholds for converting the modified Newcastle-Ottawa scales to Good, Fair, and Poor Quality:

- Good Quality - 2-3 stars in Selection category (Representativeness of Sample item must be fulfilled) AND 1-2 stars in Comparability category AND 2-3 stars in Outcome category.
- Fair Quality - 1 star in Selection category AND 1-2 stars in Comparability category AND 2-3 stars in Outcome category.
- Poor Quality - 0 star in Selection category OR 0 star in Comparability category OR 0-2 stars in Outcome category.

Wells GA, Shea B, O'Connell D, Peterson J, Welch V, Losos M, Tugwell P. The Newcastle-Ottawa Scale (NOS) for assessing the quality of nonrandomized studies in meta-analysis. Available from: http://www.ohri.ca/programs/clinical_epidemiology/oxford.asp

Modifications from original NOS include:

- 'Statistical test' was added to the Outcome category
- 'Representativeness of sample' item under Selection category and 'Statistical test' item under Outcome category must both be fulfilled for study to be considered Good Quality.
- 'Selection of the non-exposed cohort' item was removed from Selection category as most studies did not describe a non-exposed cohort, and this review sought to compare the different levels of multimorbidity within the group.
- 'Representativeness of the exposed cohort' item under Selection category was renamed 'Representativeness of the sample' since the 'non-exposed cohort' was removed above.
- 'Ascertainment of exposure' under the Selection category was renamed as 'Ascertainment of multimorbidity' to specify that multimorbidity is the exposure in this review.
- 'Study controls for age and sex' and 'Study controls for other factors' were revised to be items under Comparability category
- The thresholds for converting the scales to quality were amended accordingly due to the above modifications.

Consensus:

- The study team decided that a period of one-year was a reasonable period of follow-up under 'Was follow-up long enough for outcomes to occur?' item under Outcome category.

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Appendix 3. Coding Description for the Modified Newcastle-Ottawa Scale for Cross-sectional Studies

Category	Description
Selection	
1) Representativeness of the sample	This item assesses the representativeness of sample in the specified population, not from some general population. a) Truly representative***** (e.g., everyone from the database, random sampling) b) Somewhat representative***** (with at least 2 criteria but selection method was convincing due to random sampling) c) Selected group (e.g., only certain socio-economic groups or areas) d) No description of sampling strategy
2) Ascertainment of multimorbidity	This item assesses the method by which multimorbidity was confirmed. a) Secure record***** (e.g., Clinical records, GP questionnaire) b) Structured interview***** (e.g., interviewer-administered questionnaire) c) Written self-report (e.g., mailed survey, if items are unable to be confirmed by objective measure) d) No description / Other
3) Sample Size	If there is no description, a reported sample size of 800 and above is satisfactory. a) Justified and satisfactory***** b) Not justified
4) Non-respondents	Acceptable response rates for surveys through various methods† • In-person: 57% • Mail: 50% • Average: 33% • Email: 30% • Internet: 29% • Telephone: 18% • In-app: 13% a) Comparability between respondents and non-respondents characteristics is established, and the response rate is satisfactory.***** b) The response rate is unsatisfactory, or the comparability between respondents and non-respondents is unsatisfactory. c) No description of the response rate or the characteristics of the responders and the non-responders
Comparability	
1) Study controls for age and sex	Confounders must be adjusted for in the analysis. Statements of no differences between groups or that differences were not statistically significant are not sufficient for establishing comparability. Note: If the relative risk for the exposure of interest is adjusted for the confounders listed, then the groups will be considered to be comparable on each variable used in the adjustment. a) Yes***** b) No
2) Study controls for other factors	a) Yes***** b) No

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Outcome	
1) Statistical test	Statistical test(s) must be clearly described and appropriate before assessing the other items under the Outcome category. a) Clearly described and appropriate b) Not described, incomplete or inappropriate
2) Assessment of outcome	This item assesses the method by which the outcome of interest was confirmed. For some outcomes, reference to the medical record is sufficient to satisfy the requirement for confirmation. This may not be adequate for other outcomes where reference to specific tests or measures would be required. a) Independent or blind assessment***** b) Record linkage***** (e.g., identified through ICD codes on database records) c) Self-report (i.e., no reference to original medical records) d) No description / Other

Thresholds for converting the modified Newcastle-Ottawa scales to Good, Fair, and Poor Quality:

- Good Quality – 3-4 stars in Selection category (Representativeness of Sample item must be fulfilled) AND 1-2 stars in Comparability category AND 1 star in Outcome category.
- Fair Quality - 2 stars in Selection category AND 1-2 stars in Comparability category AND 1 star in Outcome category.
- Poor Quality – 0-1 star in Selection category OR 0 star in Comparability category OR 0 star in Outcome category.

†Lindermann N. What's the average survey response rate? [2018 benchmark]. SurveyAnyplace. Available from: https://surveyanyplace.com/average-survey-response-rate/.

Alshabanat A, Zafari Z, Albanyan O, Dairi M, FitzGerald JM. Asthma and COPD Overlap Syndrome (ACOS): a systematic review and meta-analysis. PLoS ONE. 2015;10(9):e0136065. Doi:10.1371/journal.pone.0136065.

Modifications from original NOS include:

- 'Ascertainment of exposure' item under Selection category was renamed as 'Ascertainment of multimorbidity' to specify that multimorbidity is the exposure in this review.
- The ratings for the 'Ascertainment of exposure' item under Selection category were revised to 'secure record, structured interview, written self-report, no description/other' to align with the Modified NOS for cohort studies in this review.
- 'Study controls for age and sex' and 'Study controls for other factors' items were revised to be items under Comparability category.
- 'Was follow-up long enough for outcomes to occur' item under Outcome category was renamed as 'Statistical test'. This item must be fulfilled for the study to be considered as Good Quality.
- The thresholds for converting the scales to quality were amended accordingly due to the above modifications.

Consensus:

- The study team decided on a list of acceptable response rates for various survey methods for the 'Non-respondents' item under Selection category.
- The study team decided that a sample size of 800 was reasonable under 'Sample size' item under Selection category.

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Appendix 4. Summary of risk of bias (RoB) appraisal of included cohort articles using the modified Newcastle-Ottawa Quality Assessment Scale (44 articles)

	Barile et al. (2013)[19]	Biehl et al. (2016)[22]	Boeckxstaens et al. (2015b)[24]	Brilleman et al. (2014)[25]	Brilleman and Salisbury(2013)[26]	Carey et al. (2013)[28]	Chapman et al. (2015)[29]	Charlson et al. (2014)[30]	Crane et al. (2010)[35]	Crooks et al. (2016)[36]	Crooks et al. (2015)[37]	Formiga et al. (2013)[39]	Formiga et al. (2011a)[40]	Formiga et al. (2016)[42]	Fraccaro et al. (2016)[43]	Haas et al. (2013)[48]	Hwang et al. (2015)[51]	Jennings et al. (2015)[53]	Jia et al. (2018)[54]	Jia and Lubetkin (2017)[55]	Lemke et al. (2012)[62]	Md Yusof et al. (2010)[68]	Monterde et al. (2020)[70]	Pavne et al. (2013)[77]	Pavne et al. (2014)[78]	Pavne et al. (2020)[79]	Ouail et al. (2011)[81]	Reves et al. (2014)[85]	Salisbury et al. (2011)[87]	Saver et al. (2014)[88]	Stanley and Sarfati (2017)[91]	St John et al. (2014)[92]	St John et al. (2019)[93]	Streit et al. (2014)[94]	Takahashi et al. (2011)[96]	Takahashi et al. (2016)[97]	Tvack et al. (2016)[98]	Wallace et al. (2016a)[103]	Wallace et al. (2016b)[104]	Wei et al. (2019a)[106]	Wei et al. (2019b)[107]	Wei et al. (2020a)[108]	Wei et al. (2020b)[109]	Wei and Mukamal (2018)[110]
Selection Criteria																																												
Representativeness of the sample	*	*	-	*	*	*	*	-	*	*	*	*	*	*	*	*	*	*	-	*	*	*	-	*	*	*	*	*	*	*	*	*	*	*	*	*	-	*	*	*	-	*	*	*
Ascertainment of Multimorbidity	-	*	*	*	*	*	*	-	*	*	*	*	*	*	*	*	*	*	*	-	*	*	-	*	*	*	*	*	*	*	*	*	*	*	*	*	-	*	*	*	*	*	*	-
Demonstration that outcome of interest was not present at the start of the study	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*
Comparability Criteria																																												
Study controls for age and sex	*	-	*	*	*	*	*	*	-	*	*	-	*	-	*	*	*	-	*	*	-	*	*	*	*	*	*	*	-	*	*	*	*	*	*	*	-	*	-	-	*	-	-	-
Study controls for others	*	*	-	*	*	*	*	*	*	-	*	*	*	*	*	*	*	*	-	-	*	*	*	*	*	-	*	*	*	*	-	*	*	*	*	*	*	*	-	*	-	*	-	*
Outcome Criteria																																												
Statistical test†	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	X	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Assessment of outcome	-	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*
Follow-up was long enough for outcomes to occur	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*
Adequacy of follow-up of cohorts	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	-	*	*	*	*	*	-	*	*	*	-	*	*	*	-	-	*	*
Overall RoB Judgement[a]	G	G	F	G	G	G	G	G	G	G	G	G	G	G	G	G	G	G	F	P	P	G	F	G	G	G	G	G	G	G	G	G	G	G	G	G	F	G	G	G	F	G	P	G

Note. G = Good; F = Fair; P = Poor.

†Statistical test must be clearly described and appropriate before assessing the other items under the Outcome category. [a] Good rating is given when there are 2 to 3 stars in Selection category (Representativeness of Sample must be fulfilled) AND 1 to 2 stars in Comparability category AND 2 to 3 stars in Outcome category; Fair rating is given when there are 1 star in Selection category AND 1 to 2 stars in Comparability category AND 2 to 3 stars in Outcome category; Poor rating is given when there are 0 star in Selection category OR 0 star in Comparability category OR 0 to 2 stars in Outcome category.

"*" Study satisfies the criteria; "-" Study did not satisfy the criteria; "✓" Statistical test is clearly described and appropriate; "X" Statistical test is not described, incomplete or inappropriate.

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Appendix 5. Summary of risk of bias (RoB) appraisal of included cross-sectional articles using the modified Newcastle-Ottawa Quality Assessment Scale (52 articles)

	Agborsangaya et al. (2013)[17]	Bahler et al. (2015)[18]	Barile et al. (2012)[20]	Barnett et al. (2012)[21]	Boeckxstaens et al. (2015a)[23]	Caballer-Tarazona et al. (2019)[27]	Chen et al. (2011)[31]	Chen et al. (2018)[32]	Chu (2018)[33]	Clynes et al. (2020)[34]	DiNapoli et al. (2017)[38]	Formiga et al. (2011b)[41]	Galenkamp et al. (2011)[44]	Garin et al. (2014)[45]	Glynn et al. (2011)[46]	Gunn et al. (2012)[47]	Hanmer et al. (2010)[49]	Hu et al. (2017)[50]	Isaacs et al. (2014)[52]	Jindai et al. (2016)[56]	Kim et al. (2012)[57]	Koiima et al. (2011)[58]	Kristensen et al. (2014)[59]	Lapi et al. (2015)[60]	Lawson et al. (2013)[61]	Li et al. (2016)[63]	Lonrinzi (2016)[64]	Macinko et al. (2019)[65]	Marengoni et al. (2011)[66]	McDaid et al. (2013)[67]	Milla-Perseguer (2019)[69]	Musgah et al. (2012)[71]	Mujica-Mota et al. (2015)[72]	Naessens et al. (2011)[73]	Østergaard and Foldager (2011)[74]	Palladino et al. (2019)[75]	Pati et al. (2019)[76]	Peters et al. (2018)[80]	Ranstad et al. (2014)[82]	Reinke et al. (2019)[83]	Renne and Gobbens (2018)[84]	Ryu et al. (2015)[86]	Shadmi et al. (2011)[89]	Siblev et al. (2014)[90]	Sullivan et al. (2012)[95]	Ubalde-Lopez et al. (2016)[99]	van den Bussche et al. (2014)[100]	van Oostrom et al. (2014)[101]	Vos et al. (2013)[102]	Wei et al. (2018)[105]	Wikman et al. (2011)[111]	Wister et al. (2015)[112]
Selection																																																					
Representativeness of the sample	*	*	*	*	-	*	*	-	*	-	*	*	*	*	*	*	*	-	*	-	*	-	*	*	*	*	*	*	*	*	*	*	-	-	*	*	*	*	-	*	*	*	*	*	-	*	*	*	*	*	*	*
Ascertainment of Multimorbidity	*	*	-	*	*	*	*	-	*	*	*	*	*	*	*	*	-	-	*	*	*	*	*	*	*	*	-	*	*	*	*	*	*	-	*	*	*	-	*	*	-	-	*	*	*	*	*	*	*	*
Sample size	*	*	*	*	-	*	*	*	*	-	*	-	*	*	*	*	*	*	-	*	*	*	-	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	-	*	*	*	*	*	*	*	-	*	*	*
Non-respondents	-	*	*	*	-	*	*	*	*	-	*	*	-	-	*	-	-	*	*	*	*	-	*	*	*	-	*	-	-	-	-	*	-	*	-	*	*	-	*	*	-	-	*	*	*	*	*	*	*	-
Comparability																																																				
Study controls for age and sex	*	*	*	*	*	-	*	*	*	-	*	-	*	*	*	*	*	*	-	-	-	*	*	*	*	-	*	*	*	*	*	-	*	-	-	*	*	*	*	*	*	-	*	*	*	*	*	-	*	*	*	*
Study controls for others	*	*	*	*	*	-	*	*	*	*	*	*	-	*	*	*	*	*	-	*	-	*	*	*	*	-	*	*	*	*	*	-	*	-	-	*	*	*	*	*	-	*	*	*	*	*	*	-	*	*	*	-
Outcome																																																				
Statistical test†	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	X	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Assessment of outcome	*	*	✓	*	*	*	*	-	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	-	*	*	*	*	*	*	*	*	*
Overall RoB Judgement[a]	G	G	G	G	P	P	G	P	G	P	G	G	G	G	G	F	F	F	P	G	P	P	G	G	G	P	G	G	G	G	G	P	F	P	P	G	G	F	G	G	P	G	G	P	G	G	G	G	P	G	G	G

Note. G = Good; F = Fair; P = Poor.

[†]Statistical test must be clearly described and appropriate before assessing the other items under the Outcome category. [a] Good rating is given when there are 3 to 4 stars in Selection category (Representativeness of Sample must be fulfilled) AND 1 to 2 stars in Comparability category AND 1 star in Outcome category; Fair rating is given when there are 2 stars in Selection category AND 1 to 2 stars in Comparability category AND 1 star in Outcome category; Poor rating is given when there are 0 to 1 star in Selection category OR 0 star in Comparability category OR 0 star in Outcome category.

"*" Study satisfies the criteria; "-" Study did not satisfy the criteria; "✓" Statistical test is clearly described and appropriate; "X" Statistical test is not described, incomplete or inappropriate

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	3
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	5
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	6
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	6
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	6-7
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	6-7
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	Appendix 1
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	6-7
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	7
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	7
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	7, Appendix 2,3
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	NA
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	NA

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	NA
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	NA
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	Figure 1
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	8-13
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	Appendix 4,5
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	14-22
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	NA
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	NA
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	NA
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	22-26
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	27
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	27-28
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	29

From: Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(7): e1000097. doi:10.1371/journal.pmed1000097

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